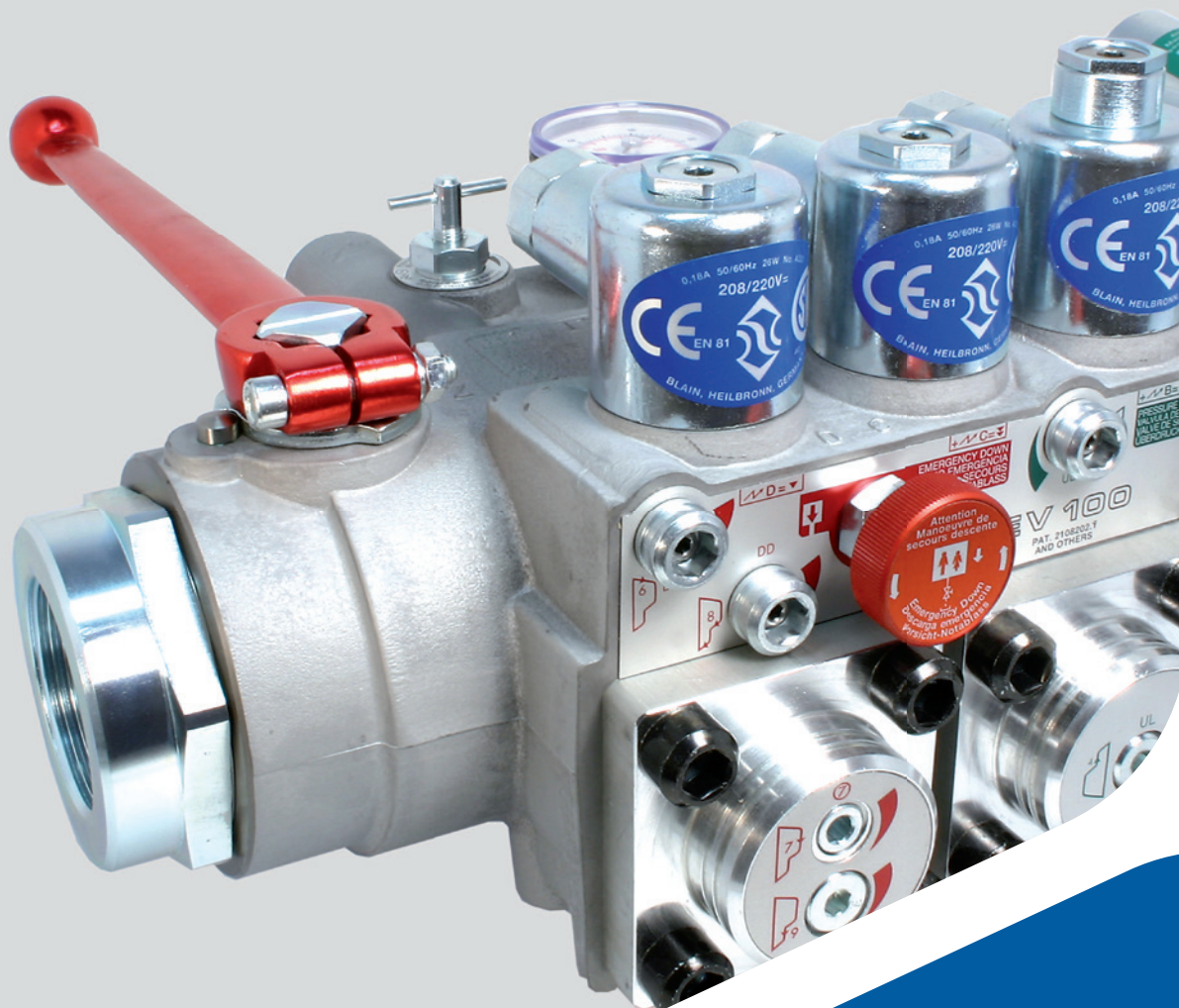


BLAIN VALVES FOR HYDRAULIC ELEVATORS

Excellence in Simplicity and Performance

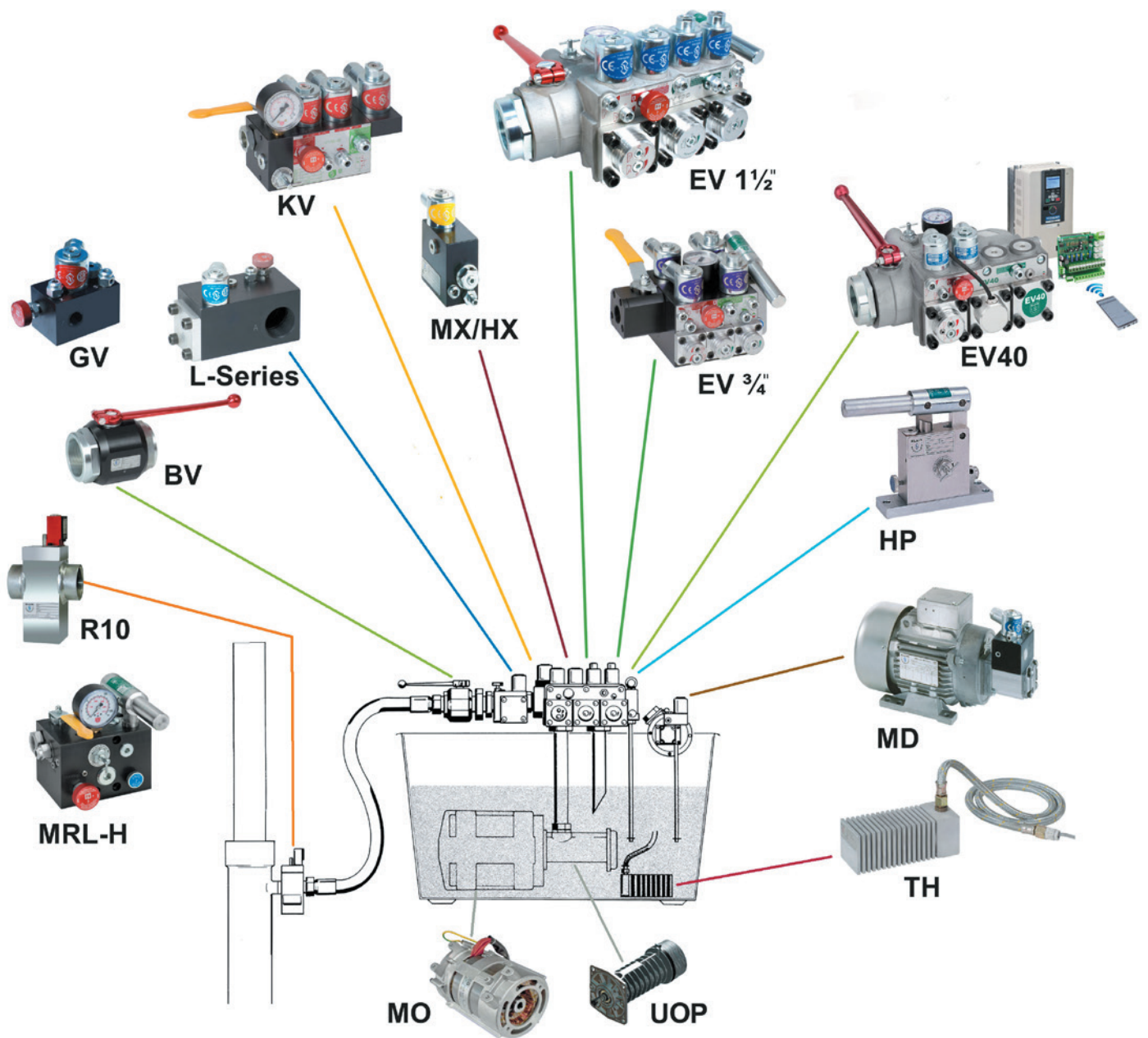


**TECHNICAL
CATALOGUE**



www.blain.de

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BLAIN PRODUCT RANGE

Illustrating hydraulic valves and other Blain equipment, serving to improve elevator safety and comfort, reduce travelling time and increase stopping accuracy. For additional information, please refer to descriptions and technical data sheets covering each product.



PRODUITS BLAIN

La gamme des produits Blain présente des soupapes hydrauliques ainsi que de nombreux accessoires, qui permettent d'augmenter la sécurité, et le confort, des ascenseurs, de réduire le temps de trajet, et d'améliorer la précision de l'arrêt. Pour des informations supplémentaires reportez vous aux descriptifs et aux notices techniques des différents produits.



BLAIN PRODUKTPALETTE

Zeigt hydraulische Ventile und anderes Blain Zubehör, welches dazu dient, die Aufzugssicherheit und den Fahrkomfort zu erhöhen, die Fahrzeit zu reduzieren und die Haltegenauigkeit zu verbessern. Zusätzliche Informationen entnehmen Sie bitte unseren Beschreibungen und technischen Datenblättern der einzelnen Produkte.



GAMA DE PRODUCTOS BLAIN

Muestra válvulas hidráulicas y otros accesorios Blain, que sirven para aumentar la seguridad así como el confort en los ascensores, reducen el tiempo del trayecto y mejoran la precisión de parada. Para más información, consulte por favor, nuestras descripciones y las página adjuntas exponen con claridad los datos técnicos de los diferentes productos.

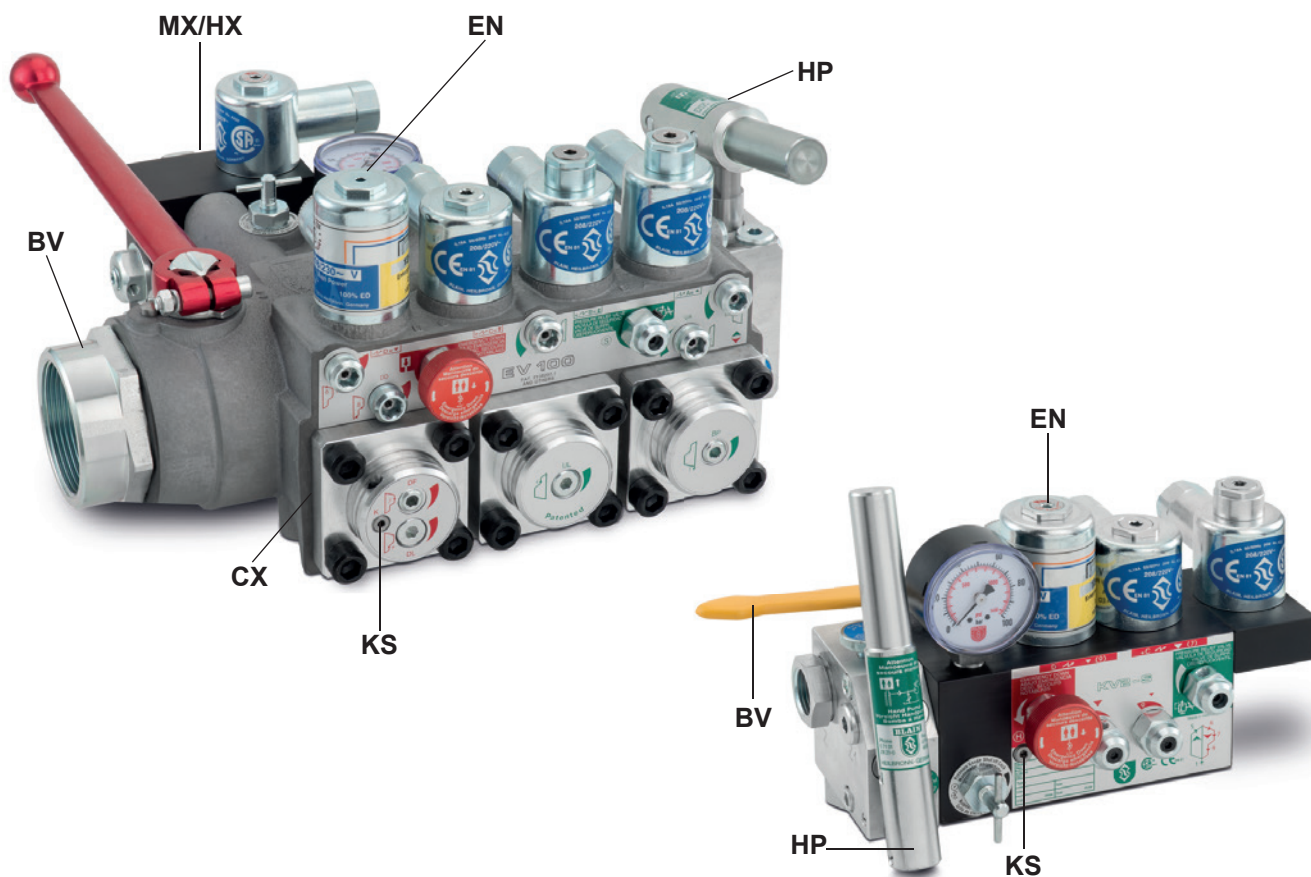


- KV - Mechanical Control Valve:** For small hydraulic elevators with speeds ranging from 0.16 m/s to 1.0 m/s (32-200 fpm), depending on the valve type.
- EV - Mechanical Control Valve:** A selection of valves with flows ranging from 10-1600 l/min (2.6-416 US gpm), from regular to highest comfort performance.
- EV40 - VVVF control valve:** A selection of valves together with inverter drive from Yaskawa for flows ranging from 10-1600 l/min (2.6-416 US gpm). EV4-vvfv program includes the widest range of vvvf solution offered to the elevator industry for high performance passenger elevators. Easy to install, EV4's are smooth, reliable and precise in operation throughout extreme load and temperature variations with in-built overload protection and different energy saving modes.
- GV - Mechanical control valve:** A selection of valve from 1-24 l/min (0.3-6.3 US gpm) for hydraulic car parking ports and material lifting platforms.
- R10 - Rupture Valve:** Decelerates the elevator in case the down speed exceeds allowable limits. Alternative connections available.
- L10 - UCM (A3) Safety Valve:** Is a pilot operated check valve which prevents down movement of the elevator should an electrical or mechanical malfunction occur in the main valve.
- MD - Micro Levelling Drive:** A drive unit consisting of a small motor, pump and valve block to provide immediate re-levelling and accurate floor stops in up and down directions.
- MRL-H - Machine room less rescue unit:** designed for servicing and rescuing operations of Machine Room-Less Hydraulic (MRL-H) elevators remotely by having easy outside access, without needing to be in the pit. MRL-H can be located up to a distance of 6 metres (19 feet) away and 5 metres (16 feet) high from the main power unit at a convenient location for easy access.
- BV - Ball Valve:** To close pressure lines. A selection of different thread connections to choose from.
- TH - Tank Heater:** Prevents the oil temperature from falling below desirable levels after a period of non-operation of hydraulic equipment.
- HP - Hand Pump:** For application with hydraulic lifting or pressing equipment and for the pressure testing of hydraulic systems in general.
- HX - Manual Down Valve:** Can be used for emergency manual lowering or in combination with the EV down valve to test the rupture valve.
- MX - Solenoid Down Valve:** For revision or inspection travel of the elevator or as an extra slow speed down valve.
- UOP - submersible screw pump:** SEIM screw pumps are specifically designed for immersion in oil to work with hydraulic lifts and meet the requirements of low noise level, high efficiency and low pulsation with flow up to 900 l/min.
- MO - submersible motor:** SB Motori submersible single and 3 phase motors are specifically designed for immersion in oil to work with submersible screw pumps and meet the requirements of low noise level and high efficiency in hydraulic lifts.



- KV - Steuerventil:** Für kleine hydraulische Personenaufzüge mit Geschwindigkeiten von 0,16 m/s bis 1,0 m/s (je nach Ausführung).
- EV - Steuerventil:** Mit Durchflüssen von 10-1600 l/min bieten sie eine Auswahl von Ventilen, von der Einfach- bis zur Höchst-Komfortfahrleistung.
- EV40 - VVVF Steuerventil:** Steuerventil mit Yaskawa Frequenzumrichter für Volumenströme von 10-1600 l/min. Das Ventil deckt einen großen Bereich von qualitativ hochwertigen hydraulischen Personenaufzügen ab. Die einfach zu installierenden EV4 Ventile arbeiten ruhig, zuverlässig und präzise auch bei extremer Last- und Temperaturschwankung und bieten zusätzlich einen eingebauten Überlastschutz und verschiedene Energiesparmodi.
- GV - Steuerventil:** Auswahl von 1-24 l/min für Hydraulik-Autopark-Systeme oder Plattformaufzüge.
- R10 - Rohrbruchventil:** Bremst den Aufzug bei überhöhter Absenkgeschwindigkeit ab. Vielzahl von Anschlussmöglichkeiten.
- L10 - UCM (A3) Sicherheitsventil:** Vorgesteuertes Rückschlagventil, verhindert ein Absenken des Aufzuges bei elektrischen oder mechanischen Fehlfunktionen des Hauptventils.
- MD - Feinnivellierungsantrieb:** Ein Kleinmotor, Pumpe und Ventilblock sorgen für sofortiges Nachholen in Hub- und Senkrichtung sowie genaues Halten des Aufzuges.
- MRL-H-Service-Einheit für maschinenraumlose Aufzüge:** Konstruiert für Wartungs- und Rettungsaufgaben aus der Ferne bei maschinenraumlosen Hydraulikaufzügen (MRL-H). Die Serviceeinheit kann bis zu 6 Meter entfernt und in einer Höhe von bis zu 5 Metern angebracht werden und ermöglicht somit einen einfachen Zugang.
- BV - Kugelhahn:** Zur Schließung der Druckleitung. Verschiedene Anschlussgrößen lieferbar.
- TH - Tankheizung:** Verhindert nach längerem Stillstand eines Aufzuges die Absenkung der Öltemperatur unter den erwünschten Wert.
- HP - Handpumpe:** Für das Anheben von hydraulischen Aufzügen, sowie zur Druckprüfung von hydraulischen Systemen allgemein.
- HX - Handbetätigtes Senkventil:** Dient zusammen mit dem Senkventil des EV Blockes zur Überprüfung des Rohrbruchventils.
- MX - Elektrogesteuertes Senkventil:** Dient zur Revisions- oder Inspektionsfahrt eines Aufzuges oder als dritte, langsame Senkgeschwindigkeit.
- UOP - Unteröl-Schraubenspindelpumpe:** SEIM Schraubenspindelpumpen sind speziell für den Unterölbetrieb in hydraulischen Aufzügen konstruiert. Sie erfüllen die Anforderungen an einen geräuscharmen Betrieb, hohe Effektivität und einen gleichmäßigen Volumenstrom mit bis zu 900 l/min.
- MO - Unterölmotor:** Die Wechsel- und Drehstrom-Unterölmotoren von SB Motori sind speziell für den Unterölbetrieb mit Schraubenspindelpumpen konstruiert und erfüllen die Anforderungen an einen geräuscharmen und effizienten Betrieb in hydraulischen Aufzügen.

Option for EV and KV valves



BV	Ball Valve
HP	Hand Pump
MX/HX	Down Valve
EN	Emergency Power Coil
KS	Slack Rope Valve
CX	Pressure Compensated Down

Pressure Line Shut Off.
To raise car manually.
Extra Down Valve Solenoid or manual.
Battery lowering in case of power failure.
Prevents slack rope condition in 2:1 systems.
Limits down speed variation with load.



BV	Kugelhahn
HP	Handpumpe
MX/HX	Senkventil
EN	Notstromspule
KS	Kolbensicherung
CX	Druckkompensierter Senkkolben

Absperrhahn für Druckleitung.
Für manuelles Anheben der Kabine.
Zusätzliches Senkventil magnetisch oder manuell.
Absenkung durch Batterie bei Stromausfall.
Verhindert Schlaffseilzustand bei 2:1 Systemen.
Begrenzt die Senkgeschwindigkeitsabweichung bei Lastunterschieden.



BV	Robinet à boisseau sphérique
HP	Pompe à main
MX/HX	Soupape de descente
EN	Bobine descente de secours
KS	Sécurité de mou de câble
CX	Soupape descente compensée par pression

Permet la fermeture de la conduite hydraulique.
Pour la montée manuelle de la cabine.
Soupape de descente supplémentaire magnétique ou manuelle.
Descente sur batterie en cas de coupure du courant principal.
Empêche un mou de câble excessif sur des systèmes 2:1.
La vitesse de descente reste presque constante malgré de charges différentes.



BV	Llave esférica
HP	Bomba a mano
MX/HX	Válvula de bajada
EN	Bobina de corriente de emergencia
KS	Válvula aflojamiento de cables
CX	Válvula de bajada compensada

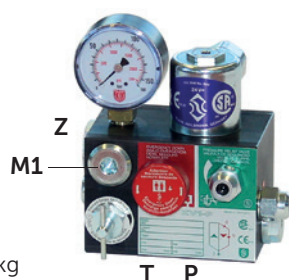
Tubería de presión, grifo de cierre.
Para elevar la cabina manualmente.
Válvula de bajada suplementaria, magnética o manual.
El acumulador acciona la bajada automática al interrumpir la corriente eléctrica.
Evita el aflojamiento excesivo de cables en sistemas 2:1.
Limita la variación de velocidad de bajada con cargas diferentes.

KV solenoid valves are designed for small hydraulic lifts operating at speeds up to 0.16 m/s (32 fpm) depending on the valve selected. The smooth and accurate ride characteristics of the KV2S valve which includes 'soft stop' in both directions, render it highly suitable for quality home lifts and lifts for the handicapped.

Coils AC:	24 V/1.8 A, 42 V/1.0 A, 110 V/0.5 A, 230 V/0.18 A, 50/60 Hz
Coils DC:	12 V/2.1 A, 24 V/1.1 A, 42 V/0.6 A, 80 V/0.3 A, 125 V/0.25 A, 196 V/0.14 A.
Operation oil temperature range:	10°C-60°C (50°F-140°F), for oil VGA46: 250cSt.-20 cSt.
Optimal oil temperature range:	25°C-55°C (77°F-131°F), for oil VGA46: 100cSt.-24 cSt.
Ambient temperature range:	0°C-70°C (32°F-158°F)
Ports:	P Pump, Z Cylinder and T Tank all G $\frac{1}{2}$ "
Max. Oil Temperature:	70°C (158°F)
Oil Viscosity:	25-60 cSt. at 40°C (104°C)
Insulation Class, AC and DC:	IP 68
Pressure Range:	8-100 bar (116-1450 psi)
Burst Pressure:	500 bar (7251 psi)
Flow Range:	5-80 l/min (1.3-21 US gpm) - see flow pressure charts

Speeds max. (EN code)

KV1P



1.8 kg

- Up** One up speed, 0.16 m/s (32 fpm) max.
Up start has built-in damping.
Up stop has no damping (pump stops).
- Down** One down speed, 0.16 m/s (32 fpm) max.
Down start has adjustable damping.
Down speed is adjustable.
Down stop has built-in damping.

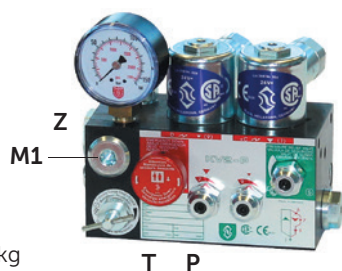
KV1S



2.3 kg

- Up** One up speed 0.16 m/s (32 fpm) max. with soft stop, or 0.4 m/s (80 fpm) max. with overtravel and relevelling.
Up start has built-in damping.
Up stop has adjustable damping (delayed pump stop required).
- Down** One down speed, 0.16 (32 fpm) max.
Down start has adjustable damping.
Down speed is adjustable.
Down stop has built-in damping.

KV2P



2.5 kg

- Up** One up speed, 0.16 m/s (32 fpm) max.
Up start has built-in damping.
Up stop has no damping (pump stops).
- Down** Two down speeds, 1 m/s (200 fpm) max.
Down start has adjustable damping.
Fast down speed and levelling speeds are adjustable.
Slow down and stop have built-in damping.

KV2S



3.2 kg

- Up** One up speed, 0.16 m/s (32 fpm) max. with soft stop, or 0.4 m/s (80 fpm) max. with overtravel and relevelling.
Up start has built-in damping.
Up stop has adjustable damping (delayed pump stop required).
- Down** Two down speeds, 1 m/s (200 fpm) max.
Down start has adjustable damping.
Fast down speed and levelling speeds are adjustable.
Slow down and stop have built-in damping.



Control Elements

- A** Solenoid 'Up Stop'
- C** Solenoid 'Down Deceleration'
- D** Solenoid 'Down Stop'
- U** Bypass Valve
- H** Manual Lowering
- M1** Test port

- V** Check Valve
- X** Down Valve
- Y** Down Level Valve
- F** Main Filter
- S** Relief Valve

Adjustments UP

- 1** Bypass
- 5** Up Soft Stop
- Up Acceleration built-in*

Adjustments DOWN

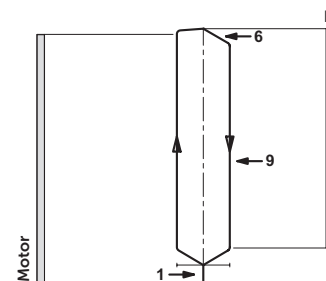
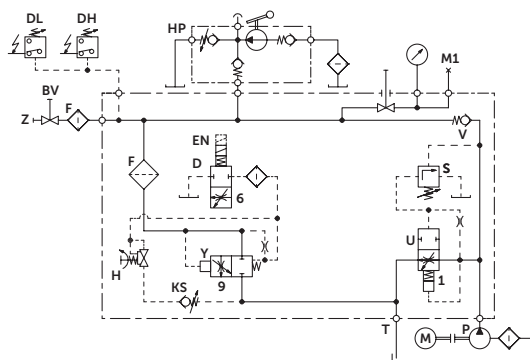
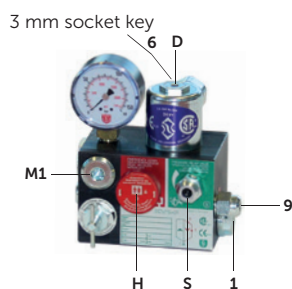
- 6** Down Acceleration
- 7** Down Full Speed
- 9** Down Levelling Speed
- Down Deceleration built-in*



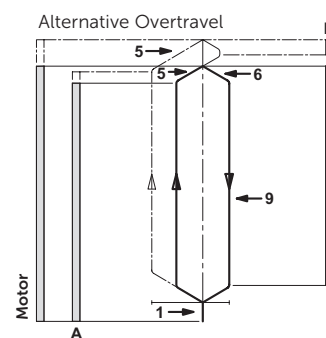
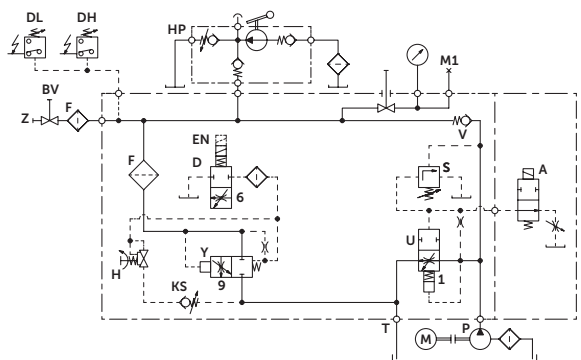
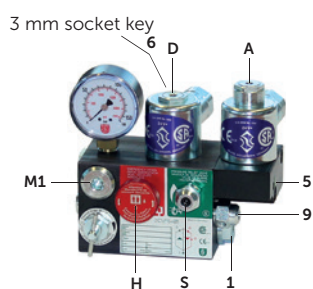
Hydraulic Circuit

Electrical Sequence

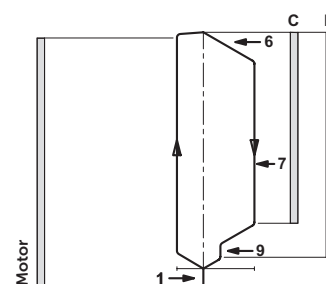
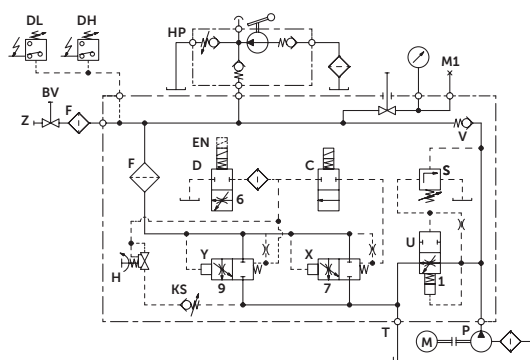
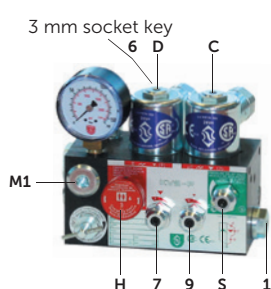
KV1P



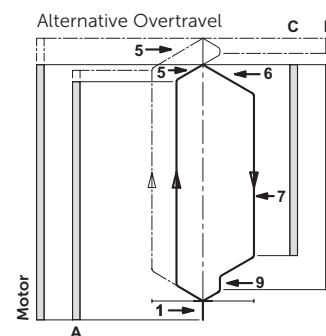
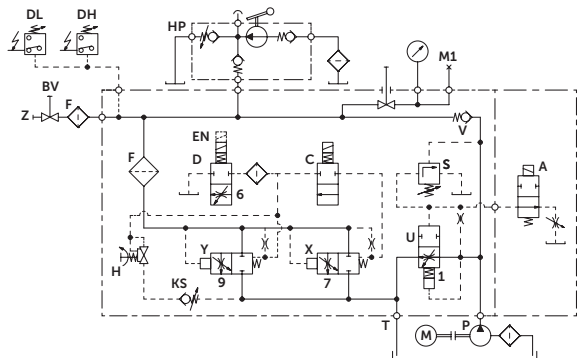
KV1S



KV2P



KV2S





Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off, cylinder line is closed and residual pressure in the valve is reduced to zero.



Adjustments UP

Valves are already adjusted and tested. Check electrical operation before changing valve settings. Test that the correct coil is energized, by removing the nut and raising the coil slightly to feel pull.

Nominal Settings: Adjustment **1** level with flange face. Adjustment **5** (KV1S & KV2S) level with flange face.

KV1P

- 1. Up Bypass:** When the pump is started, the unloaded car should remain stationary at the floor for a period of about 1 second before starting upwards. The length of this delay is according to the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
Up Stop: At floor level, the pump-motor is de-energized. The stop may be abrupt depending on load and speed of approach. No adjustment possible.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.

KV1S

- 1. Up Bypass:** When the pump is started, the unloaded car should remain stationary at the floor for a period of about 1 second before starting upwards. The length of this delay is according to the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
- 5. Up Stop:** At floor level coil **A** is de-energized. Through a time relay the pump must run approx. 1 second longer to allow the car to stop smoothly by valve operation according to the setting of adjustment **5**. 'In' (clockwise) provides a softer stop, 'out' (c-clockwise) a quicker stop. Pre-adjustment: With coil A disconnected and the pump running, **5** should be turned in until the car starts to move up, then slowly turned out until the car stops.
Alternative Up Stop: At relatively higher speeds and with the time relay arrangements as with 'up stop' above, the car may travel to just above floor level. In overtravelling the floor, down levelling coil **D** is energized, lowering the car smoothly back down to floor level where **D** is de-energized.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.

KV2P

- 1. Up Bypass:** When the pump is started, the unloaded car should remain stationary at the floor for a period of about 1 second before starting upwards. The length of this delay is according to the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
Up Stop: At floor level, the pump-motor is de-energized. The stop may be abrupt depending on load and speed of approach. No adjustment possible.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.

KV2S

- 1. Up Bypass:** When the pump is started, the unloaded car should remain stationary at the floor for a period of about 1 second before starting upwards. The length of this delay is according to the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
- 5. Up Stop:** At floor level coil **A** is de-energized. Through a time relay the pump must run approx. 1 second longer to allow the car to stop smoothly by valve operation according to the setting of adjustment **5**. 'In' (clockwise) provides a softer stop, 'out' (c-clockwise) a quicker stop. Pre-adjustment: With coil **A** disconnected and the pump running, **5** should be turned in until the car starts to move up, then slowly backed off again until the car stops.
Alternative Up Stop: At relatively higher speeds and with the time relay arrangements as with 'up stop' above, the car may travel to just above floor level. In overtravelling the floor, down levelling solenoid **D** is energized, lowering the car smoothly back down to floor level where **D** is de-energized.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.



Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off, cylinder line is closed and residual pressure in the valve is reduced to zero.



Adjustments DOWN

Valves are already tested and adjusted. Check electrical operation before changing valves settings. Test that the correct coil is energized by removing nut and raising the coil slightly to feel pull.

KV Nominal Settings: Adjustments **7 & 9** level with flange face.

KV1P / KV1S

6. Down Acceleration: When coil **D** is energized, the car will accelerate downwards according to the setting of adjustment **6**. 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration. Pre-adjustment: Turn adj. **6** all the way in and then energize coil **D**. Turn **6** slowly back out until the car accelerates downwards.

9. Down Speed: With coil **D** energized as above, the down speed of the car is according to the setting of adjustment **9**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

Down Stop: At floor level, coil **D** is de-energized causing the car to stop. No adjustment necessary.

H Manual Lowering: 'out' (c-clockwise) allows the car to be lowered by hand. Closes automatically on release.

KV2P / KV2S

6. Down Acceleration: When coil **D** is energized, the car will accelerate downwards according to the setting of adjustment **6**. 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration. Pre-adjustment: Turn adj. **6** all the way in and then energize coil **D**. Turn **6** slowly back out until the car accelerates downwards.

7. Down Speed: With coils **C** and **D** energized as above, the down speed of the car is according to the setting of adjustment **7**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

Down deceleration: When coil **C** is de-energized whilst coil **D** remains energized, the car will decelerate according to the built-in damping. No further adjustment will be required.

9. Down Levelling: With coil **C** de-energized and coil **D** remaining energized, the car will travel at its down levelling speed according to the setting of adjustment **9**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster down levelling speed.

Down Stop: At floor level, coil **D** is de-energized causing the car to stop. No adjustment necessary.

H Manual Lowering: 'out' (c-clockwise) allows the car to be lowered by hand. Closes automatically on release.

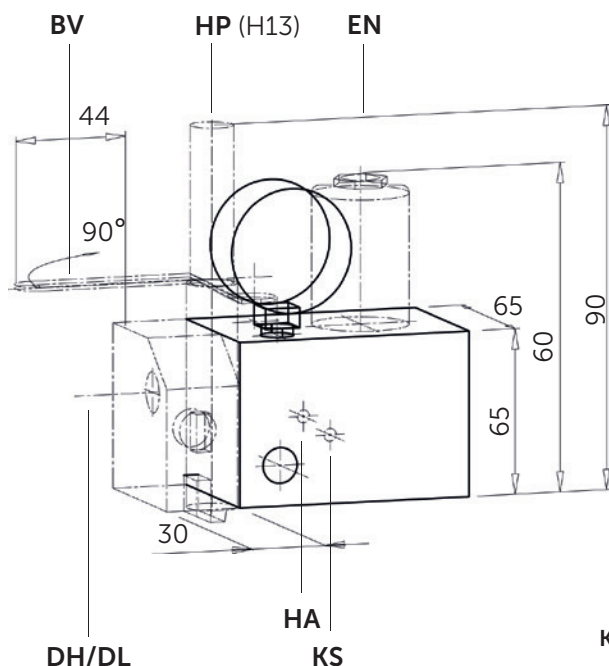
KS Slack Rope Valve: Coil **D** must be de-energized! The KS, is adjusted with a 3 mm Allen key by turning the screw **K** 'in' for higher pressure and 'out' for lower pressure. With **K** turned all the way 'in', then half a turn back out, the unloaded car should descend when Manual Lowering **H** is opened. Should the car not descend, **K** must be turned out until the car just begins to descend, then turned out a further half turn to ensure that with cold oil, the car can be lowered as required.

Optional

KV Optional Equipment

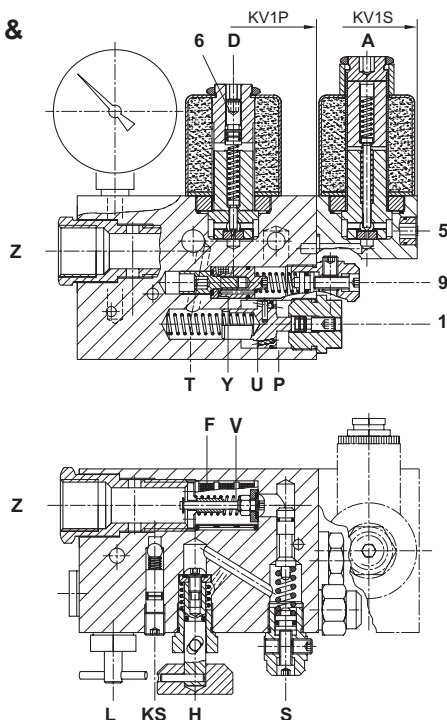
- BV** Ball Valve built in
- EN** Emergency Power Solenoid
- HP** Hand Pump H 13
- KS** Slack Rope Valve
- DH** Pressure Switch 10-100 bar
- DL** Pressure Switch 1-10 bar
- CSA** CSA Solenoids
- HA** Emergency Manual Down Speed Adj.

The possible options are shown with KV1P Valve.
The same Options can be applied to all other KV Valve types.

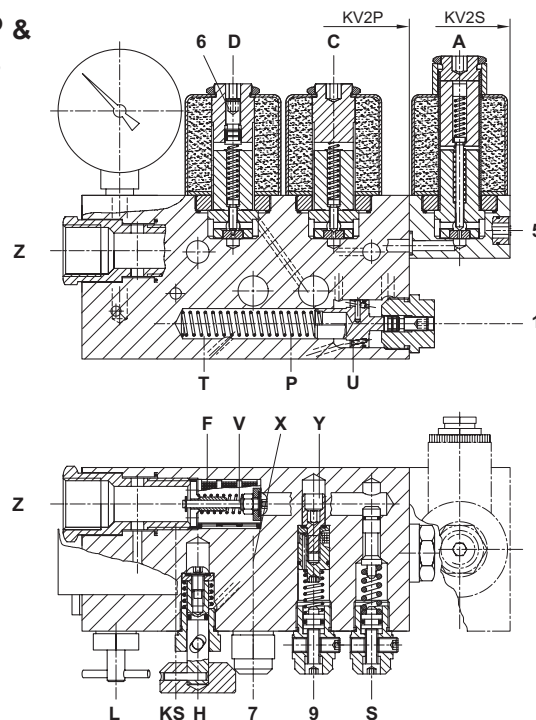


KV Example with Options

KV1P & KV1S



KV2P & KV2S



Control Elements

- A Solenoid 'Up Stop'
- C Solenoid 'Down Deceleration'
- D Solenoid 'Down Stop'
- U Bypass Valve
- V Check Valve
- X Down Valve
- Y Down Levelling Valve
- H Manual Lowering
- L Gauge Shut Off Cock
- F Main Filter
- M1 Test port

Connections

- P Pump
- T Tank - return
- Z Cylinder

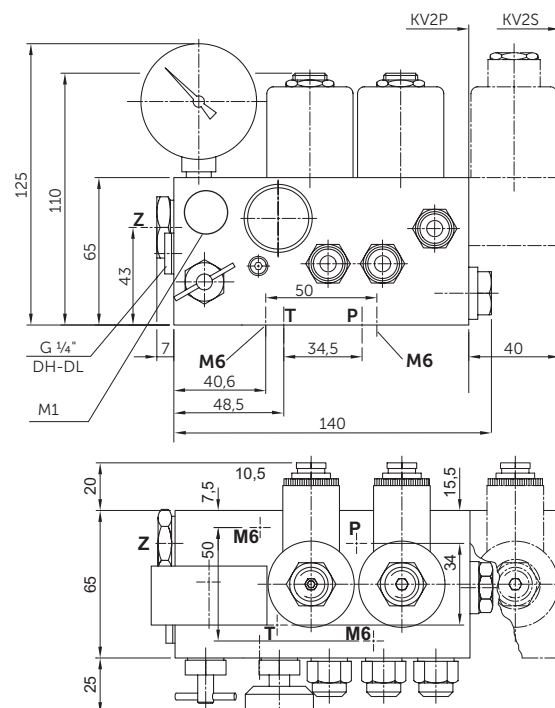
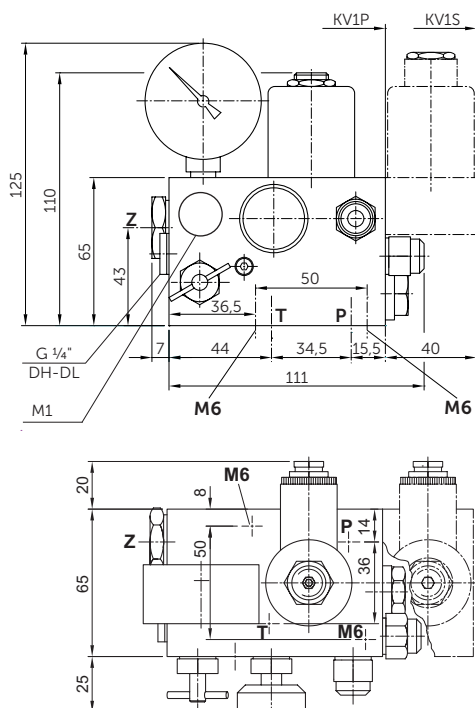
Adjustments

- 1 Bypass
- 5 UP Soft Stop
- 6 Down Acceleration
- 7 Down Full speed
- 9 Down Levelling Speed
- S Relief Valve



Important: Length of 1/2" thread on pipe connections should not be longer than 14 mm!

Measurements



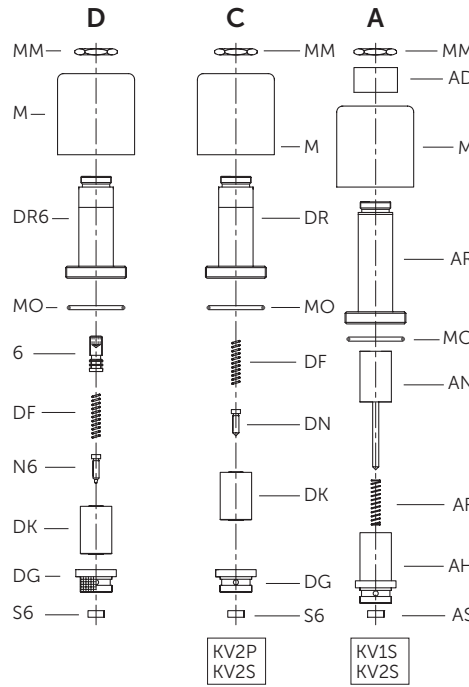


KV Spare Parts List

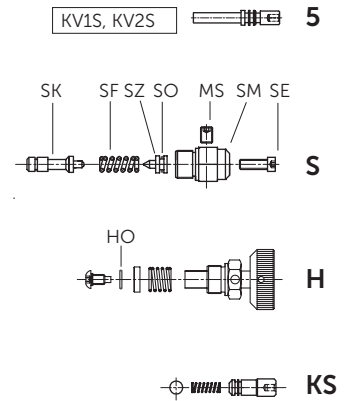
KV

Pos.	No.	Item	
1	1F	Flange - Bypass	
	1E	Adjustment - Bypass	
	EO	O-Ring - Adjustment (3,5x1,5 - P)	
	U	Flow Guide - Bypass	
	UO	O-Ring - Bypass (17x1 - V)	
	UF	Spring - Bypass	
5	5	Adjustment - Up Stop	
6	6	Adjustment - Down Acceleration	
7+9	7E	Adjustment - Down Valve	
	9F	Spring - Down Valve	
	YO	O-Ring - Flow Guide (10x1 - V)	
	XO	Seal - Flow Guide (5,28x1.78 - V)	
	XT	O-Ring Disc	
	FI	Filter - Down Valve	
	X	Down Flow Guide (Brass)	
	Y	Down Levelling Flow Guide (Steel) - KV2	
	Y	Down Flow Guide (Steel) - KV1	
	S	SE	Adjustment Screw - Relief Valve
SM		Hexagonal - Relief Valve	
MS		Locking Screw	
SO		O-Ring - Nipple	
SZ		Nipple - Relief Valve	
SF		Spring - Relief Valve	
SK		Piston - Relief Valve	
H	H	Manual Lowering - Self Closing	
HO	HO	Seal - Man. Lowering (O-Ring 5.28x1.78 - V)	
HA	HA	Adjustable Manual Lowering	
KS	KS	Slack Rope Valve	
A	MM	Nut Solenoid	
	AD	Collar Solenoid	
	M	Coil Solenoid (indicate voltage)	
	AR	Tube Solenoid 'Up'	
	MO	O-Ring Solenoid	
	AN	Needle Solenoid 'Up'	
	AF	Spring Solenoid 'Up'	
	AH	Seat Housing 'Up'	
	AS	Seat Solenoid 'Up'	
	C+D	M	Coil Solenoid (indicate voltage)
C		DR	Tube - Solenoid 'Down', w/o adj. 6
D		DR6	Tube Solenoid 'Down', with adj. 6
MO		O-Ring Solenoid	
DF		Spring Solenoid 'Down'	
C		DN	Needle Solenoid 'Down'
D		N6	Needle Solenoid 'Down' (Nipple)
C		HN	Needle Solenoid 'Down'
DK		Core Solenoid 'Down'	
DG		Seat Housing 'Down'(Solen.D with screen)	
C	S6	Seat Solenoid 'Down'	
	C	CO	O-Ring Seat Housing
	C	CO	O-Ring Seat Housing
Z	ZA	Cylinder Thread Connection	
	V	Check Valve	
	VO	O-Ring Check Valve (5,28x1,78 - V)	
	F	Main Filter	
L	L	Gauge Shut Off Cock	

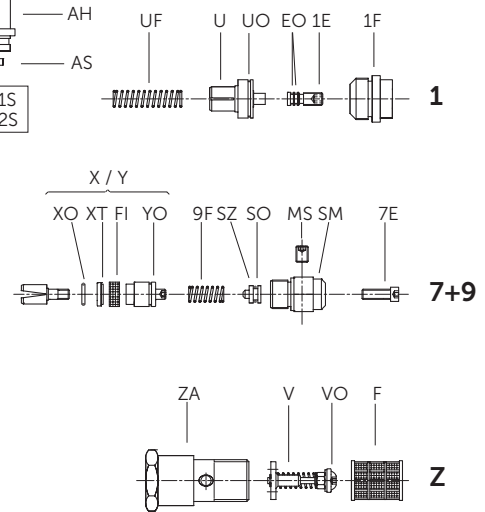
Solenoid Valves



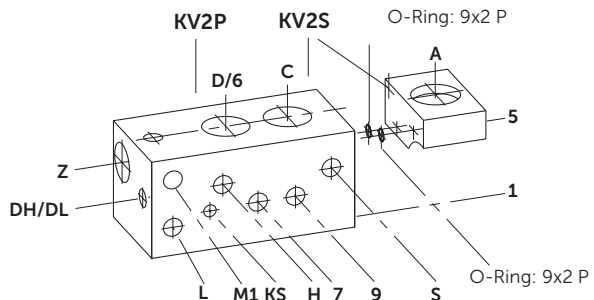
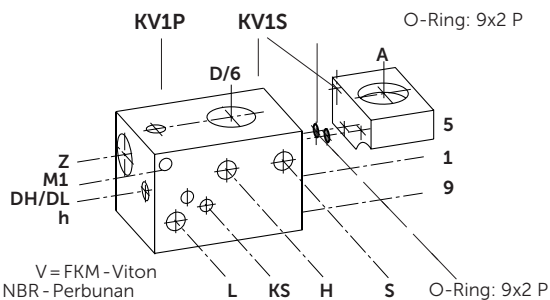
Adjustments



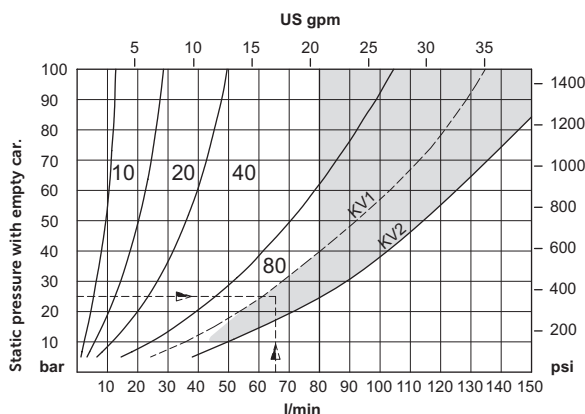
Flow Valves



In case of down leakage, replace and test in the following order:
S6, N6, HO, V complete, XO,
(2x XO with KV2).



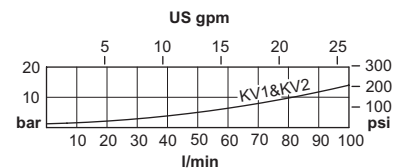
Flow Guide Selection Charts



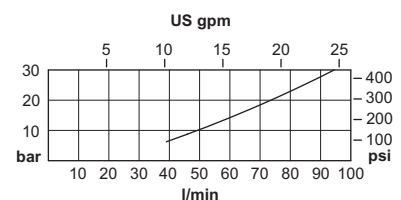
For pressure-flow condition within shaded area, use 3/4" piping to avoid unnecessary pressure loss.
Pump flows above 80 l/min are not recommended

Example order
KV2S, 65 l/min, 25 bar (empty),
220 AC
or: KV2S/80/220 AC

Pressure Drop P - Z



Lowest Relief Pressure



The BLAIN EV program includes the widest range of options offered to the elevator industry for high performance passenger service. Easy to install, EV's are smooth, reliable and precise in operation throughout extreme load and temperature variations.

3/4" EV



EV 100

1 1/2" & 2" EV



EV 100

2 1/2" EV



EV 100

Description

Available port sizes are 3/4", 1 1/2", 2" and 2 1/2" pipe threads, depending on flow. EV's start on less than minimum load and can be used for across the line or wye-delta starting. According to customers' information, valves are factory adjusted ready for operation and very simple to readjust if so desired. The patented up levelling system combined with compensated pilot control ensure stability of elevator operation and accuracy of stopping independent of wide temperature variations.

EV valves include the following features essential to efficient installation and trouble free service:



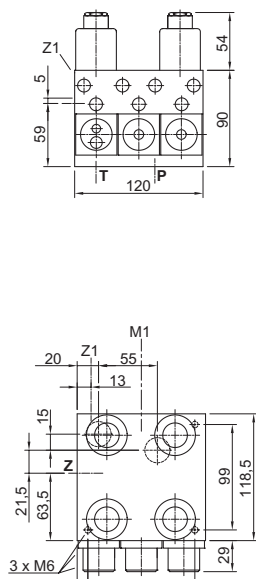
Simple Responsive Adjustment
Temperature and Pressure Compensation
Solenoid with Connecting Cables
Pressure Gauge and Shut Off Cock
Self Closing Manual Lowering

Self Cleaning Pilot Line Filters
Self Cleaning Main Line Filter (Z-T)
Built-in Turbulence Suppressors
70 HRc Rockwell Hardened Bore Surfaces
100% Continuous Duty Solenoids

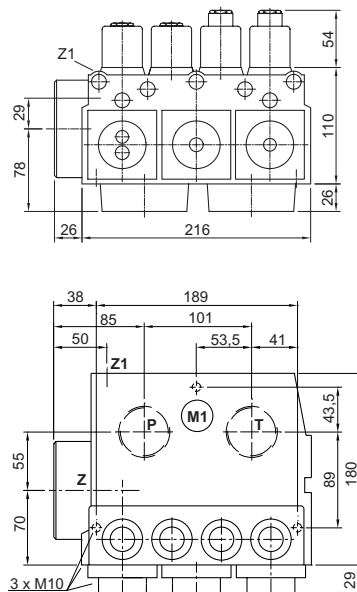
Technical Data:

	3/4" EV	1 1/2" & 2" EV	2 1/2" EV
Flow Range:	l/min 10-125 (2-33 US gpm)	30-800 (8-211 US gpm)	500-1530 (132-404 US gpm)
Pressure Range (valve):	bar 8-100 (116-1450 psi)	8-100 (116-1450 psi)	8-68 (116-986 psi)
Press. Range CSA (valve):	bar 8-100 (116-1450 psi)	8-70 (116-1015 psi)	8-47 (116-682 psi)
Burst Pressure Z:	bar 575 (8339 psi)	505 (7324 psi)	340 (4931 psi)
Pressure Drop P-Z:	bar 6 (87 psi) at 125 l/min	4 (58 psi) at 800 l/min	4 (58 psi) at 1530 l/min
Weight:	kg 5 (11 lbs)	10 (22 lbs)	14 (31 lbs)
Coils AC:	24 V/1.8 A, 42 V/1.0 A, 110 V/0.43 A, 230 V/0.18 A, 50/60 Hz.		
Coils DC:	12 V/2.0 A, 24 V/1.1 A, 42 V/0.5 A, 48 V/0.6 A, 80 V/0.3 A, 110 V/0.25 A, 196 V/0.14 A.		
Oil Viscosity:	25-60 cSt. at 40°C (104°F).		
Operation oil temperature range:	10°C-60°C (50°F-140°F), for oil VGA46: 250cSt.-20 cSt.		
Optimal oil temperature range:	25°C-55°C (77°F-131°F), for oil VGA46: 100cSt.-24 cSt.		
Ambient temperature range:	0°C-70°C (32°F-158°F)		
Max. Oil Temperature:	70°C (158°F)		
Insulation Class, AC and DC:	IP 68		

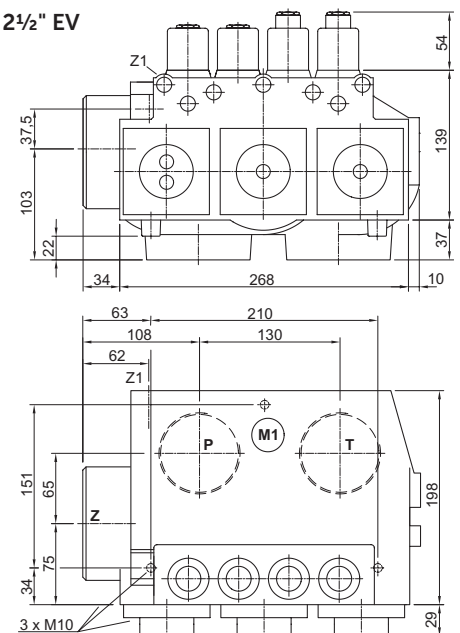
3/4" EV



1 1/2" & 2" EV



2 1/2" EV



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Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators

EV Control Valve Types

Optional Equipment

EN	Emergency Power Solenoid	DH	High Pressure Switch
CSA	CSA Solenoids	DL	Low Pressure Switch
KS	Slack Rope Valve	CX	Pressure Compensated Down Valve
BV	Main Shut-Off Valve	MX	Auxiliary Down
HP	Hand Pump		



EV 0

3/4"

1 1/2" & 2"

2 1/2"

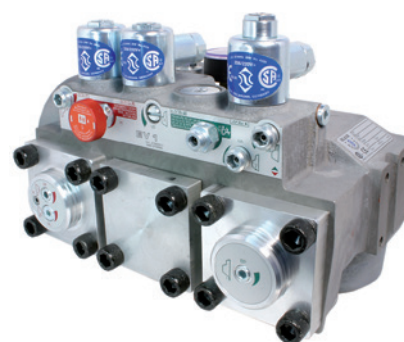


Up Up to 0.16 m/s (32 fpm). 1 Up Speed.
Up Start is smooth and adjustable.
Up Stop by de-energising the pump-motor.

Down Up to 1.0 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
All down functions are smooth and adjustable.

USA Patent No. 4,601,366
Pats & Pats Pend: France, Germany,
Italy, Japan, Switzerland & U.K.

EV 1



Up Up to 0.16 m/s (32 fpm). 1 Up Speed.
Up to 0.4 m/s (80 fpm) by overtravelling and levelling back down.
Up Start is smooth and adjustable.
Up Stop is smooth and exact through valve operation whereby
the pump must run approx. 1 sec. longer through a time relay.

Down Up to 1.0 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
All down functions are smooth and adjustable.

USA Patent No. 4,601,366
Pats & Pats Pend: France, Germany,
Italy, Japan, Switzerland & U.K.

EV 10



Up Up to 1.0 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
Up Start and Slow Down are smooth and adjustable.
Up Levelling speed is adjustable.
Up Stop is by de-energising the pump-motor.

Down Up to 1.0 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
All down functions are smooth and adjustable.

USA Patent No. 4,637,495
Pats & Pats Pend: France, Germany,
Italy, Japan, Switzerland & U.K.

EV 100



Up Up to 1.0 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
All 'up' functions are smooth and adjustable.
Up Levelling speed is adjustable.
Up Stop is smooth and exact through valve operation whereby
the pump must run approx. 1 sec. longer through a time relay.

Down Up to 1.0 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
All down functions are smooth and adjustable.

USA Patent No. 4,637,495
Pats & Pats Pend: France, Germany,
Italy, Japan, Switzerland & U.K.



Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off, cylinder line is closed and residual pressure in the valve is reduced to zero.



Adjustments UP

Valves are already adjusted and tested. Check electrical operation before changing valve settings.

Test that the correct coil is energized, by removing nut and raising the coil slightly to feel pull.

Initial settings for lifting operation: Setting **1** flush with flange, set bypass pressure (see *Quick Settings* document); Setting **4** flush with flange, then turn $\frac{1}{2}$ rotation 'out'; turn pressure relief valve **S** completely 'in', then $1\frac{1}{2}$ rotations 'out'; turn adj. **2, 3 & 5** completely 'in', then: for EV $\frac{3}{4}$ ": turn adj. $2\frac{1}{2}$ rotations 'out'; for EV $1\frac{1}{2}$ " – $2\frac{1}{2}$ ": turn adj. **2** 2 rotations 'out'; turn adj. **3 & 5** $2\frac{1}{2}$ rotations 'out'.

EV 0

- 1. By Pass:** When the pump is started, the unloaded car should remain stationary at the floor for a period of 1 to 2 seconds before starting upwards. The length of this delay is determined by the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
- 2. Up Acceleration:** With the pump running, the car will accelerate according to the setting of adjustment **2**. 'In' (clockwise) provides a softer acceleration, 'out' (c-clockwise) a quicker acceleration.
Up Stop: The pump-motor is de-energized. There is no adjustment.
Alternative Up Stop with Over-travel: The pump-motor is de-energized at floor level. Through the flywheel action of the pump-motor drive the car will travel to just above floor level. In overtravelling the floor, down levelling coil **D** is energized, lowering the car smoothly back down to floor level where **D** is de-energized.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.

EV 1

- 1. By Pass:** When the pump is started and coil **A** energized, the unloaded car should remain stationary at the floor for a period of 1 to 2 seconds before starting upwards. The length of this delay is determined by the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
- 2. Up Acceleration:** With the pump running and coil **A** energized as in 1, the car will accelerate according to the setting of adjustment **2**. 'In' (clockwise) provides a softer acceleration, 'out' (c-clockwise) a quicker acceleration.
- 5. Up Stop:** At floor level, coil **A** is de-energized. Through a time relay the pump should run approx. 1 second longer to allow the car to stop smoothly by valve operation according to the setting of adjustment **5**. 'In' (clockwise) provides a softer stop, 'out' (c-clockwise) a quicker stop.
Alternative Up Stop: At relatively higher speeds, the car will travel to just above floor level. In overtravelling the floor, down levelling coil **D** is energized, lowering the car smoothly back down to floor level where **D** is de-energized.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.

EV 10

- 1. By Pass:** When the pump is started and coil **B** energized, the unloaded car should remain stationary at the floor for a period of 1 to 2 seconds before starting upwards. The length of this delay is determined by the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
- 2. Up Acceleration:** With the pump running and coil **B** energized as in 1, the car will accelerate according to the setting of adjustment **2**. 'In' (clockwise) provides a softer acceleration, 'out' (c-clockwise) a quicker acceleration.
- 3. Up Deceleration:** When coil **B** is de-energized, the car will decelerate according to the setting of adjustment **3**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration.
- 4. Up Levelling:** With coil **B** de-energized as in 3, the car will proceed at its levelling speed according to the setting of adjustment **4**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster up levelling speed.
Up stop: The pump-motor is de-energized. There is no adjustment.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.

EV 100

- 1. By Pass:** When the pump is started and coils **A** and **B** energized, the unloaded car should remain stationary at the floor for a period of 1 to 2 seconds before starting upwards. The length of this delay is determined by the setting of adjustment **1**. 'In' (clockwise) shortens the delay, 'out' (c-clockwise) lengthens the delay.
- 2. Up Acceleration:** With the pump running and coils **A** and **B** energized as in 1, the car will accelerate according to the setting of adjustment **2**. 'In' (clockwise) provides a softer acceleration, 'out' (c-clockwise) a quicker acceleration.
- 3. Up Deceleration:** When coil **B** is de-energized, whilst coil **A** remains energized, the car will decelerate according to the setting of adjustment **3**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration.
- 4. Up Levelling:** With coil **A** energized and coil **B** de-energized as in 3., the car will proceed at its levelling speed according to the setting of adjustment **4**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster up levelling speed.
- 5. Up Stop:** At floor level, coil **A** is de-energized with coil **B** remaining de-energized. Through a time relay the pump should run approx. 1 second longer to allow the car to stop smoothly by valve operation according to the setting of adjustment **5**. 'In' (clockwise) provides a softer stop, 'out' (c-clockwise) a quicker stop.
- S Relief Valve:** 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting. After turning 'out', open manual lowering **H** for an instant.
Important: When testing relief valve, close ball valve gradually.



Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off, cylinder line is closed and residual pressure in the valve is reduced to zero.



Adjustments DOWN

Valves are already adjusted and tested. Check electrical operation before changing valve settings.

Test that the correct coil is energized, by removing nut and raising the coil slightly to feel pull.

Standard settings: adj. **7 & 9** level with flange faces, then turn out adj. **9** for $\frac{1}{2}$ a turn; turn in adj. **6 & 8** completely, then for EV $\frac{3}{4}$ ": turn out adj. **6** for $2\frac{1}{2}$ turns and turn out adj. **8** for **1** turn; for EV $1\frac{1}{2}$ " - $2\frac{1}{2}$ ": turn adj. **6** for 2 to $2\frac{1}{2}$ turns out and adj. **8** for $1\frac{1}{2}$ turns out.

6. Down Acceleration: When coils **C** and **D** are energized, the car will accelerate downwards according to the setting of adjustment **6**. 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration.

7. Down Speed: With coils **C** and **D** energized as in 6 above, the full down speed of the car is according to the setting of adjustment **7**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

8. Down Deceleration: When coil **C** is de-energized whilst coil **D** remains energized, the car will decelerate according to the setting of adjustment **8**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration. **Attention: Do not close all the way in! Closing adjustment 8 completely (clockwise) may cause the car to fall on the buffers.**

9. Down Levelling: With coil **C** de-energized and coil **D** energized as in 8 above, the car will proceed at its down levelling speed according to the setting of adjustment **9**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster down levelling speed.

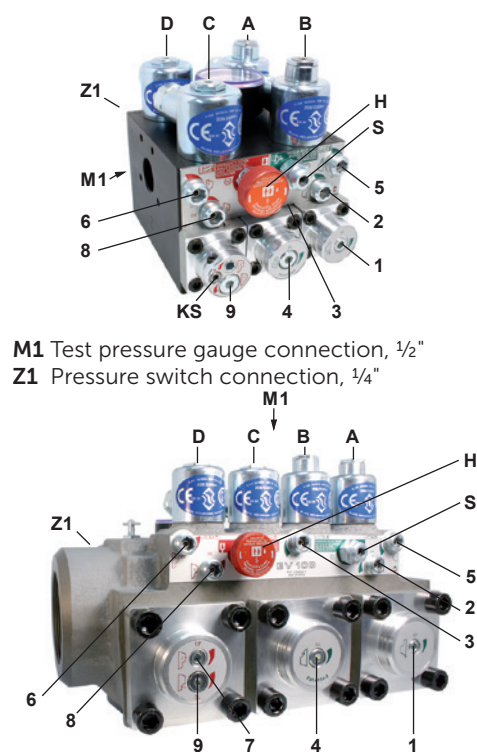
Down Stop: When coil **D** is de-energized with coil **C** remaining de-energized, the car will stop according to the setting of adjustment **8** and no further adjustment is required.

KS Slack Rope Valve: Both coils **C** and **D** must be de-energized beforehand! Loosen the small grub screw on the top of the **K** on the left hand side. The **KS** is adjusted with a 3 mm Allen key by turning the screw **K** 'in' for higher pressure and 'out' for lower pressure. With **K** turned all the way 'in', then half a turn back out, the unloaded car should descend when Manual Lowering **H** is opened. Should the car not descend, **K** must be turned out until the car just begins to descend, then turned out a further half turn to ensure that with cold oil, the car can be lowered as required.

Positions of Adjustments



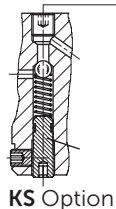
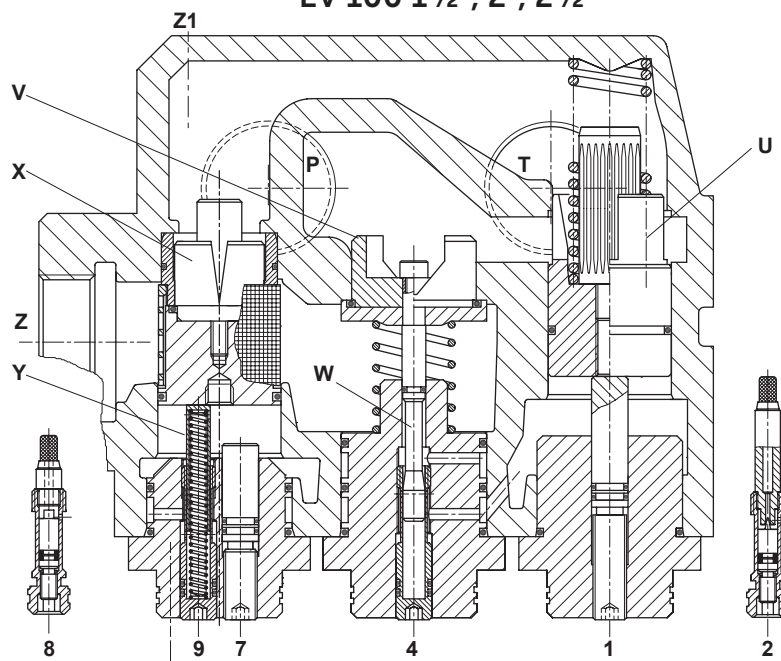
Important: Length of $\frac{3}{4}$ " thread on pump connections should not be longer than 14 mm!



M1 Test pressure gauge connection, $\frac{1}{2}$ "

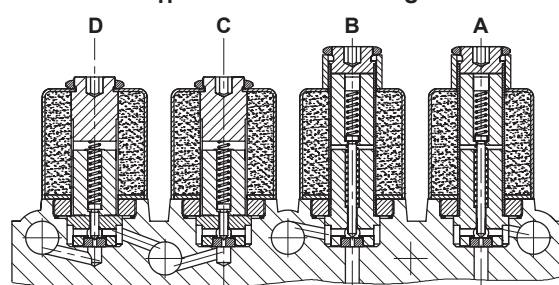
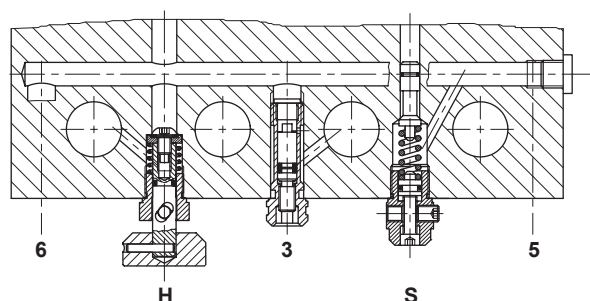
Z1 Pressure switch connection, $\frac{1}{4}$ "

EV 100 $1\frac{1}{2}$ ", 2", $2\frac{1}{2}$ "



KS Option

Horizontal Sections



Vertical Section

Adjustments UP

- 1 By Pass
- 2 Up Acceleration
- 3 Up Deceleration
- 4 Up Levelling Speed
- 5 Up Stop

Adjustments DOWN

- 6 Down Acceleration
- 7 Down Full Speed
- 8 Down Deceleration
- 9 Down Levelling Speed

Control Elements

- A Solenoid (Up Stop)
- B Solenoid (Up Deceleration)
- C Solenoid (Down Deceleration)
- D Solenoid (Down Stop)
- H Manual Lowering
- S Relief Valve
- U By Pass Valve
- V Check Valve
- W Levelling Valve (Up)
- X Full Speed Valve (Down)
- Y Levelling Valve (Down)

Valve Types

- EV 0
- EV 1
- EV 10
- EV 100

Elements Omitted

- A, B, W, 3, 4 & 5
- B, W, 3 & 4
- A & 5
- as shown



Control Elements

- | | |
|---------------------------------------|----------------------------------|
| A Solenoid (Up Stop) | U By Pass Valve |
| B Solenoid (Up Deceleration) | V Check Valve |
| C Solenoid (Down Deceleration) | W Levelling Valve (Up) |
| D Solenoid (Down Stop) | X Full Speed Valve (Down) |
| H Manual Lowering | Y Levelling Valve (Down) |
| S Relief Valve | F Filter |
| M1 Test port | |

Adjustments UP

- 1** By Pass
- 2** Up Acceleration
- 3** Up Deceleration
- 4** Up Levelling Speed
- 5** Up Stop

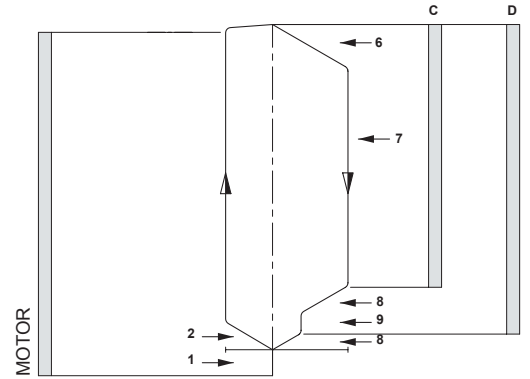
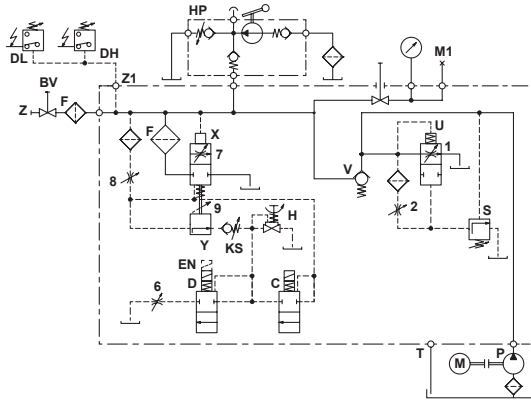
Adjustments DOWN

- 6** Down Acceleration
- 7** Down Full Speed
- 8** Down Deceleration
- 9** Down Levelling Speed

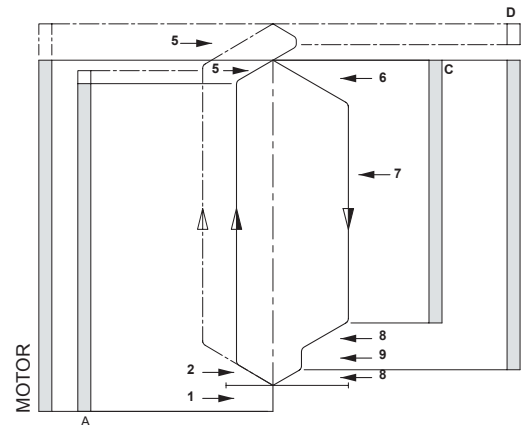
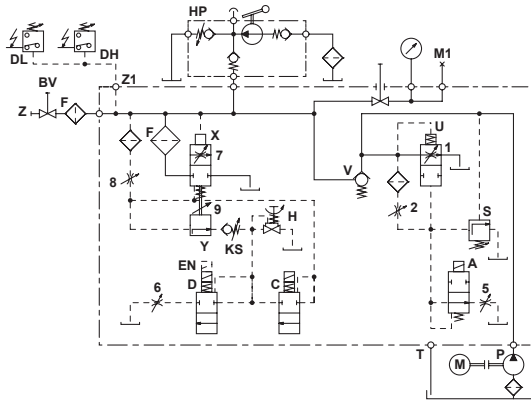
Hydraulic Circuit

Electrical Sequence

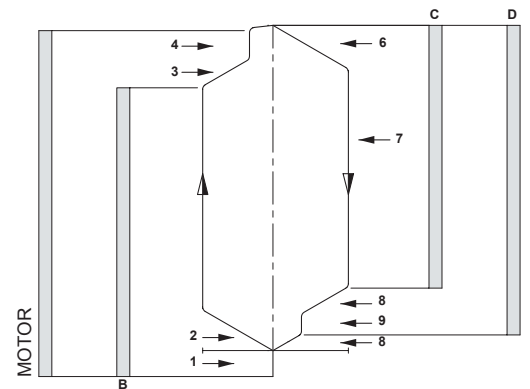
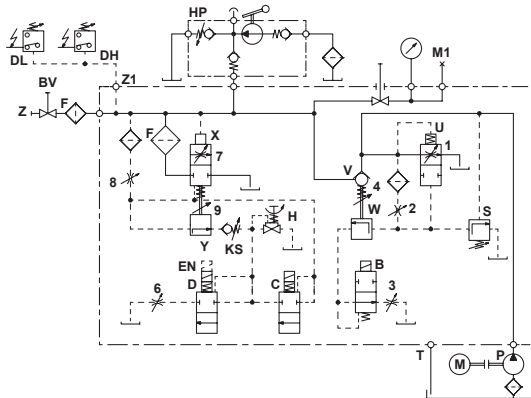
EV 0



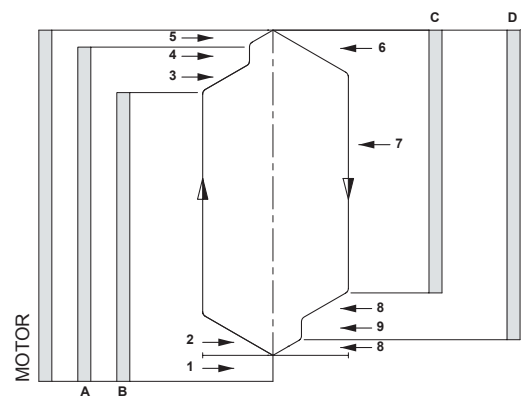
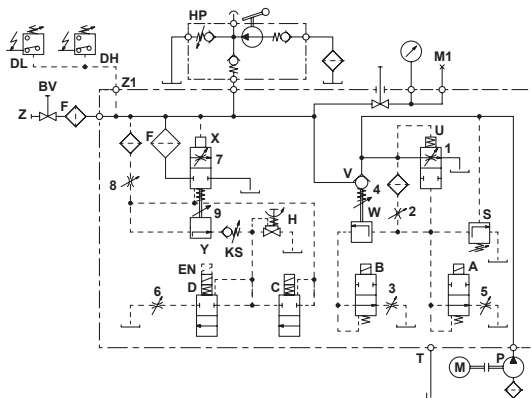
EV 1



EV 10



EV 100





EV Spare Parts List

EV

Pos.	No.	Item
1	FS	Lock Screw - Flange
	FO	O-Ring - Flange
	1F	Flange - By Pass
	EO	O-Ring - Adjustment
	1E	Adjustment - By Pass
	UO	O-Ring - By Pass Valve
	U	By Pass Valve
	UD	Noise Suppressor
UF	Spring - By Pass	
2	2	Adjustment - Up Acceleration
3	3	Adjustment - Up Deceleration
4	EO	O-Ring - Adjustment
	4E	Adjustment - Up Levelling
	4F	Flange - Check Valve
	FO	O-Ring - Flange
	VF	Spring - Check Valve
	W	Up-Levelling Valve
	WO	O-Ring - Up Levelling Valve
	VO	Seal - Check Valve
V	Check Valve	
W6	Screw - Check Valve	
5	3	Adjustment - Up Stop
6	3	Adjustment - Down Acceleration
7	7F	Flange - Down Valve
	FO	O-Ring - Flange
	7O	O-Ring - Adjustment
	7E	Adjustment - Down Valve
	UO	O-Ring - Down Valve
	XO	Seal - Down Valve
	X	Down Valve
	XD	Noise Suppressor
F	Main Filter	
8	8	Adjustment - Down Deceleration
9	9E	Adjustment - Down Levelling
9	EO	O-Ring - Adjustment
	9F	Spring - Down Valve
	Y	Down Levelling Valve
H	H	Manual Lowering - Self Closing
HO	HO	Seal - Manual Lowering
S	SE	Adjustment - Screw
	SM	Hexagonal
	MS	Grub Screw
	SO	O-Ring - Nipple
	SZ	Nipple
SF	Spring	
SK	Piston	
A+B	MM	Nut - Solenoid
	AD	Collar - Solenoid
	M	Coil - Solenoid (indicate voltage)
	AR	Tube - Solenoid 'Up'
	MO	O-Ring - Solenoid
	AN	Needle - 'Up'
	AF	Spring - Solenoid 'Up'
	AH	Seat Housing - 'Up'
C+D	AS	Seat - Solenoid 'Up'
	MM	Nut - Solenoid
	M	Coil - Solenoid (indicate voltage)
	DR	Tube - Solenoid 'Down'
	MO	O-Ring - Solenoid
	DF	Spring - Solenoid 'Down'
	DN	Needle - 'Down'
	DK	Core - Solenoid
DG	Seat Housing with Screen - 'Down'	
FD	Filter Solenoid	
DS	Seat - Solenoid 'Down'	

Some parts occur more than once in different positions of the valve.

O-Ring-Size

No.	O-Ring-Size	O-Ring-Size
FO	26x2P	47x2.5P
EO	9x2P	9x2P
UO	26x2V	39.34x2.62V
WO	5.28x1.78V	5.28x1.78V
VO	23x2.5V	42x3V
7O	5.28x1.78P	9x2P
XO	13x2V	30x3V
HO	5.28x1.78V	5.28x1.78V
SO	5.28x1.78P	5.28x1.78P
MO	26x2P	26x2P

* FO by 4F 2 1/2" is 67x2.5P
** 90 Shore

O-ring: V = FKM - Viton
P = NBR - Perbunan

Solenoid Valves

C+D **A+B**

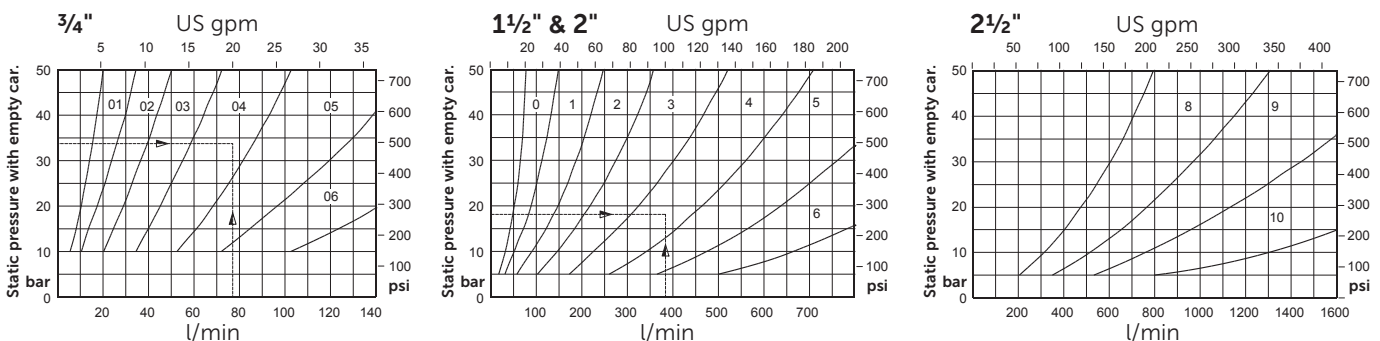
Adjustments

Flow Valves

In case of internal leakage, replace and test in the following order: (DS) & (DN), (XO), (VO), (WO), (FO) + (HO).

⚠ Taper threads: Do not exceed 8 turns of piping into the valve connections.

Flow Guide Selection Charts



To order EV: Valve size (inch), valve type, state pump flow, empty car pressure (or flow guide size) and coil voltage.
Example order: 1 1/2"EV100, 380l/min, 18bar (empty), 110 AC or 1 1/2"EV100/4/110AC

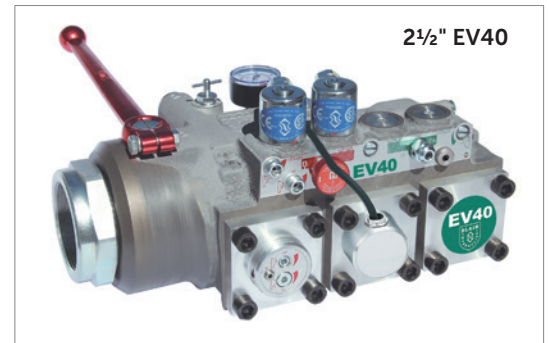
The BLAIN EV40-vvfv program includes the widest range of vvfv solution offered to the elevator industry for high performance passenger elevators. Easy to install, EV40's are smooth, reliable and precise in operation throughout extreme load and temperature variations with inbuilt overload protection and different energy saving modes. The EV40 system uses the L1000H or GA700 VVVF drive to control up travel, while down travel is managed by the EV40 valve itself. In this way, the EV40-vvfv solution offers a most cost-effective and energy-efficient alternative.



3/4" EV40



1 1/2" & 2" EV40



2 1/2" EV40

Description

Available port sizes are 3/4", 1 1/2", 2", and 2 1/2" pipe threads, depending on flow requirements. The **EV40-F** eliminates high inrush currents and does not require wye-delta switching. Based on customer elevator data, valves are factory-adjusted, ready for operation, and very easy to readjust if needed. The **L1000H** or **GA700 YASKAWA** drives, combined with feedback systems, are designed to compensate for elevator speed fluctuation, regardless of oil temperature or car load conditions.

Caution: The EV4 valve is to be used only together with L1000H or GA700 YASKAWA drives and not as a standalone control valve. EV40 valves include the following features essential for efficient installation and trouble free service:



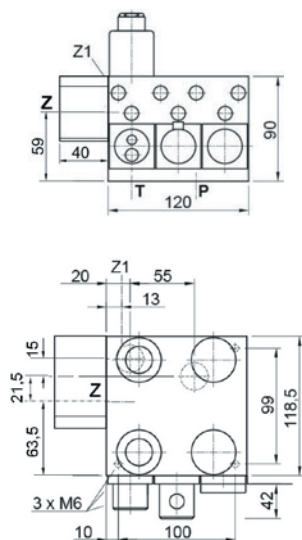
Simple Responsive Adjustment
Temperature and Pressure Compensations
Pressure Gauge and Shut Off Cock
Self Closing Manual Lowering
Self Cleaning Pilot Line Filters

Self Cleaning Main Line Filter (Z-T)
Built-in Turbulence Suppressors
70 HRC Rockwell Hardened Bore Surfaces
100% Continuous Duty Solenoids
Compact and aesthetic design

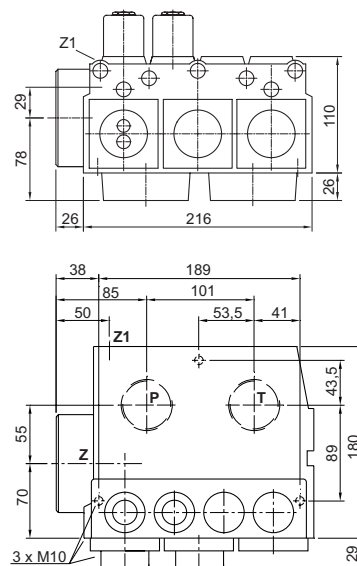
Technical Data:

		3/4" EV40	1 1/2" & 2" EV40	2 1/2" EV40
Flow Range:	l/min (US gpm)	10-125 (2-33)	30-800 (8-212)	500-1530 (130-405)
Pressure Range (valve):	bar (psi)	8-70 (116-1015)	8-70 (116-1015)	8-68 (116-986)
Press. Range CSA (valve):	bar (psi)	8-55 (117-797)	8-55 (117-797)	8-55 (117-797)
Burst Pressure Z:	bar (psi)	575 (8340)	505 (7324)	340 (4931)
Pressure Drop P-Z:	bar (psi)	6 (87) at 125 l/min	4 (58) at 800 l/min	4 (58) at 1530 l/min
Weight:	kg (lbs)	5 (11)	10 (22)	14 (31)
Coils AC:		24 V/1.8 A, 42 V/1.0 A, 110 V/0.43 A, 230 V/0.18 A, 50/60 Hz.		
Coils DC:		12 V/2.0 A, 24 V/1.1 A, 42 V/0.5 A, 48 V/0.6 A, 80 V/0.3 A, 110 V/0.25 A, 196 V/0.14 A.		
Oil Viscosity:		25-75 cSt. at 40°C (104°F).		
Operation oil temperature range:		10°C-60°C (50°F-140°F), for oil VGA46: 250cSt.-20 cSt.		
Optimal oil temperature range:		20°C-55°C (68°F-131°F), for oil VGA46: 140cSt.-24 cSt.		
Ambient temperature range:		0°C-50°C (32°F-122°F)		
Insulation Class, AC and DC:		IP 68		
Max. Oil Temperature:		70°C (158°F)		

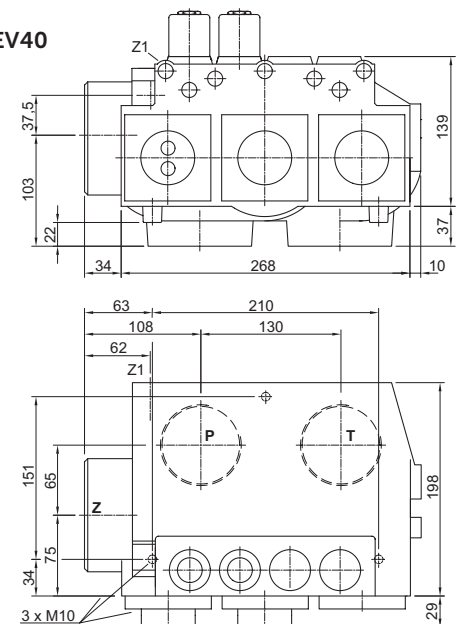
3/4" EV40



1 1/2" & 2" EV40



2 1/2" EV40



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74078 Heilbronn
Germany

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www.blain.de
info@blain.de



GmbH

Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators

Optional Equipment

EN Emergency Power Coil
CSA CSA Coils
KS Slack Rope Valve
BV Main Shut-Off Valve
HP Hand Pump

DH High Pressure Switch
DL Low Pressure Switch
CX Pressure Compensated Down Valve
MX Auxiliary Down
iL10-S UCM valve

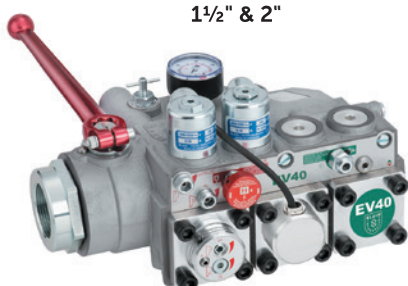


EV40

3/4"



1 1/2" & 2"



2 1/2"



Up Up to 1m/s (200 fpm). 1 full speed and 1 levelling speed.
 All up-direction input data and setup are done via a mobile phone.

Down Up to 1 m/s (200 fpm). 1 Full Speed and 1 Levelling Speed.
 All down functions are smooth and adjustable.

Control Elements

C Solenoid (Down Deceleration)
D Solenoid (Down Stop)
H Manual Lowering
S Relief Valve
U By Pass Valve
V Check Valve
X Full Speed Valve (Down)
Y Levelling Valve (Down)
F Filter

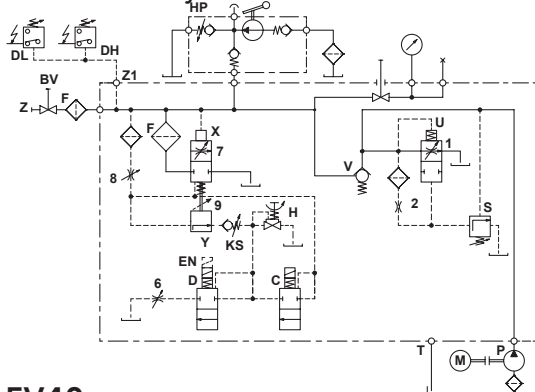
Adjustments UP

None
 (Fixed Orifice)

Adjustments DOWN

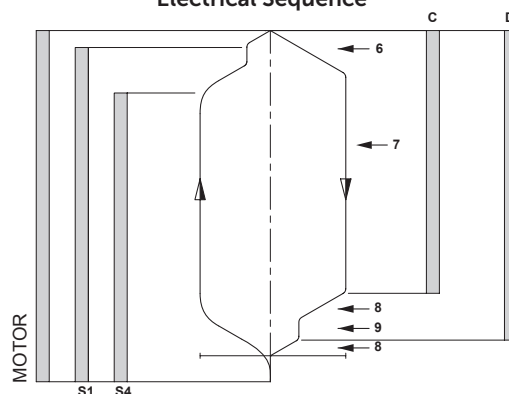
6 Down Acceleration
7 Down Full Speed
8 Down Deceleration
9 Down Levelling Speed

Hydraulic Circuit



EV40

Electrical Sequence



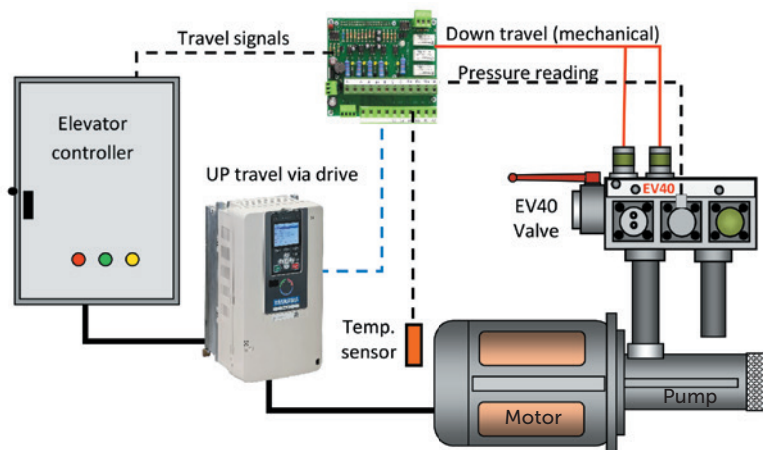
UP direction control



Caution: Refer to the detailed installation and set-up procedures in the EV40-F Handbook and the YASKAWA L1000H or GA700 Technical Manual.

The upward direction is controlled by the YASKAWA L1000H or GA700 drives. Using its software, the EV40-F system reads the car's load and current oil temperature, then processes the data to determine the appropriate motor speeds for nominal, inspection, and levelling travel.

The system can be set up quickly and easily via a smartphone, without needing to adjust any drive parameters. An advanced teach run with an empty car is sufficient for self-configuration, ensuring smooth take-off and accurate stopping at the floor.





Warning: Only qualified personnel should adjust or service the EV40 valve and the L1000H or GA700 drives. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical controller is switched off, cylinder line is closed and residual pressure in the valve is reduced to zero.



DOWN travel adjustments

Valves are already adjusted and tested. Check electrical operation before changing valve settings. Test that the correct coil is energised, by removing nut and raising the coil slightly to feel pull.

Standard settings: Adjustments 7 and 9 should be set level with the flange face. Then, turn adjustment 9 out by half a turn. Turn adjustments 6 and 8 fully inward. For EV40 3/4": Turn adjustment 6 out by 2 1/2 turns and adjustment 8 out by 1 turn. For EV40 1 1/2" – 2 1/2": Turn adjustment 6 out by 2 to 2 1/2 turns and adjustment 8 out by 1 1/2 turns.

6. Down Acceleration: When coils **C** and **D** are energized, the car will accelerate downwards according to the setting of adjustment **6**. 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration.

7. Down Speed: With coils **C** and **D** energized as in **6** above, the full down speed of the car is according to the setting of adjustment **7**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

8. Down Deceleration: When coil **C** is de-energized whilst coil **D** remains energized, the car will decelerate according to the setting of adjustment **8**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration.

Attention: Do not close all the way in! Closing adjustment 8 completely (clockwise) may cause the car to fall on the buffers.

9. Down Levelling: With coil **C** de-energized and coil **D** energized as in **8** above, the car will proceed at its down levelling speed according to the setting of adjustment **9**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster down levelling speed.

Down Stop: When coil **D** is de-energized with coil **C** remaining de-energized, the car will stop according to the setting of adjustment **8** and no further adjustment will be required.

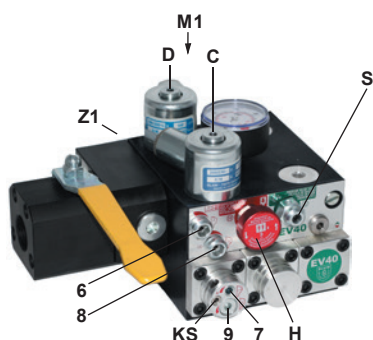
KS Slack Rope Valve: Both coils **C** and **D** must be de-energized beforehand! Loosen the small grub screw located on the left-hand side of the down flange. The **KS** is adjusted using a 3 mm Allen key. Turn **in** for higher pressure and **out** for lower pressure. With **KS** turned fully in, then backed out half a turn, the unloaded car should descend when Manual Lowering **H** is opened. If the car does not descend, turn **KS** further out slowly until the car just begins to descend. Then turn it out an additional half turn to ensure the car can be lowered properly, even with cold oil.

Adjustments pressure relief valve

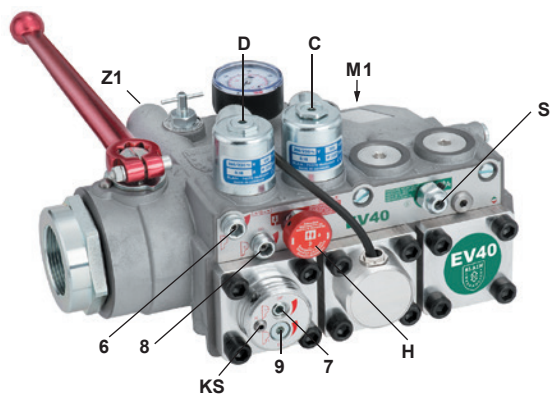
Valves are pre-adjusted for proper functionality. Verify electrical operation before making any changes to the inverter settings. Refer to the EV40 manual for the required parameter settings related to upward travel.

S Relief Valve: Turning 'in' (cw) increases and turning 'out' (ccw) decreases the **maximum pressure setting**. After turning the valve **out**, briefly open Manual Lowering **H** to read the actual pressure.

Important: When testing the relief valve, always close the ball valve gradually to avoid sudden pressure spikes.



M1 Second pressure gauge connection, 1/2"
Z1 Pressure switch connection, 1/4"



Adjustments DOWN

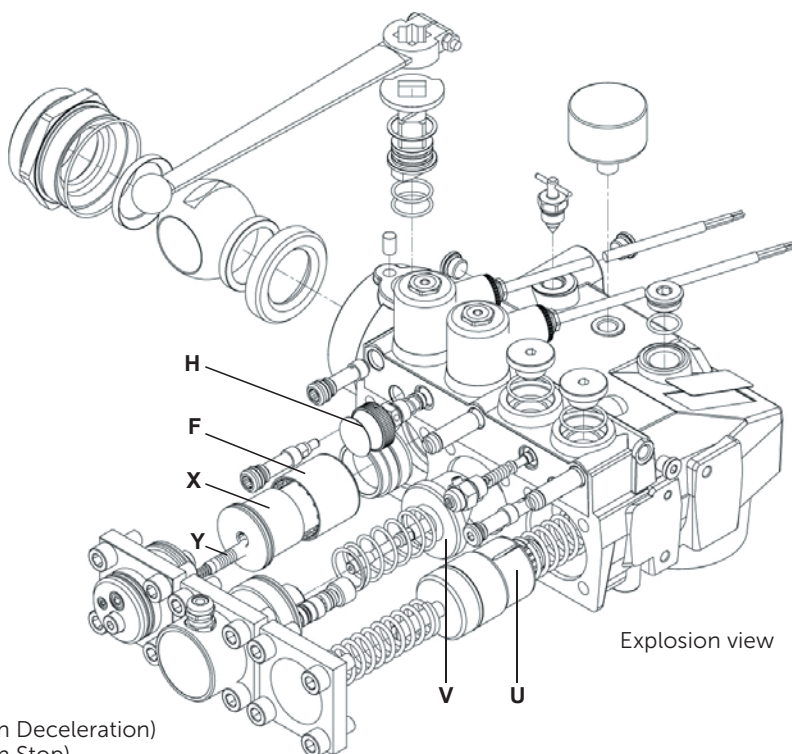
- 6** Down Acceleration
- 7** Down Full Speed
- 8** Down Deceleration
- 9** Down Levelling Speed

Control Elements

- C** Solenoid (Down Deceleration)
- D** Solenoid (Down Stop)
- H** Manual Lowering
- S** Relief Valve
- U** By Pass Valve
- V** Check Valve
- X** Full Speed Valve (Down)
- Y** Levelling Valve (Down)
- 2** Fix Orifice



Important: Length of 3/4" thread on pump connector should not be longer than 14 mm!



Explosion view



EV40 Spare Parts List

EV40-F

Pos. No.	Item
1	FS Lock Screw - Flange
	FO O-Ring - Flange
	1F4 Flange - By Pass
	UO O-Ring - By Pass Valve
	U4 By Pass Valve
	UD Noise Suppressor
	UF1 Spring - By Pass
	UF2 Spring - By Pass
2	US Dead Stop
	2 Fixed orifice
3	Plug
4	4F4 Flange - Check Valve
	FO O-Ring - Flange
	VF Spring - Check Valve
	VO Seal - Check Valve
	V Check Valve
	W Up-Levelling Valve
	WO O-Ring - Up Levelling Valve
	VO Seal - Check Valve
5	W6 Screw - Check Valve
	3 Plug
6	Adjustment - Down Acceleration
7	7F Flange - Down Valve
	FO O-Ring - Flange
	7O O-Ring - Adjustment
	7E Adjustment - Down Valve
	UO O-Ring - Down Valve
	XO Seal - Down Valve
	X Down Valve
	XD Noise Suppressor
8	F Main Filter
	8 Adjustment - Down Deceleration
9	EO O-Ring - Adjustment
	9E Adjustment - Down Levelling
	9F Spring - Down Valve
	Y Down Levelling Valve
H	Manual Lowering - Self Closing
HO	Seal - Manual Lowering
S	SE Adjustment - Screw
	SM Hexagonal
	MS Grub Screw
	SO O-Ring - Nipple
	SZ Nipple
C+D	SF Spring
	SK Piston
	MM Nut - Solenoid
	M Coil - Solenoid (indicate voltage)
	DR Tube - Solenoid 'Down'
C+D	MO O-Ring - Solenoid
	DF Spring - Solenoid 'Down'
	DN Needle - 'Down'
	DK Core - Solenoid
	DG Seat Housing with Screen-'Down'
C+D	FD Filter Solenoid
	DS Seat - Solenoid 'Down'

Some parts occur more than once in different positions of the valve.

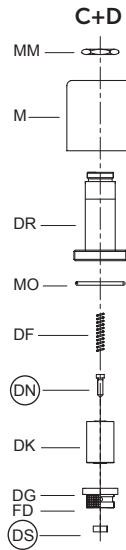
O-Ring-Size			
No.	3/4"	1 1/2"	2 1/2"
FO	26x2P	47x2.5P	58x3P *
EO	9x2P	9x2P	9x2P
UO	26x2V	39.34x2.62V	58x3V
WO	5.28x1.78V	5.28x1.78V	5.28x1.78V
VO	23x2.5V	42x3V	60x3V **
7O	5.28x1.78P	9x2P	9x2P
XO	13x2V	30x3V	47x3V
HO	5.28x1.78V	5.28x1.78V	5.28x1.78V
SO	5.28x1.78P	5.28x1.78P	5.28x1.78P
MO	26x2P	26x2P	26x2P

* FO by 4F 2 1/2" is 67x2.5P
** 90 Shore

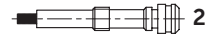
O-Ring: V=FKM - Viton
P=NBR - Perbunan

US is only for EV40 1 1/2" and above sizes!

Solenoid Valves



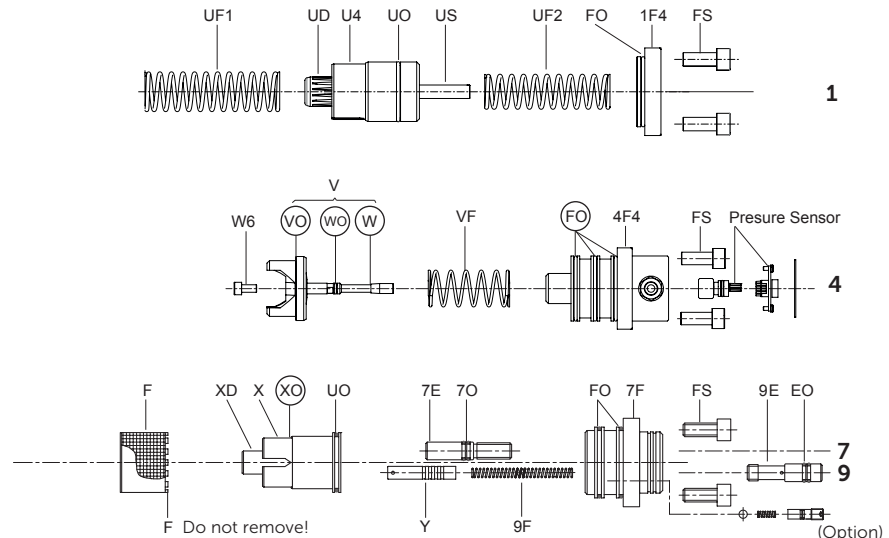
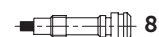
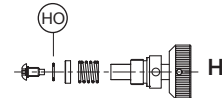
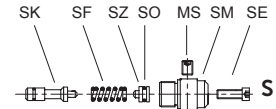
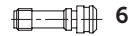
Fix orifice



Plug



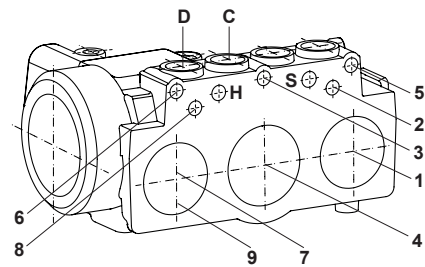
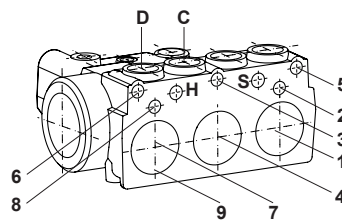
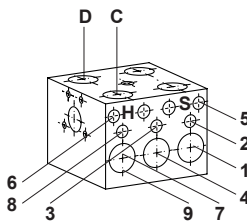
Adjustments



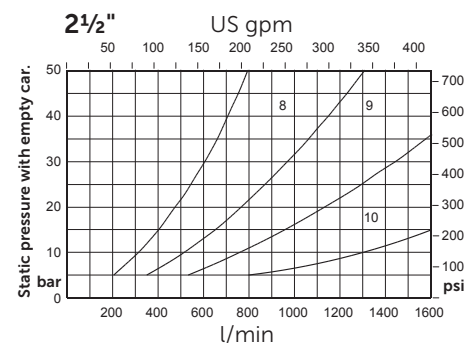
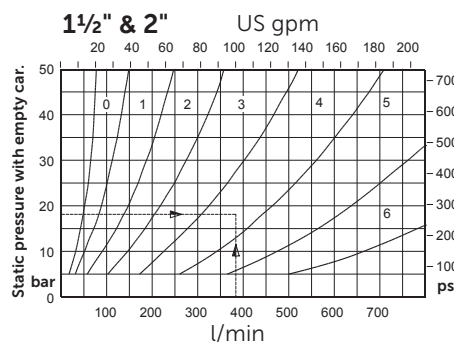
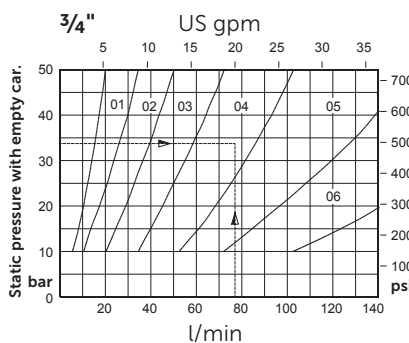
In case of internal leakage, replace and test in the following order: (DS) & (DN), (XO), (VO), (WO), (FO) + (HO).



Taper threads: Do not exceed 8 turns of piping into the valve connections.



Flow Guide Selection Charts

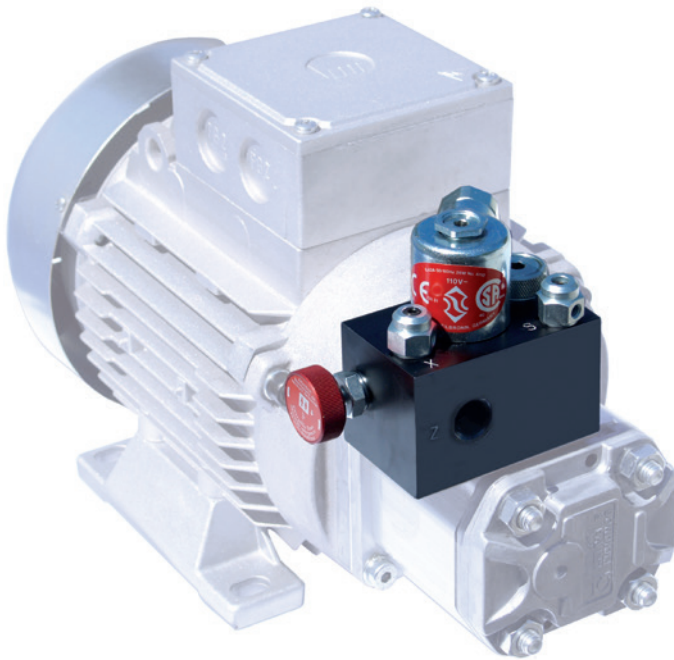


To order EV40: Size (inch), state pump flow, empty car pressure (or flow guide size) and coil voltage.

Example order: 1 1/2"EV40, 380l/min, 18bar (empty), 110AC or 1 1/2"EV40/4/110AC

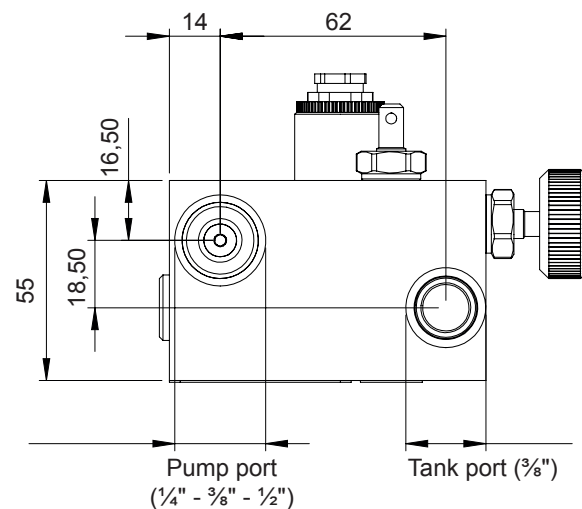
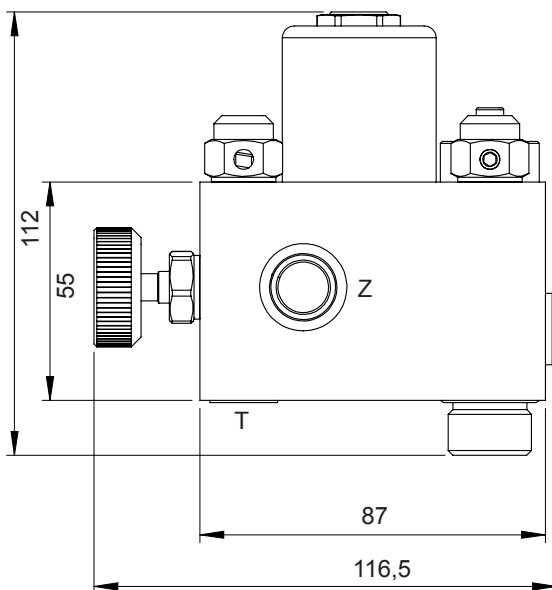
For hydraulic car-parking ports and material lifting platforms

The Blain car parking platform valve can be used in car parking applications where the platform needs to be raised above the ground to accommodate another car below the port. Alternatively this valve is also ideal for lifting material, cargo and suitable for dumb-waiters and goods lifts. The valve offers single up speed and adjustable down speed.



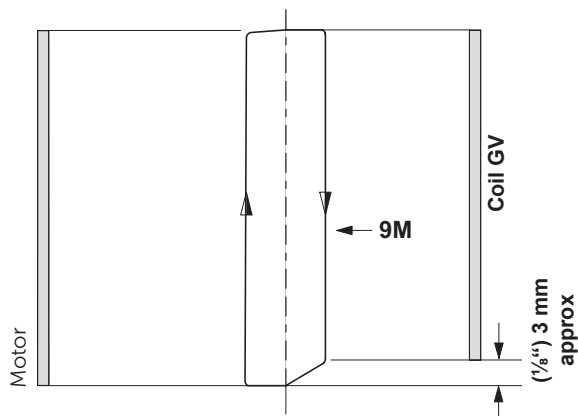
Flow Range:	1-24 l/min (0.3 – 6.3 US gpm)
Oil Viscosity:	25-60 cSt. at 40°C (104°F)
Coils AC:	24 V/1.8 A, 42 V/1.0 A, 110 V/0.5 A, 230 V/0.18 A, 50/60 Hz
Coils DC:	12 V/2.1 A, 24 V/1.1 A, 42 V/0.6 A, 80 V/0.3 A, 125 V/0.25 A, 196 V/0.14 A
Ports:	P Pump, Z Cylinder and T Tank
Pressure Range:	3-130 bar (44-1885 psi)
Burst Pressure:	600 bar (8702 psi)
Insulation Class, AC and DC:	IP 68

Check valve, pressure relief valve and manual lowering are standard options which are integrated in the GV valve. Additionally a shut off valve is also provided to shut the flow of oil from the pump to cylinder and vice-versa. The valve offers a standard $\frac{3}{8}$ " cylinder and tank connection, whereas the pump connection is available in sizes $\frac{1}{4}$ ", $\frac{3}{8}$ " and $\frac{1}{2}$ ".

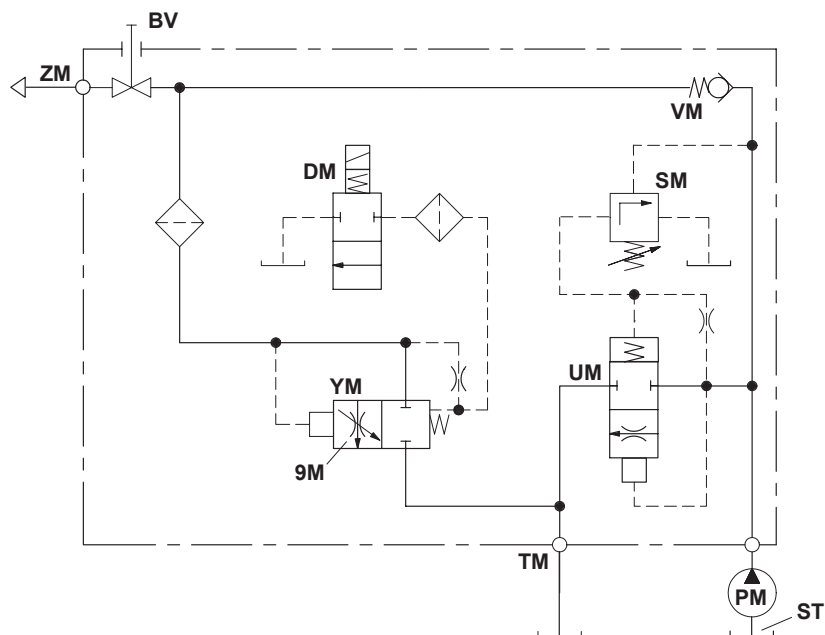




Electrical Sequence



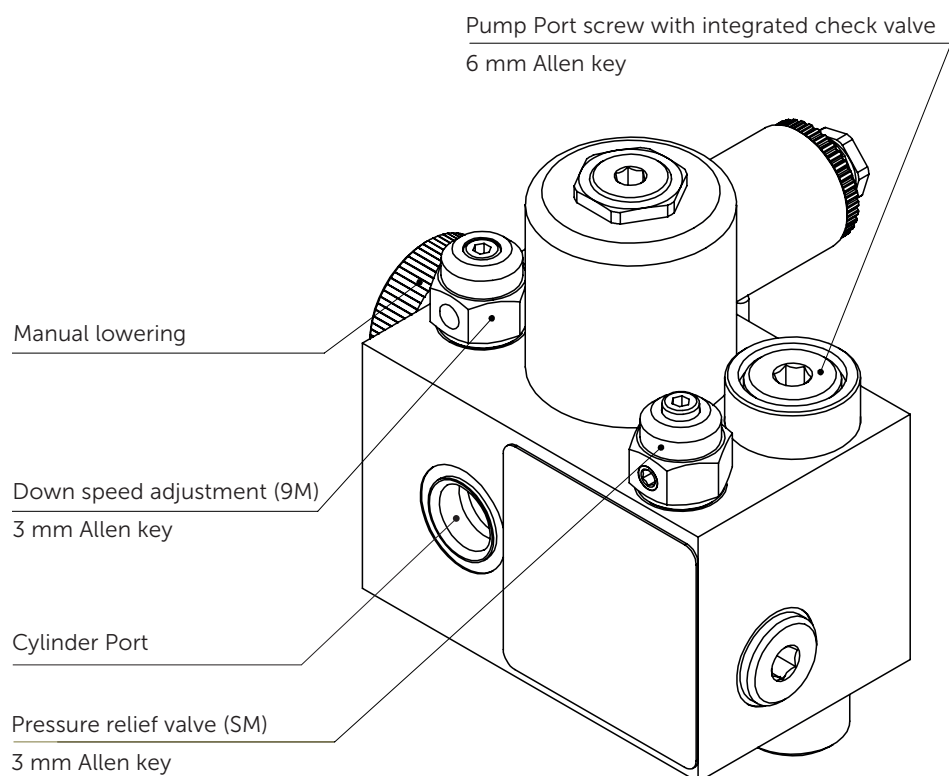
Hydraulic Circuit



DM Solenoid Down
9M Down Speed Adjustment
SM Pressure Relief Valve Adjustment
VM Check Valve

BV Shut Off Valve
ST Suction Tube
ZM Cylinder Connection
TM Return Connection

UM Pilot Operated Relief Valve
YM Pilot Operated Down Valve
PM Pump Motor



GV Adjustments

Up Speed is determined by the pump output and the effective diameter of the cylinder system.

SM Relief Valve. 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting.

9M Down Speed. When coil **DM** is energized, the car will travel downwards at a speed according to the setting of adjustment **9M**, (3 mm Allen key). Turning 'in' (clockwise) provides a slower, turning 'out' (c-clockwise), a faster down speed.

While ordering

Please specify pump flow, coil voltage and maximum operating pressure.

R10 Elevator Rupture Valve Instruction Manual



Aufzugs-Rohrbruchventil Betriebsanleitung



(GB)

The rupture valve **R10** can be mounted in any position directly onto the cylinder inlet connection **Z**. The 'T' port is connected to the Tank. Fitted incorrectly the R10 valve will not function. The cylinder side of the R10 can be identified by locating a small orifice behind the inside screw thread.

In the event of failure in the main cylinder line or where the down speed exceeds allowable limits, the R10 valve closes, bringing the car to a smooth stop.



R 10 AA Standard



R 10L-R

Optional

R 10L-AA

R 10L AA



R 10 AA + Option (DK + ES)

(D)

Das Rohrbruchventil **R10** wird direkt am Zylinder montiert. Es ist sicherzustellen, dass die mit **Z** markierte Seite zum Zylinder, hier befindet sich eine eingeschraubte Düse im Anschluss, und die mit **T** zum Tank zeigt. Ein falscher Einbau verhindert das Schließen des R10. Eine Einbaulage ist nicht zu beachten.

Sinkt der Aufzug wegen eines Rohrbruchs in der Zylinderleitung oder sonstiger Übergeschwindigkeit ab, wird der Aufzug vom R10 durch gedämpftes Schließen zum Stillstand gebracht.

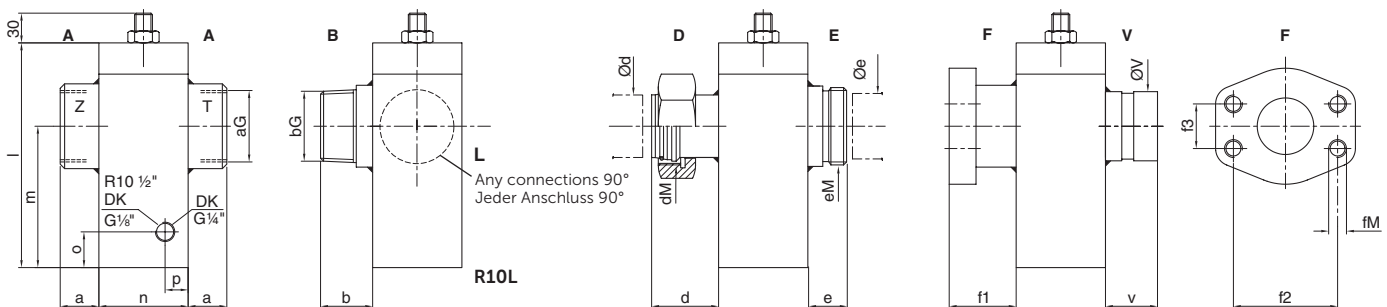


Warning: Only qualified personnel should service hydraulic valves. Unauthorised manipulation may result in injury or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off and residual pressure in the system is reduced to zero.



Warnung: Verletzungsgefahr! Neueinstellungen und Wartung dürfen nur durch qualifiziertes Aufzugspersonal durchgeführt werden. Vor der Wartung innerer Teile ist sicherzustellen, dass der elektrische Strom des Aufzuges abgeschaltet und der Druck im Ventil auf null reduziert worden ist.

Alternative connections • Alternative Anschlüsse

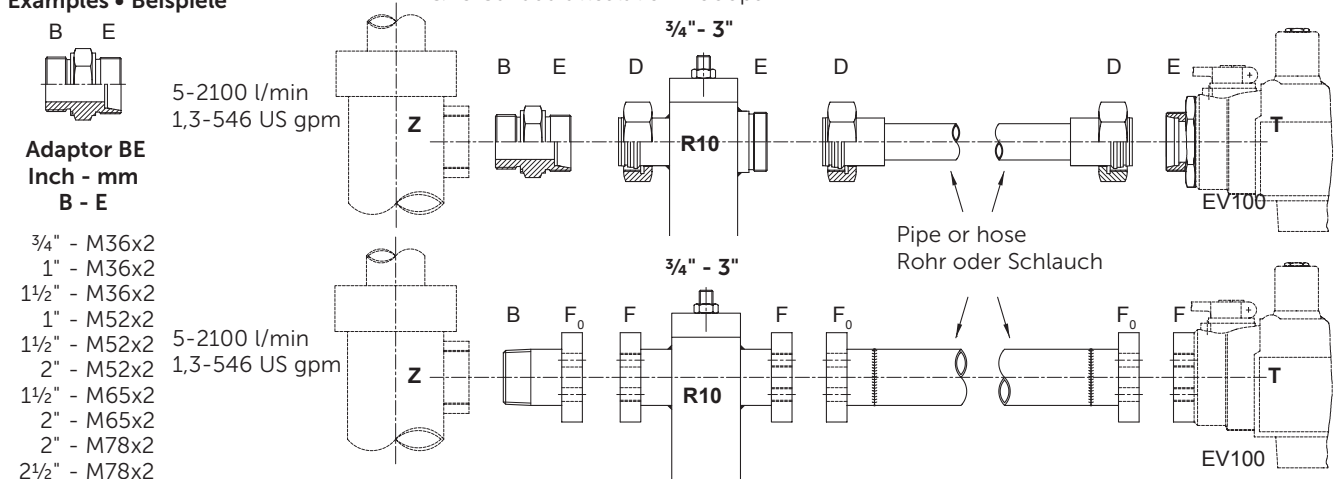


Recommended size • Empfohlene Größe

R10 Qc		R10 P max		A		B		UK CA cert. No.	D		E		F SAE				V Victaulic		l	m	n	o	p	DK Tube d	kg (AA)
l/min	US gpm	bar EN81	psi CSA	G o. NPT a	G o. NPT b				dM	Ød d	eM	Øe e	DN	fM	f1	f2	f3	Øv in Øv mm V in V mm							
4 - 90	1,0 - 23,8	100	1350	1/2"	18			UK-RV 009										1,05 26,7 1,5 38	88	53	40	9	11	6	0,9
5 - 100	1,3 - 26	100	1150*	3/4"	18	3/4"	32	UK-RV 010	M 52 x 2	42 44	M 36 x 2	28 26	19	M 10	36	47,6	22,2	1,31 33,4 1,5 38	105	62	50	11	11	6	1,6
50 - 175	13 - 46	100	1150*	1"	21	1"	32	UK-RV 011	M 52 x 2	42 44	M 36 x 2	28 26	25	M 10	38	52,4	26,2	1,31 33,4 1,5 38	105	62	50	11	11	6	1,6
100 - 425	26 - 112	100	1350	1 1/2"	26	1 1/2"	35	UK-RV 012	M 52 x 2	42 44	M 52 x 2	42 26	38	M 12	44	70	35,7	1,90 48,3 1,5 38	143	94	60	11	11	6	3,4
250 - 800	66 - 211	100	950	2"	28	2"	38	UK-RV 013	M 65 x 2	56 44	M 65 x 2	56 40	51	M 12	45	77,8	42,9	2,37 60,3 1,5 38	167	108	80	20	17	8	7
700 - 1250	185 - 330	80	880	2 1/2"	30	2 1/2"	45	UK-RV 014	M 78 x 2	63 50	M 78 x 2	63 40	64	M 12	50	89	50,8	2,87 73,0 1,5 38	196	121	100	19	18	8	13
1200 - 2100	317 - 554	60	690	3"	34	3"	45		M 78 x 2	63 50	M 78 x 2	63 40	76	M 16	50	106,4	62	3,50 88,9 1,5 38	240	149	120	22	21	10	21

Examples • Beispiele

*QPS Canada attestation: 1500psi



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Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators



Elevator Rupture Valve Instruction Manual

(GB)



Warning: Only qualified personnel should service hydraulic valves. Unauthorised manipulation may result in injury or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off and residual pressure in the system is reduced to zero.

Operation

Adjustment 1 'Closing flow' Q_c. By screwing inwards, the valve closes with increased oil flow. Screwing outwards, the valve closes with decreased oil flow. Loosening the lock nut (on the flange) before and tightened afterwards should not be forgotten in order to prevent any thread damage inside the flange. The Deceleration is less than 1 G and for safety reasons this is a non-adjustable built-in feature of the R10 valve. Adjustment should only be carried out when the R10 has not been triggered (active state). The rupture valve reopens through an 'Up' command or with the use of a hand pump (if fitted) if it has closed.

2 'Lowering Speed' (optional). By screwing inwards, the lowering speed is increased.

Service and Repair

Servicing is not necessary on the R10 valve. Inspection for leakage is recommended at least once a year. Should external leakage be discovered, the O-Rings on the screw of Adjustment 1 and between the flange and R10 body must be changed. If internal leakage is detected, then the complete R10 valve must be changed. For doing that the lift must be lowered up to a suitable position, secured and the oil removed from the system.

Testing

Testing the R10 must be tested under payload conditions to ensure closing of the valve between nominal speed plus 0.3 m/s (60 fpm). When the down flow is approximately equal to the adjusted closing flow, the R10 will take several seconds to close. When the down flow is in much excess of the adjusted closing flow, the R10 will close in 1-2 seconds.

Option

Option 2. For evacuation purpose, the adjustment '2' of the activated R10 can be turned slowly inside to lower the cabin to the bottom floor.

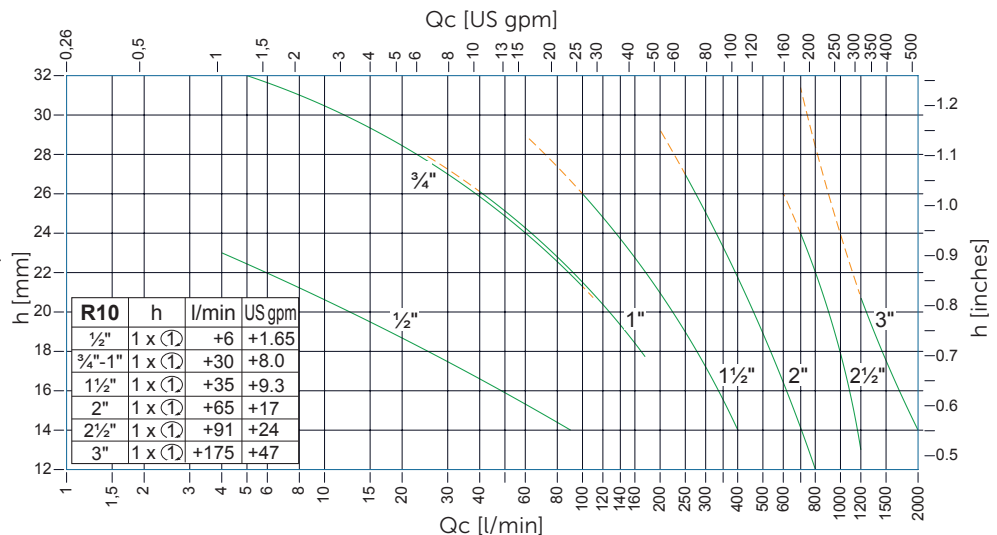
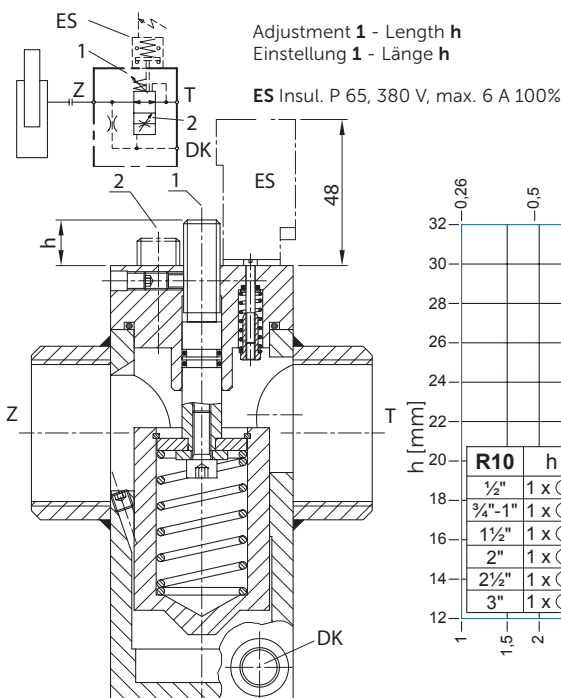
DK

For a multiple cylinder system, the connecting of pilot ports 'DK' ensures that all R10 valves close almost simultaneously.

Please refer the table 'Recommended Size' in column 'DK Tube' for selecting the inside connecting diameters between the various R10 for different valve sizes. This connection is pressurised and should only be serviced after the line has been de-pressurised.

ES

The rupture valve can be fitted with an electrical limit switch 'ES' which is actuated when the valve closes and serves to initiate a signal to the Lift Controller to either indicate a warning signal or to shut down the elevator.



Aufzugs-Rohrbruchventil R10 Betriebsanleitung

(D)



Warnung: Verletzungsgefahr! Neueinstellungen und Wartung dürfen nur durch qualifiziertes Aufzugspersonal durchgeführt werden. Vor der Wartung innerer Teile ist sicherzustellen, dass der elektrische Strom des Aufzuges abgeschaltet und der Druck im Ventil auf null reduziert worden ist.

Einstellungen

Das Hineindrehen der Stellschraube 1 bewirkt eine größere Schließmenge Q_c, das Herausdrehen eine geringere. Die Kontermutter ist vorher zu lösen und anschließend wieder zu befestigen. Löst das R10 aus und schließt den Kolben, ist die Abbremsung weniger als 1 G und aus Sicherheitsgründen nicht verstellbar. Das Nachstellen der Stellschraube ist nur bei nicht aktivierten R10 durchzuführen. Durch eine Hubfahrt bzw. das Betätigen einer Handpumpe erfolgt ein Druckausgleich zwischen Zylinder- und Tank-Anschluss des R10 und es öffnet sich selbsttätig.

Bei geschlossenem R10 erlaubt die optionale Absenk-Einstellung 2 ein kontrolliertes Absinken des Aufzuges. Hineindrehen bewirkt eine schnellere Absenkgeschwindigkeit; heraus eine langsamere.

Wartung und Reparatur

Eine Wartung des R10 ist nicht notwendig. Inspektionen zur Leckage sind in regelmäßigen Intervallen, mindestens 1x pro Jahr, durchzuführen. Bei einer festgestellten äußeren Undichtheit sind die O-Ringe am Deckel bzw. die der Stellschraube zu wechseln. Bei innerer Undichtheit muss das R10-Ventil getauscht werden. Dazu ist der Aufzug jeweils aufzuhängen und die Zylinderleitung zu entleeren.

Prüfung

Zur Vergewisserung, dass sich das R10 Ventil zwischen der Nenngeschwindigkeit und 0,3 m/s über Nenngeschwindigkeit schließt, wird es üblicherweise mit Last geprüft. Entspricht der Durchfluss abwärts nur ungefähr der Schließmenge, schließt das R10 erst nach einigen Sekunden. Ist der Durchfluss abwärts aber weit mehr als die eingestellte Schließmenge, schließt es bereits in 1-2 Sekunden.

Optionen

Option 2. Zur Evakuierung von Personen kann im geschlossene Zustand das R10 Ventil mittels Einstellung 2 so eingestellt werden, dass der Aufzug langsam in die unterste Haltestelle absinkt.

DK

Bei Aufzügen mit mehreren parallel wirkenden Hebern sichert eine Steuerleitungsverbindung zwischen den Anschlüssen DK das fast gleichzeitige Schließen dieser Rohrbruch-Ventile. Die notwendigen Innen-Ø der Verbindungsleitung werden in der Spalte 'DK Tube' der Tabelle dargestellt. Diese Steuerleitung steht unter Druck. Wartungsarbeiten an dieser sind nur im drucklos Zustand durchzuführen.

ES

Ein auf dem Deckel montierter elektrischer Endschalter ES wechselt beim Schließen des R10 die Schalterstellung. Diese kann zur weiteren Signalverarbeitung im Schaltschrank genutzt werden, um so ein Warnsignal auszulösen oder den Aufzug abzuschalten.

	Range	Bereich
-----	Adjustable	Einstellbar
-----	Permitted*	Zulässig*

*AR 2014/33/EU



Instruction Manual - L10 Pressure Lock Valve (UCM/A3 Valve)

Certified by TÜV SÜD Germany



1/2"



3/4"



1 1/2"



2"



2 1/2"

Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment.



Prior to servicing internal parts, ensure that the electrical power is switched off, ball valve is closed and residual pressure in the valve is reduced to zero. Very high pressure spikes could result in deformation and oil splashing, this could cause serious injuries.

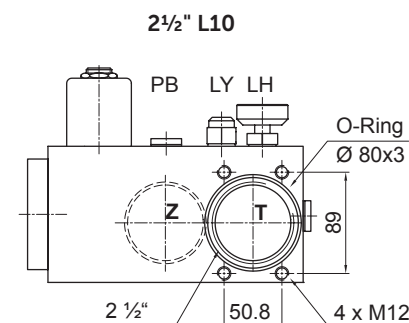
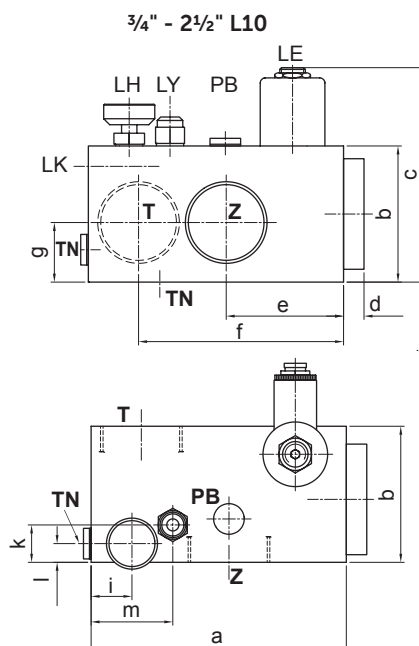
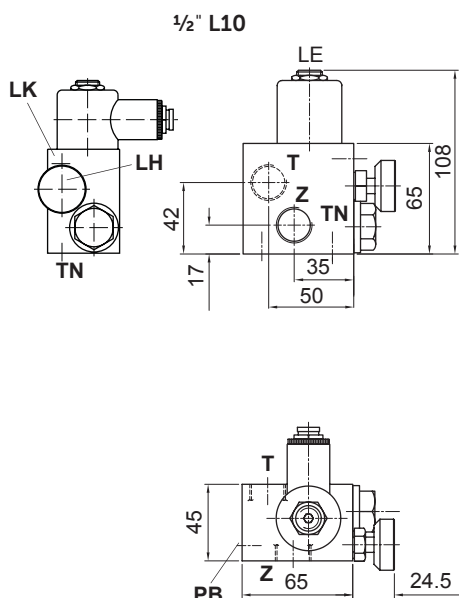
L10 Description: The L10 Pressure Lock Valve is a solenoid operated check valve designed for hydraulic elevators and includes a self closing manual lowering valve. Its purpose is to allow free flow of oil from the pump unit to the cylinder for up travel and to prevent flow in the reverse direction from the cylinder to pump until an electrical signal is given to the coil. The L10 can be mounted in any position without causing any operational problems.

Installed in the main cylinder line directly adjacent to the main elevator control valve, the L10 can be employed as a safety back up valve to the down system of the main control valve to prevent unwanted down movement of the elevator should an electrical or mechanical malfunction occur in the main control valve (UCM case).

Another application of the L10 is to reduce the amount of bounce in a hydraulic elevator system due to the compressibility factor of oil between the cylinder and the control valve, by mounting the L10 directly onto the cylinder connection.

A Slack Rope Valve LK for roped elevators (e.g. 2:1 indirect transmission) is optional. It prevents the slack rope condition caused by the lowering of the ram when the car is suspended in the safeties or resting on the buffers.

Technical Data:		1/2" L10	3/4" L10	1 1/2" L10	2" L10	2 1/2" L10
Flow Range:	l/min	80	125	400	800	1400
Pressure Range:	bar	10-100	10-100	10-100	10-80	10-70
Permitted pressure (according AR 2014/33/EU):	bar	10-100	10-100	10-59	12-59	10-59
Burst Pressure:	bar	500	500	500	450	365
Tank Connection for LH	TN	1/4"	1/4"	1/2"	1/2"	1/2"
Weight:	kg	0,8	1,4	2,5	4,2	7,0
PB Cylinder pressure port:	G	1/4"	1/4"	1/4"	1/4"	1/4"
UKCA certification No.		UK-UCM 004/1	UK-UCM 005/2	UK-UCM 006/2	UK-UCM 007/2	-



L10	3/4"	1 1/2"	2"	2 1/2"
a	110	150	175	210
b	65	80	100	120
c	108	134	154	174
d	8	12	12	16
e	60	69	75	89
f	92	120	136	160
g	31	35	45	55

possible AC coil voltages: 24 V/1.8 A; 42 V/1.0 A; 110 V/0.43 A; 230 V/0.18 A; 50/60 Hz

possible DC coil voltages: 12 V/2.0 A; 24 V/1.1 A; 42 V/0.5 A; 48 V/0.6 A; 80 V/0.3 A; 110 V/0.25A; 196 V/0.14 A

T→Z Free Flow. Solenoid LE not energized.

Z→T Flow only when Solenoid LE energized.

Attention: For manual lowering connect port TN with tank.

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Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators



Pressure Lock Valve (UCM/A3 Valve)

L10

Rest Position: When L10 is at stand-by, the coil **LE** is de-energized and the main flow guide **LV** closed, preventing flow from cylinder to tank.

Up Travel: During up travel with the pump running, oil flows through port **T**, through the flow guide **LV** and out through port **Z** to the main cylinder. Coil **LE** is not energized.

Down Travel: For the car to have a down travel, the L10 should be energized approximately 0.5s earlier than the main control valve (e.g. EV100). This enables the oil to escape from the pilot chamber and allows the main piston **LV** to open. Else, the pressure between the L10 and the control valve connecting line drops considerably and to make the elevator operational, the connecting line needs to be re-pressurized again. This can be done by starting the elevator for a short time in the up direction or by pumping oil using a hand pump. Re-pressurizing may also be needed if the main control valve has an internal leakage. Opening of the **LV** allows the flow of oil from the cylinder to the tank (from **Z** to **T**) via the L10 and the main control valve.

The coil **LE** on the L10 valve is de-energized after the down coil of the main control valve (e.g. coil D of EV100). In this way, the piston **LV** in L10 and the down piston **X** in the main control valve close completely.

Pressure drop: The pressure loss of the L10 valve depends on the size of the valve and the flow rate. The size and type of connectors used also influence the pressure losses. The pressure loss of the L10 valve should be taken into account while the main control valves empty car pressure is calculated.

Emergency down: The emergency manual lowering **LH** on the L10 is to be operated to bring the car down in emergency. The down speed of the car is determined by the setting of **LY**. As the **LH** is open, oil from the cylinder flows back to the tank through a return tank-line attached to the tank port **TN**. The return tank-line should not be smaller in size than the tank port **TN**, else the emergency manual lowering may not function properly.

The slack rope valve **LK** prevents the sinking of the RAM when the manual lowering **LH** is operated in a 2:1 roped elevator to prevent a tangled rope condition.

Air-bleed: After connecting the L10 valve or right after servicing the L10 valve needs to air-bleeded to ensure its functionality. It is sufficient to operate the emergency lowering valve or loosening the coil tube slightly until oil is visible and tightening it again.

Adjustments

Manual Down Speed LY (3/4", 1 1/2", 2" and 2 1/2" valves): 'In' (clockwise) provides a slower, 'out' a faster down lowering speed.

Slack Rope Valve LK: The **LK** is adjusted with a 3 mm Allen key by turning the screw **LK** 'in' for higher pressure and 'out' for lower pressure. With **LK** turned all the way 'in', then half a turn back out, the unloaded car should descend when the **LE** coil is energized. Should the car not descend, **LK** must turned out until the car just begins to descend, then turned out a further half turn to ensure that with cold oil, the car can be lowered as required.

Functional test

In order to check the functionality of the L10 pressure lock valve, the coil **LE** can be de-energized during down travel. Alternatively the L10 can also be tested by unscrewing the retaining nut **MM** and manually lifting the coil **M**.

Caution! Once the coil **M** is removed from the coil tube **DR**, the energized coil will begin to overheat after about 10 seconds, holding it out longer may result in burning of coil.

The test may result in pressure drop in the connecting line between L10 and the control valve. In order to make the elevator operational, the connecting line needs to be pressurized again. This can be done by using the handpump to move the cabin over a small distance in up direction. Pressurising the line might also be necessary in cases where the pressure loss between the L10 and control valve is high for e.g. due to internal leakage.

Filter FH (optional)

Severely contaminated oil can lead to internal leakage in the area of the emergency lowering **LH** or **LY**. This can be prevented by installing the **L10** with the optional filter **FH**. Retrofitting the filter is not possible in the older L10 valves.

Status of lift	Power supply to coil of L10
Up travel and relevelling	power off
Down travel and relevelling	power on
Stop with door closed	power can be switched off to save standby power
Unintended up travel with open doors	motor off once the movement sensor gets triggered
Unintended down travel with open door	power off once the movement sensor gets triggered
Emergency lowering	power on
Emergency manual lowering	manual actuation
Hand pump operation	power off

No.	Parts List
LF	Flange
LFO	O-Ring- Flange
LB	Ball
LVF	Spring - Flow Guide
LFG	Flow Guide
LVO	Seal - Flow Guide
LVB	Body - Flow Guide
LUO	O-Ring - Flow Guide
LH	Manual Down - Self Closing
LY	Manual Down Speed Adjuster
HO	Seal - Manual Low. (5,28x1,78)
MM	Nut
M	Coil (indicate voltage)
DR	Solenoid - Tube
MO	O-Ring
DF	Spring
DN	Needle
DK	Core
DG	Seat Housing (with screen)
FD	Filter
DS	Seat

Maintenance

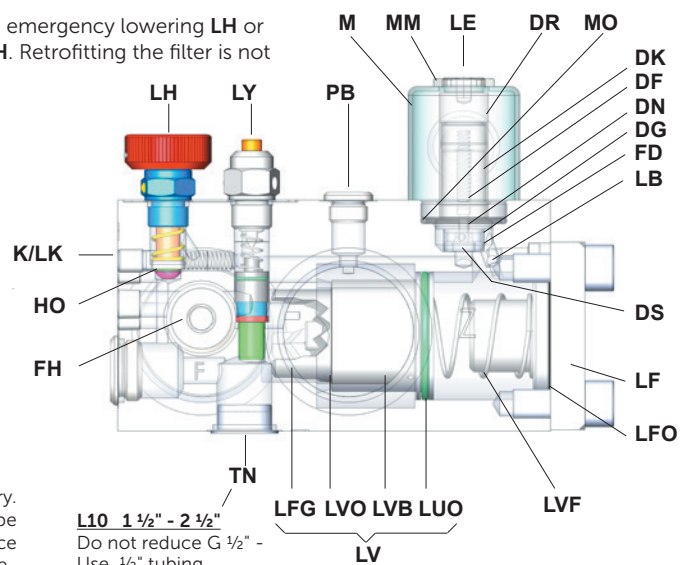
Maintenance of the L10 is not necessary. Inspection of internal leakage should be done in regular intervals, at least once a year. If internal leakage has been detected, check the parts **DN**, **DS** and **FD** of the Solenoid **LE** first. Then inspect the O-rings of **LV**, **LY** and **LH**. The pressure of the valve has to be released before. The filter **FH** should be checked and cleaned during maintenance.

Control Elements

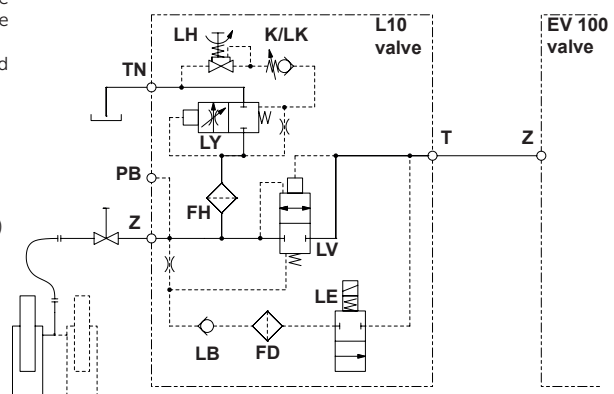
LV Check Valve
LH Manual Lowering
LK Slack Rope Valve (option)
LE Solenoid
PB Pressure Gauge (cylinder pressure)
LY Manual Down Speed Adjust. (not with 1/2" L10)

Connections

T Control Valve Connection
Z Cylinder Side Connection
TN Tank Return Line



Hydraulic Circuit



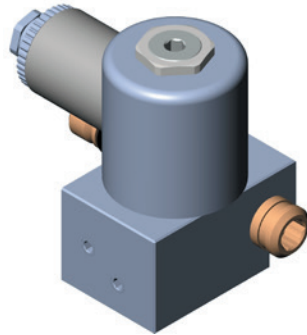


Instructions manual - UCM Valve iL10-S

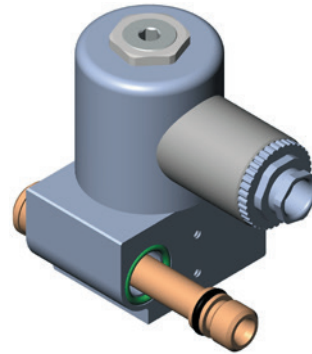
Certified by TÜV SÜD Germany



**NEW
PRODUCT**



iL10-S front view



iL10-S back view



Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off, ball valve is closed and residual pressure in the valve is reduced to zero. Very high-pressure spikes could result in deformation and oil splashing, which could cause serious injuries.

iL10-S Description: The **iL10-S** valves are solenoid operated lock valves designed as a safety option against Unintended Car Movement (UCM) for hydraulic elevators. The **iL10-S** model possesses only the adjustment 6 (Adj.6). Down movement of the lift is only possible by energizing coil D on the EV control valve and coil **A3** on the **iL10-S** valves. The upward direction is not affected.

Existing Blain valves of the EV, SEV, EV4 and EV40 assemblies in sizes $\frac{3}{4}$ ", 1", 1.5", 2" and 2.5" can easily be retrofitted with UCM safety using the **iL10-S** valve. The **iL10-S** model has a special **adjustment 6**. The type of adjustment 6 varies depending on the size of the control valve. Specify the actual valve size when ordering the **iL10-S** for renovation purposes. (See the table on page 4.)

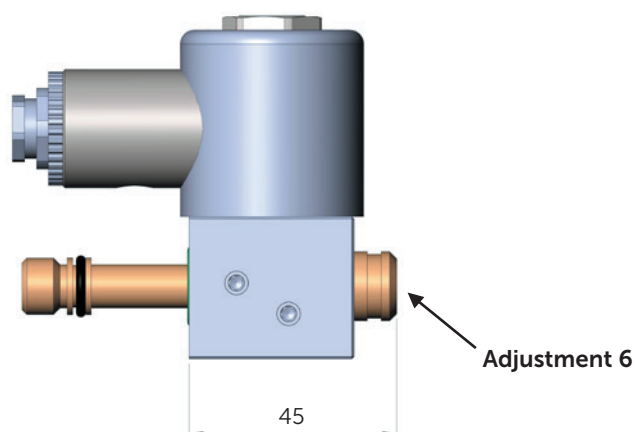
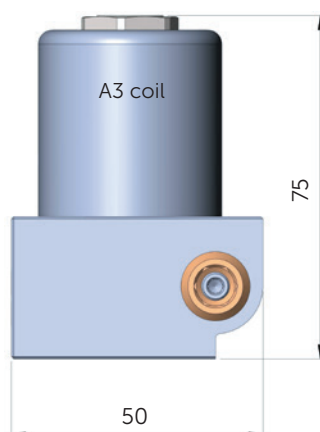
Technical data	unit	$\frac{3}{4}$ " iL10-S	1" - 2" iL10-S	2.5" iL10-S
Flow range:	l/min	125	800	1530
Pressure range:	bar	10-95	8-75	8-75
Burstpressure:	bar	p > 500	p > 500	340
Oil viscosity range:	cSt	~20 to 300 (~55°C to 10°C for ISO VG46)		
Oil temperature:	°C	10°C to 55°C		
Weight:	kg	0.6		

Coil voltages:

Alternating current (IP68) 50/60 Hz 24V/1.8A; 42V/1A; 110V/0.43A; 230V/0.18A

Direct current (IP 68) 12V/2A; 24V/1.1A; 42V/0.5A; 48V/0.6A; 80V/0.3A; 110V/0.25A; 196V/0.14A

Dimensions:



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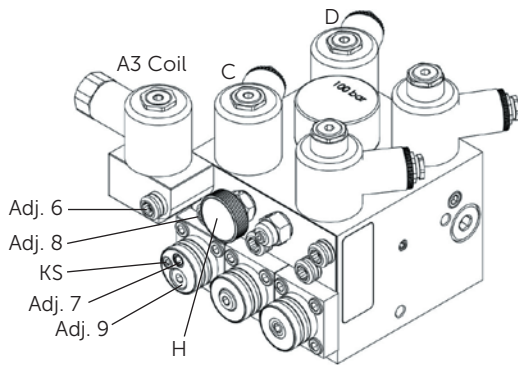


Figure 1: iL10-S valve assembly on 3/4" EV valve

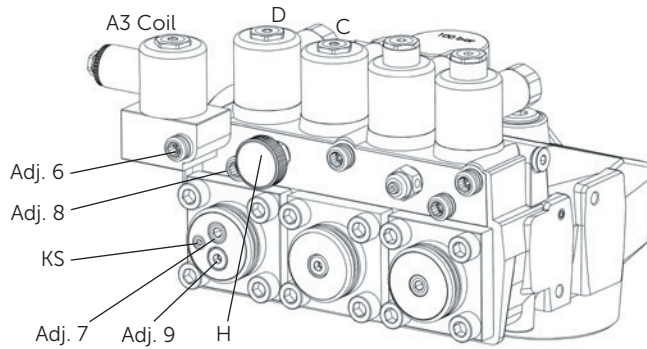


Figure 2: iL10-S valve assembly on 1.5"/2" EV valve

Adjustments

6	Adjustment-down acceleration	A3	iL10-S Solenoid coil
7	Adjustment-down speed	H	Manual lowering
8	Adjustment-down deceleration	KS	Slack rope valve (option)
9	Adjustment-down levelling		

iL10-S UCM Action

Rest Position: When **iL10-S** is in stand-by, the solenoid **A3** de-energized and the down valve **X** (inside the control valve) is kept closed by the cylinder pressure and a spring force preventing flow from cylinder to the tank.

Up Travel: During up travel with the pump running, oil flows through the check valve **V** and out through the port **Z** to the main cylinder. The coils of solenoids **A3**, **C** and **D** are not energized.

Down Travel: In order to have a down travel, the **A3** coil of **iL10-S** should be energized approximately 0.5s earlier than the coils on the main control valve (e.g. **D** & **C** of the EV100). When **A3** and the down coils **D** & **C** are energized the down valve **X** opens allowing the oil flow from cylinder to the tank (from **Z** to **T**), that is, lift starts a down travel. As the elevator reaches the stop switch at slow speed (**C** coil is de-energized), first the down coil **D** of main control valve is de-energized. About 0.5s later, the coil **A3** is de-energized on the **iL10-S** valve. In this way, it is made sure that the down valve **X** is at its closed position when the coil **A3** is switched off.

Pressure loss: The **iL10-S** valves does not cause extra pressure loss in the main hydraulic line.

Emergency down: To lower the cabin in an emergency situation, the self-closing emergency lowering valve **H** on the control valve is operated. When it is opened, the control oil behind the down flow guide escapes into the tank and the down flow guide opens. The lowering speed is determined by setting **9**.

The **KS** option prevents a slack rope condition in a 2:1 roped elevator. If the cabin rests on the buffer after manual lowering with **H**, the **KS** option prevents the piston rod from sinking further and thus the slack rope condition.

Adjustments

Control valve adjustment: **iL10-S** valves function effectively when the down travel adjustments on the main control valve are done acceptably. Too long, unacceptable deceleration time may increase stopping distance. The main control valve is adjusted similar to the EV type valves. When solenoids **C**, **D** and **A3** are energised, the car will accelerate downwards according to the setting of the adjustments:

6. Down Acceleration: 'In' (clockwise) provides a softer down acceleration, 'out' (c-clockwise) a quicker acceleration.

7. Down Speed: With solenoids **C**, **D** and **A3** energised, the full down speed of the car is according to the setting of adjustment **7**. 'In' (clockwise) provides a slower down speed, 'out' (c-clockwise) a faster down speed.

8. Down Deceleration: When solenoid **C** is de-energised whilst solenoid **D** and **A3** remain energised, the car will decelerate into slow speed according to the setting of adjustment **8**. 'In' (clockwise) provides a softer deceleration, 'out' (c-clockwise) a quicker deceleration.

Attention: Do not close all the way in! Closing Adjustment **8** completely (clockwise) may cause the car to fall on the buffers. If setting **8** is changed significantly, settings **6** and **9** must be readjusted.

9. Down Levelling: If only coils **D** and **A3** energised, the car will proceed at its down levelling speed according to the setting of adjustment **9**. 'In' (clockwise) provides a slower, 'out' (c-clockwise) a faster down levelling speed.

Down Stop: When solenoid coil **D** is also de-energised (coil **C** is already de-energised), the car will stop according to the setting of Adjustment **8**. Coil **A3** should be de-energized after a delay of 0.5s.

Manual Down Speed: When the Manual Lowering Knob **H** is opened, the elevator travels down with levelling speed according to the Adjustment **9**.

Slack Rope Valve KS: With **KS** turned all the way 'in', then half a turn back out, the unloaded car should descend when Manual Lowering **H** is opened. Should the car not descend, **KS** must be backed off (CCW) until the car just begins to descend, then backed off a further turn to ensure that with cold oil, the car can be lowered as required. If the cabin is sitting on the buffer, the piston must not move any further downwards when **H** is opened. The **KS** setting can be adjusted by turning the **KS** screw in (cw) to increase the pressure, and out (ccw) to decrease the pressure.

Solenoid seat: Depending on the valve size and maximum working pressure, seat diameter size is changed in the **iL10-S** valves. This is shown in Table on page 4.

Instructions manual - UCM Valve iL10-S

Certified by TÜV SÜD Germany



Maintenance

Regular maintenance of the **iL10-S** is not required; however, an annual inspection is recommended. If internal leakage is detected, the components **DS** and **DN** in solenoid valve **D** (EV control block) and in solenoid valve **A3** (iL10-S) should be checked first. Subsequently, the condition of the O-rings at **V**, **X**, and **H** on the EV control block should be examined.

During the inspection, the condition of the two **FD** filters in both solenoid valve **D** and solenoid valve **A3** should be checked and cleaned if necessary. The filter for the adjustment **8** should also be inspected and cleaned as needed.

Functional test

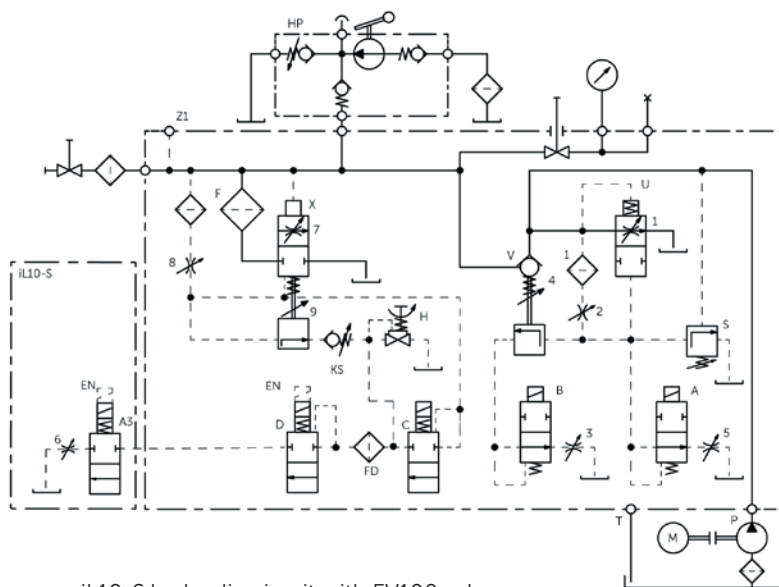
In order to check the functionality of the **iL10-S** valve, the coil **A3** can be de-energized during down travel. Alternatively, the **iL10-S** can also be tested by unscrewing the retaining nut **MM** and manually lifting the coil **A3**.

Caution! If the solenoid coil **M** is removed from the solenoid tube **DR** while under voltage for more than 10 seconds, this may cause deformation of the coil body or burning out of the solenoid coil **M**.

En primer lugar, se debe ajustar la válvula de control principal para garantizar que el ascensor proporcione un descenso suave con tiempos de desaceleración aceptables. Para probar la función UCM (movimiento involuntario de la cabina), se desactiva la bobina A3 de la válvula iL10-S durante el descenso del ascensor, o se retira manualmente la bobina A3 de la válvula solenoide. A continuación, el ascensor debe detenerse en un recorrido de 800 mm.

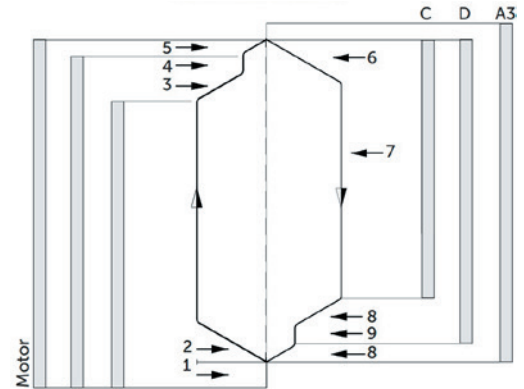
Status of the elevator	Power to coil A3	Power to coil D	Power to coil C
Up travel and relevelling	power off		
Down travel	power on		power on
Down relevelling			power off
Stop with closed doors	power off		
Emergency lowering	power on		power off
Emergency manual lowering	power off		
Hand pump operation	power off		

Hydraulic circuit

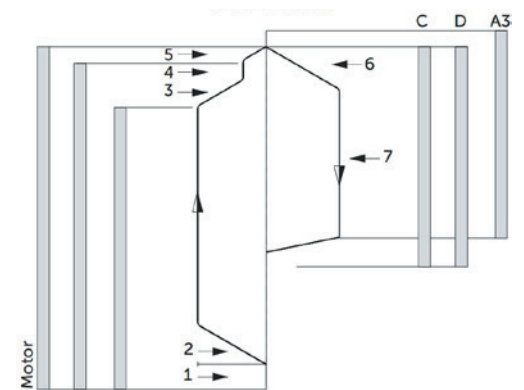


iL10-S hydraulic circuit with EV100 valve

Electrical sequence



normal up and down travel

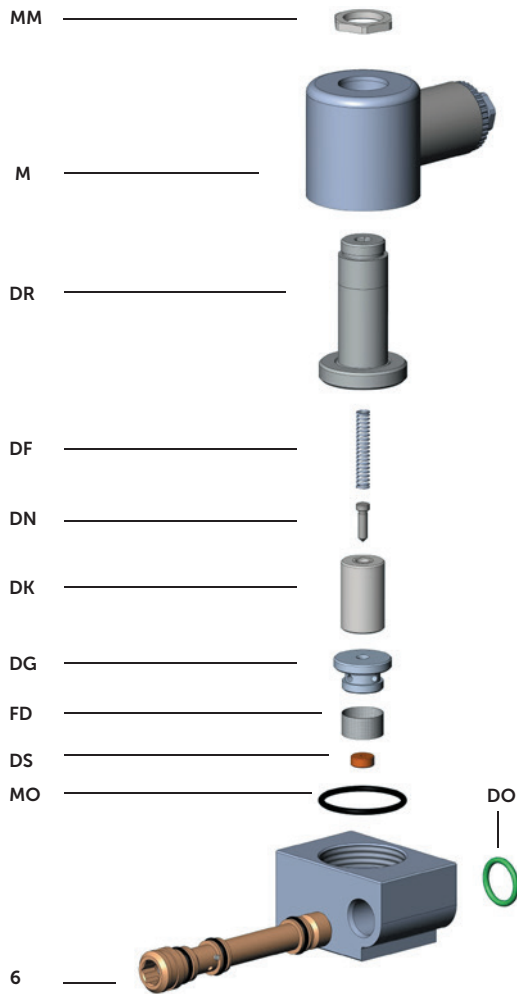


normal up travel, UCM case in down direction

Instructions manual - UCM Valve iL10-S

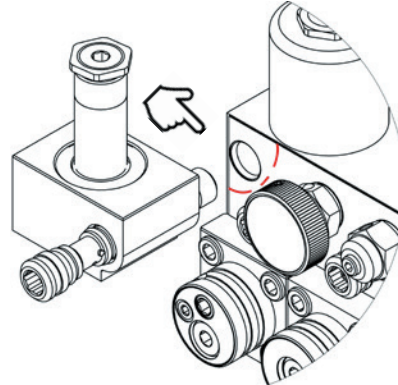
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iL10-S Explosion view

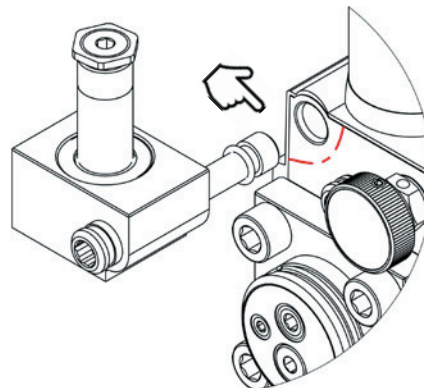


Caution!

After assembling the **iL10-S** on the EV block, if any external leakage is detected, remove the iL10-S block and cut away the aluminium front shield to ensure that no part of the shield remains between the iL10-S and EV valves. Then, clean the surface of the EV block and reassemble the iL10-S valve.



iL10-S with 3/4" EV100 valve



iL10-S with 1.5" EV100 valve

While ordering iL10-S

When placing an order, please specify the valve size (e.g., 2" EV100 or a pump flow rate of 250 l/min) and the working pressure range (e.g., 28bar to 55 bar). Depending on the pressure range and pump flow, different size of adjustment 6 and inserts are used. The seat discs (DS) vary depending on the operating pressure.

MM	Solenoid nut
M	Coil
DR	Solenoid tube
DF	Solenoid spring
DN	Needle
DK	Iron core
DG	Seat housing
FD	Filter solenoid
DS	Solenoid seat
MO	O-ring (26x2)
DO	O-ring (13x2)
6	Adjustment 6

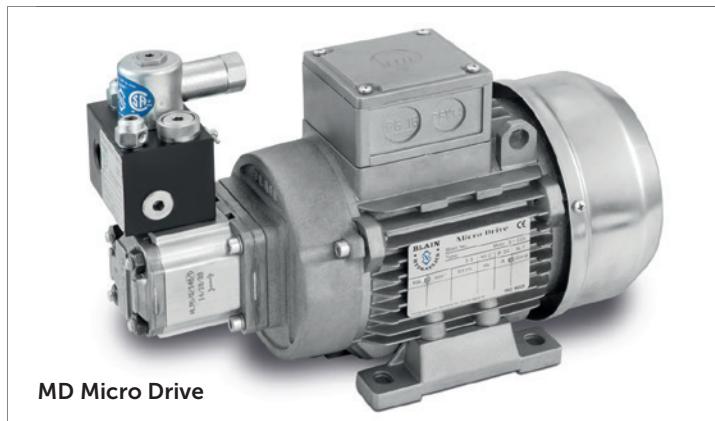
Valve size	Maximal flow	SKU Adjustment 6
EV 3/4"	125 l/min	501177
EV 1.5"-2"	800l/min	501178
EV 2.5"	1500 l/min	501179

size	DS hole diameter	
	p_max ≤ 50 bar	p_max > 50 bar
EV 3/4" / EV 1.5"-2"	2.0 mm	1.6 mm
EV 2.5"	p_max ≤ 34 bar	p_max > 34 bar
	2.9 mm	2.5 mm

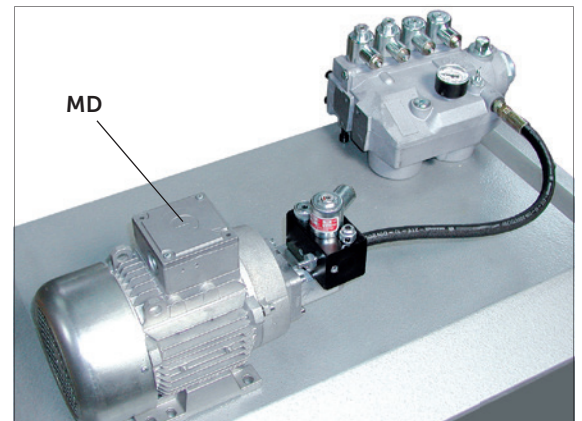
Select size of solenoid seat (DS) according to the maximum system pressure (cabin with payload).

For Hydraulic Elevators

The Blain Micro Drive for hydraulic elevators consists of a small motor, pump and valve unit in one assembly. Exact floor stops and releveling operations are achieved with low electrical power requirement, low noise levels and no unnecessary heating of the oil. The MD unit is mounted on or under the cover of the main hydraulic power unit, using the same oil source. It can also be used to slowly move the car independently of the main drive during installation or in an emergency.



MD Micro Drive



For Traction Elevators

The Micro Drive is equally effective as a separate unit for accurate releveling of traction elevators, where a small cylinder is integrated into the cable suspension. This eliminates the need to start the main drive when the car needs to be moved only a few centimetres.

Micro Levelling Speeds

Micro levelling at approximately 1 cm/s (2 fpm) is recommended at which speed positioning of the car is highly accurate and the releveling movement, usually lasting less than a second, is barely if at all noticeable by the passenger. Floor stopping accuracy of better than ± 5 mm (0.2 inch) can be expected, provided the floor switch system is correspondingly accurate. Outside of 2 cm (0.8 inch) from the floor landing, the main elevator releveling system must be activated.

Increased safety

Exact landing of the car eliminates the danger of passengers tripping while entering or leaving the car.

Faster Floor to Floor Times

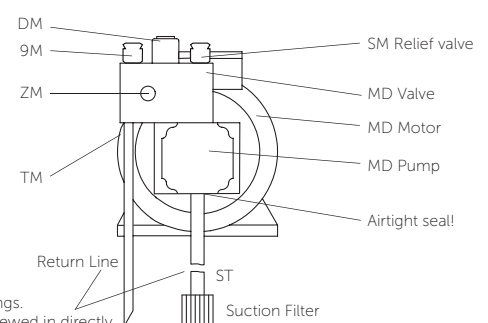
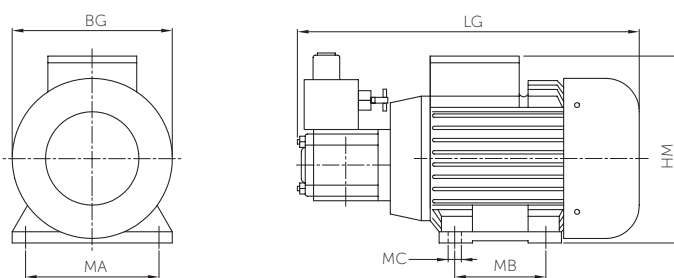
Because the Micro Drive assures final stopping accuracy, the main levelling approach speed to the floor can be increased to approximately 10 cm/s (20 fpm) to shorten travelling time.

Quicker Releveling

When load is added or removed from the car causing it to shift its position, the MD Unit releveling correction begins instantaneously and usually taking less than one second to complete. There is no unwanted delay due to bypassing of oil as it is the case with the main hydraulic drive.

Quieter Releveling

Since the main power unit does not need to be started, releveling with the Micro Drive unit is quiet and shorter in duration than when employing the main hydraulic drive. This is especially important in private homes or offices, where the releveling sound of the main elevator drive can be disturbing; especially at night-time.



TM & ST: Do not use hydr. fittings.
Use only low pressure pipe screwed in directly.

Measurements

Type	l/min		US gpm	KW	max bar	max psi	Ports			LG mm	BG mm	HM mm	MA mm	MB mm	MC mm	weight	
	50 Hz	60 Hz					ST Suction	TM Tank	ZM Cylinder							kg	lbs
MD 0103	1.0	0.3	0.37	130	1870	1/4"	3/8"	3/8"	3/8"	220	120	175	100	80	7	6	13.5
MD 0205	1.9	0.6	0.55	130	1870	3/8"	3/8"	3/8"	3/8"	290	140	190	110	90	7	8.5	19
MD 0411	4.2	1.3	1.10	130	1870	3/8"	3/8"	3/8"	3/8"	325	160	200	120	100	10	10	22
MD 0611	6.4	2.0	1.10	130	1870	3/8"	3/8"	3/8"	3/8"	330	160	200	120	100	10	10.5	23
MD 1022	10.6	3.3	2.20	130	1870	1/2"	3/8"	3/8"	3/8"	355	175	215	140	125	10	16	35.5
MD 1535	15.4	4.8	3.50	130	1870	1/2"	3/8"	3/8"	3/8"	410	195	245	160	140	12	25	55
MD 2450	24.0	7.5	5.00	130	1870	3/4"	3/8"	3/8"	3/8"	450	220	265	190	140	14	32.5	72

US units

mm ÷ 25.4 = inches

Blain Hydraulics GmbH
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Germany

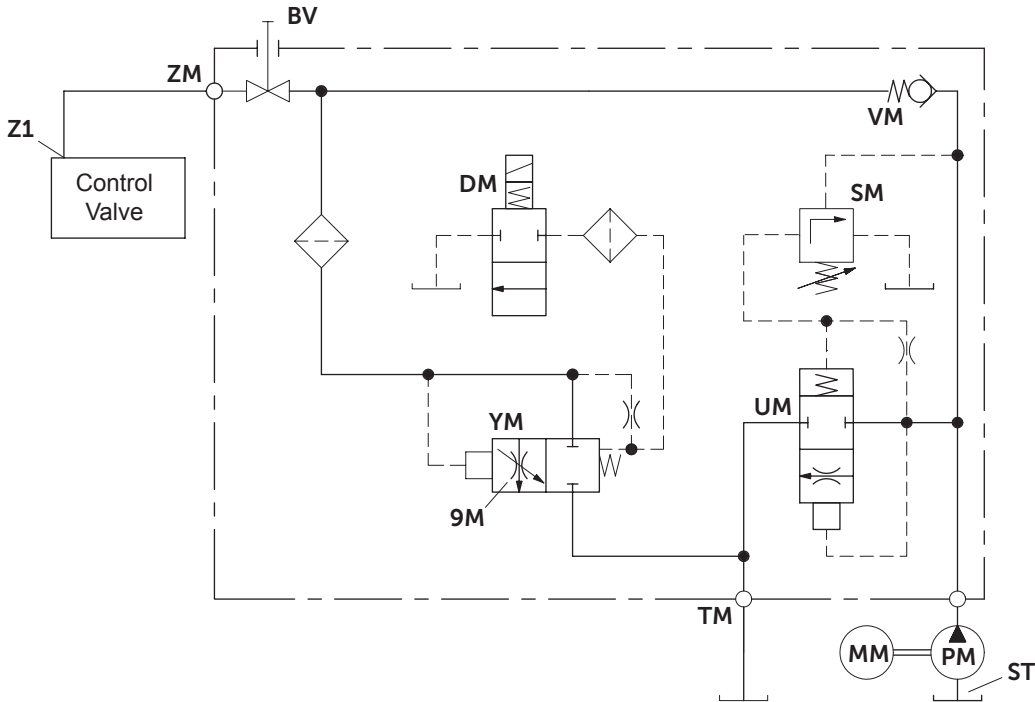
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info@blain.de



Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators



Hydraulic Circuit



DM Solenoid Down
9M Down Speed Adjustment
SM Pressure Relief Valve Adjustment
VM Check Valve

BV Shut Off Valve
ST Suction Tube
ZM Cylinder Connection
TM Return Connection
MM MD-Motor

PM MD-Pump
UM Pilot Operated Relief Valve
YM Pilot Operated Down Valve

Selection Chart

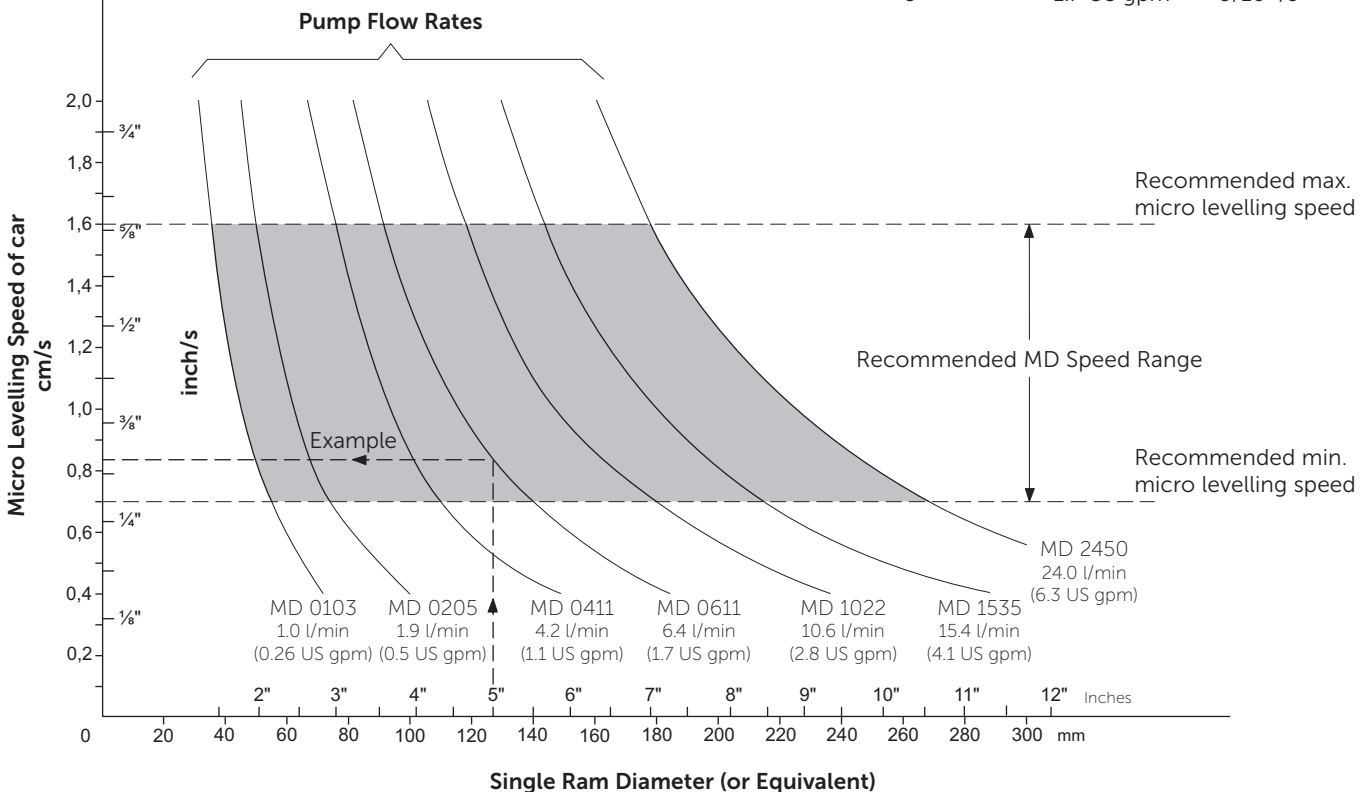
50 Hz Motors (For 60 Hz connection, add 20% to pump flow rates and MD levelling speeds)

Example:

Ram
127 mm
5"

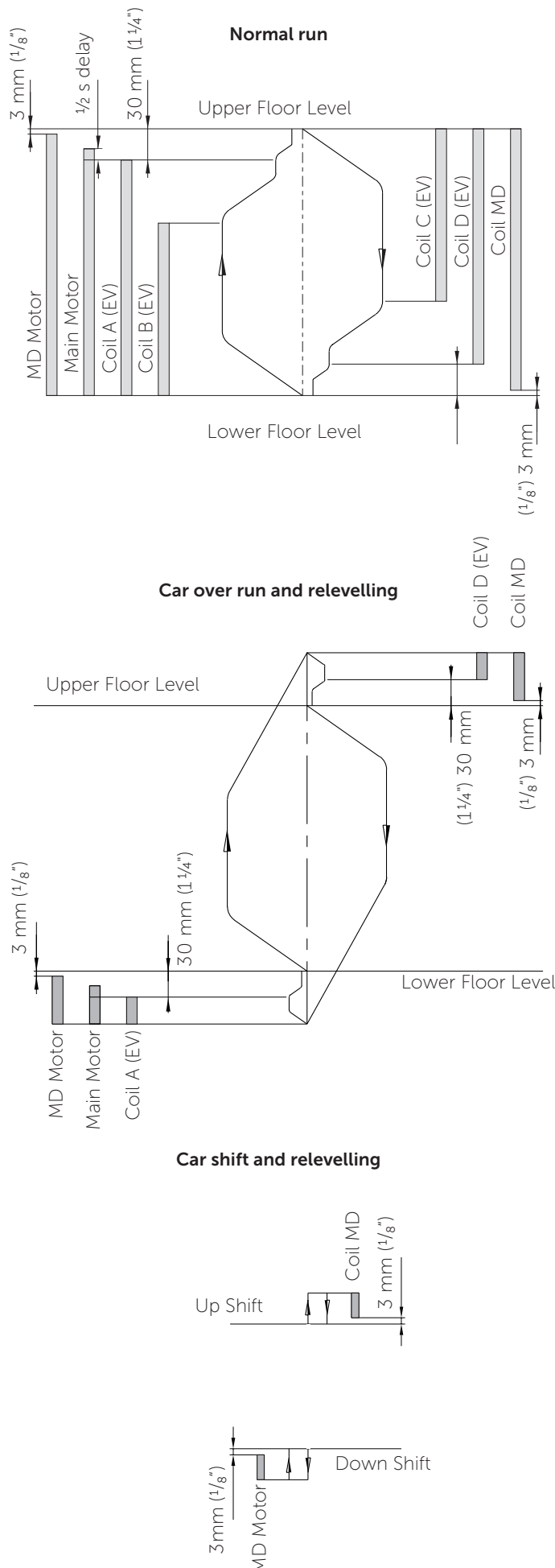
Pump
6.4 l/min
1.7 US gpm

Speed
~0.84 cm/s
~5/16 "/s



The Equivalent Ram Diameter is the calculated single ram diameter which would produce the same car speed as the alternative cylinder system.

Electrical Sequence



Normal Run

1. At the beginning of a normal up run, the friction of the system (stiction) is overcome by the Micro Drive. This reduces any breakaway jerk.
2. The standard control valve levelling speed no longer determines the stopping accuracy of the elevator. Therefore it can be increased to approximately 10 cm/s (20 fpm), effectively smoothing out the ride and at the same time, shortening the floor to floor travelling time.
3. The stopping accuracy of the car, determined by the Micro Drive, is improved to less than 5 mm (0,2 inch) from floor level.

Over Run

If the car travels past the floor level, the shaftway switches would activate to re-level the car through the MD unit, in less time than it takes the car doors to open and more accurately than with the standard levelling system. Should the MD correction not be completed within 2-3 seconds, the main levelling system automatically actuates.

Car Shift

When the position of the car shifts away from floor level due to loading or unloading, cooling of the oil or valve leakage, the Micro Drive performs a releveilling correction when the shift exceeds 3 mm.



Important

An accurate levelling switch installation is a precondition for satisfactory results.

All distances mentioned are for reference only and can vary from installation to installation.



Installation and Parts List

Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off and residual pressure in the valve is reduced to zero.

MD Installation

1. Use the same oil reservoir for the **MD** unit as for the main pump drive.
2. Do not use hydraulic fittings for suction and return lines. Their smaller bores prevent efficient oil flow. For the suction line, use low pressure, taper threaded tube, screwed directly into the suction port **ST** of the pump with airtight sealing and with the lower end including filter, submerged at **all times** under the reservoir oil level. To avoid sucking air, ensure that the thread connection into the pump is perfectly sealed.
3. For the return line to tank, also use low pressure, taper threaded tube, screwed directly into the tank port **TM** of the valve.
4. For the cylinder line connection, high pressure hose may be used to join port **ZM** of the valve to the main cylinder line.

Electrical Installation

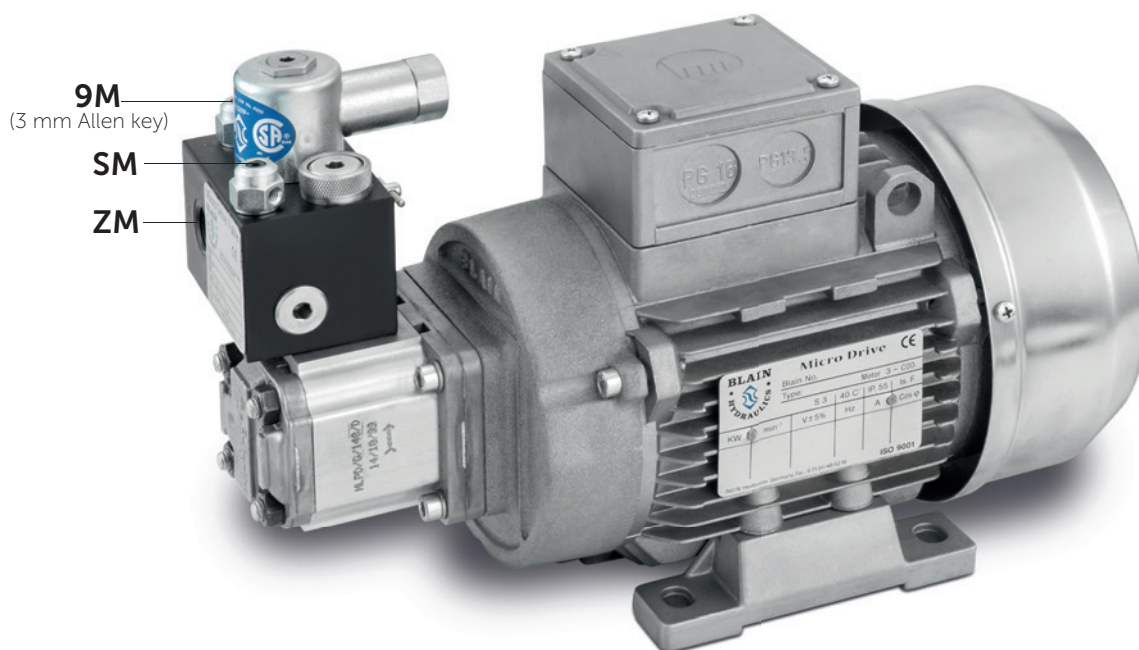
5. The existing safe operation of the elevator according to the applicable local code must not be changed. Separate **MD** relevelling switches set to trigger **MD** relevelling at between 2.5 to 5 mm (0.1 - 0.2 inches) above and below flow level which are actuated before the main relevelling switches.
6. Electrical connections are 3 phase direct starting.

MD Adjustments

Up Speed is determined by the pump output and the effective diameter of the cylinder system.

SM Relief Valve. 'In' (clockwise) produces a higher, 'out' (c-clockwise) a lower maximum pressure setting.

9M Down Speed (Micro Drives MD1022 and 2450). When coil **DM** is energised, the car will travel downwards at a speed according to the setting of adjustment **9M**, (3mm Allen key). Turning 'in' (clockwise) provides a slower, turning 'out' (c-clockwise), a faster down speed.

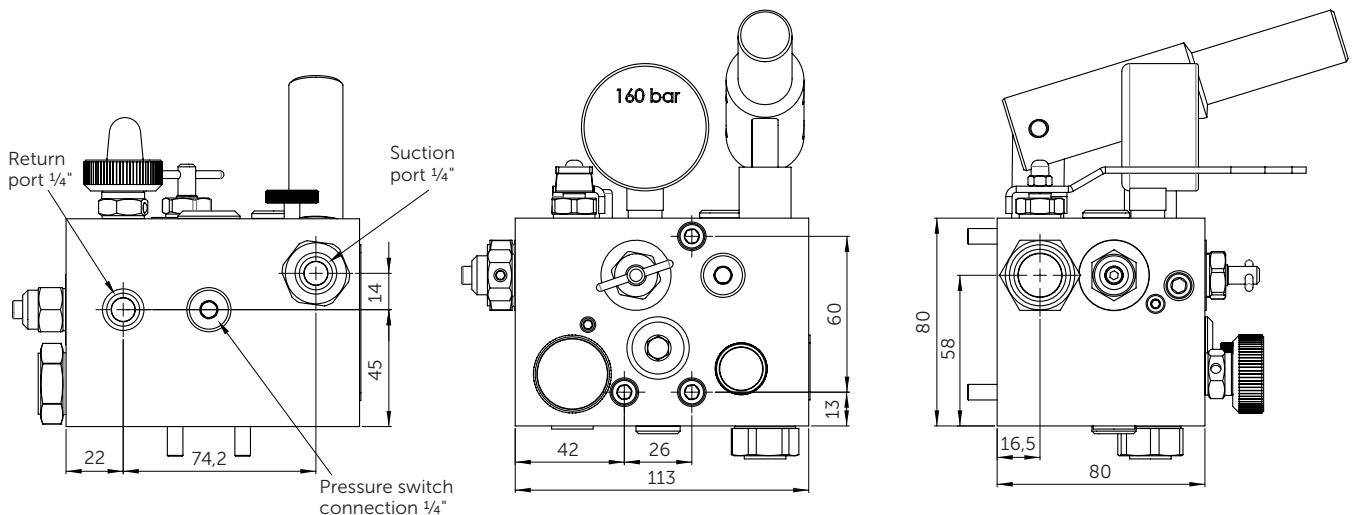


Micro Drive Unit
(1.0 to 24 l/min) (0.3 to 7.5 US gpm)

The MRL-H has been designed for servicing and rescuing operations of machine roomless (MRL) hydraulic elevators remotely by having easy outside access, without needing to be in the pit. Many functional valves such as an adjustable self-closing manual lowering valve, hand pump, slack rope valve, pressure relief valve, ball valve as well as a manometer have been added to a compact body. MRL-H can be located up to 6 metre (19 feet) away and 5 metre (16 feet) high from the main power unit to a convenient location for easy access. MRL-H can be optionally delivered with pipes and necessary accessories upon request.



Pressure switch port (2 times): 1/4" BSP
Output port: 1/2" BSP
Suction port: 1/4" BSP
Return port: 1/4" BSP
Pressure range: 0-100 bar (0-1450 psi)
Oil Viscosity: (25-60) cSt. bei 40°C (15-35 cSt. at 120°F)
max. suction height: 5 m (with PVC-pipe inside-Ø6 mm);
 check valve recommended at suction point for pipe length longer than 1 m
Options:
 check valve
 PVC-pipe set (2x6m)



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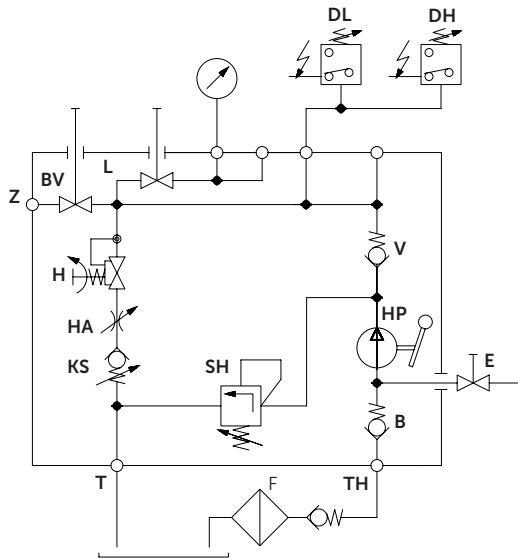


GmbH

Designer and Manufacturer of the highest
 quality control valves & safety components
 for hydraulic elevators



Hydraulic Circuit

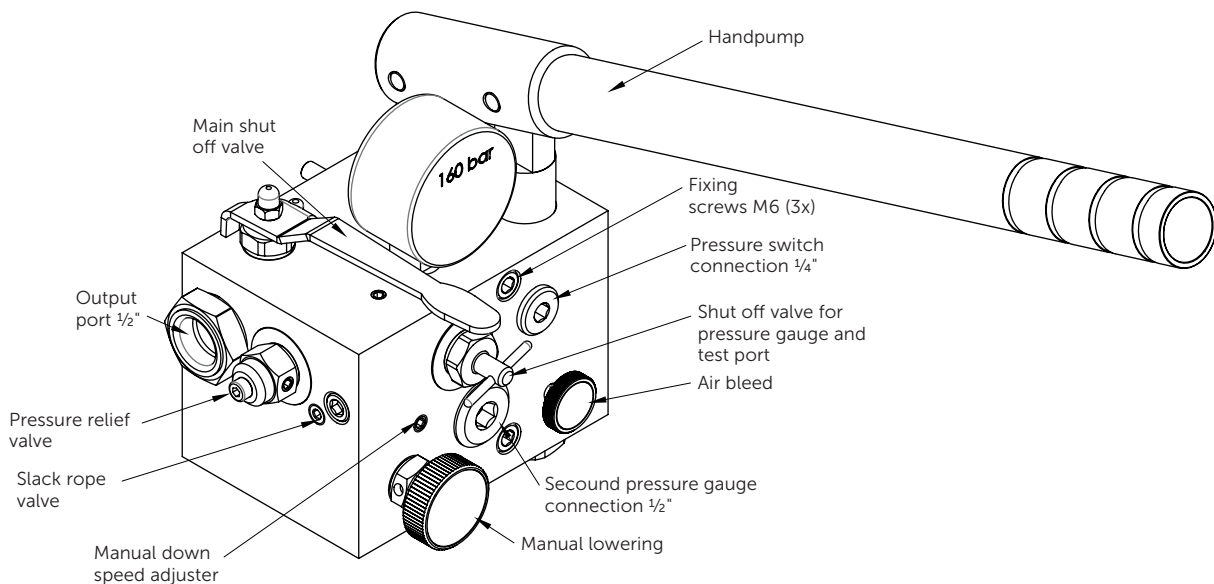


- Z** Pressure port
- T** Tank port
- TH** Suction port
- F** Filter
- B** Check-Valve (suction)
- V** Check-Valve (Pressure line)
- BV** Main shut-off valve (built in)
- KS** Slack rope valve
- H** Manual lowering
- L** Shut-off valve for pressure gauge and test port
- HA** Manual down speed adjustment
- HP** Hand pump
- E** Air bleed
- SH** Relief valve
- DL** Pressure switch 1-10 bar
- DH** Pressure switch 1-100 bar

Connections: The output port (1/2") connected to the cylinder line, the oil suction (1/4") and return (1/4") ports to the main tank. Use polyamide oil suction and return pipes with a inside diameter of 6 mm and a outside diameter of 8 mm minimum.

To lower the lift: Open the main shut off valve. Turn the manual lowering anticlockwise. If the speed is too slow adjust the speed adjuster accordingly.

To raise the lift: Open the main shut off valve on the MRL-H unit. To release trapped air inside the unit, open the air-bleed 1/2 or 3/4 turn and pump oil until oil appears at the bleed screw. Tighten the bleed screw and continue pumping to raise the lift.



Main shut off valve: Allows pressure line to be kept closed during non-service times.

Slack rope valve: Avoids the ram from being lowered by opening the manual lowering valve below ram pressure/weight.

Manual lowering valve: Descends the car to lower position manually.

Manual lowering speed adjustment: It allows manual lowering speed adjustment away from the main power unit.

Pressure relief valve: By default it is set to 100 bar (adjustable).

Hand pump: Allows servicing people to raise the car to the upper floor.

Type AA - Female Threads
Typ AA - Innengewinde
Type AA - Filet femelle
Tipo AA - Rosca hembra

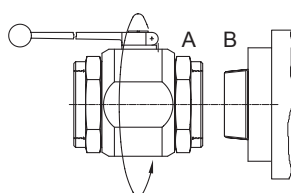
Techn. Data

Typ	Q max.	P max.
BV3 (1½")	800 l/min 211 US gpm	100 bar 1450 psi
BV5 (2½")	1600 l/min 423 US gpm	70 bar 1015 psi

Type ED - Swivel Nut
Typ ED - Überwurfmutter
Type ED - Ecrou raccord
Tipo ED - Tuerca loca



2.5 kg

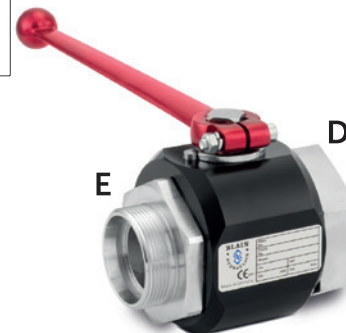


Standard A - A

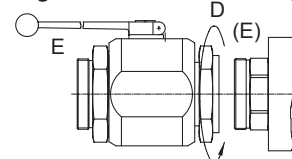
Pipe Thread
Seal Tape Required
Rohrgewinde
Dichtband notwendig
Filetage au pas du gaz Bande d'étanchéité nécessaire
Rosca para tubos
Se necesita cinta de junta

Option E - D

Faster Connections
Perfect Sealing
Schnelle Montage
Perfekte Dichtung
Montage rapide
Joint parfait
Montaje rápido
Estanqueidad perfecta



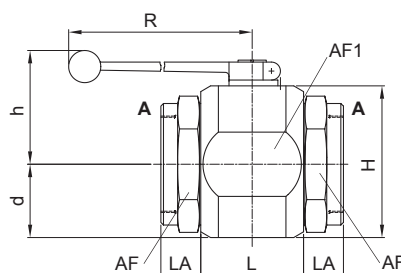
2.5 kg



A

Typ	A	AF*	(BSP)		(NPT)	
			Typ No.	LA	Typ No.	LA
B3	1"	70	B3G1	19	B3N1	28
	1¼"	70	B3G1.25	21	B3N1.25	28
	1½"	70	B3G1.5	24	B3N1.5	34
	2"	70	B3G2	30	B3N2	34
B5	2"	95	B5G2	31	B5N2	31
	2½"	95	B5G2.5	31	B5N2.5	35

Standard



Dimensions

Typ	DN	L	H	AF1	d	h	R
B3	38	65	90	86	43	70	240
B5	55	80	118	114	57	82	280

DN = Ø Inside · Ø Innen
Ø Intérieur · Ø Interior

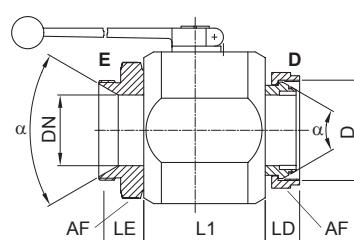
E

DIN 2353 (24°)

DIN 3863 (60°)

Typ	E	α	LE	*AF	Typ No.
B3	M36x2	24°	24.5	70	B3E36
	M45x2	24°	26.5	70	B3E45
	M52x2	24°	26.5	70	B3E52
	M65x2	60°	27	70	B3E65
B5	M78x2	60°	35	95	B5E78

Option



D

DIN 2353 (24°)

DIN 3863 (60°)

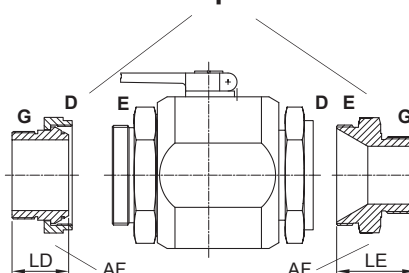
Typ	D	α	L1	LD	*AF	Typ No.
B3	M52x2	24°	66	35	60	D52
	M65x2	60°	66	25	75	D65
B5	M78x2	60°	94	24	90	D78

Adaptor GD

Typ	D	B	*AF	LD	Typ No.
B3	M65x2	G1½"	70	47	GD65.G1.5
	M65x2	Ø57 Weld	70	45	WD65.57
B5	M78x2	G2"	90	48	GD78.G2
	M78x2	Ø70 Weld	90	44	WD78.70

*AF - Across Flats
AF - Schlüsselweite
AF - Ouverture de clé
AF - Ancho de llave

Adaptors



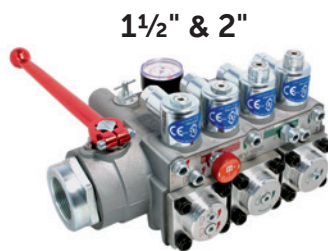
Adaptor GE

Size	E	B	*AF	LE	Typ No.
B3	M52x2	G1"	70	55	GE52.G1
	M52x2	G1 ¼"	70	54	GE52.G1.25
	M52x2	G1 ½"	70	52	GE52.G1.5
	M52x2	G2"	70	60	GE52.G2
	M65x2	G1 ½"	70	60	GE65.G1.5
B5	M65x2	G2"	70	52	GE65.G2
	M78x2	G2"	80	59	GE78.G2
	M78x2	G2 ½"	80	59	GE78.G2.5
	M78x2	NPT2 ½"	80	63	GE78.N2.5

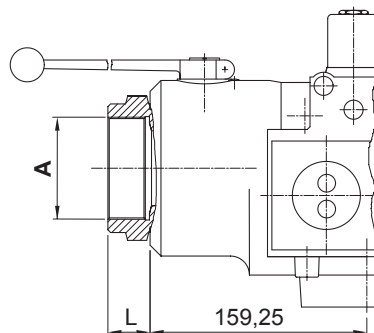
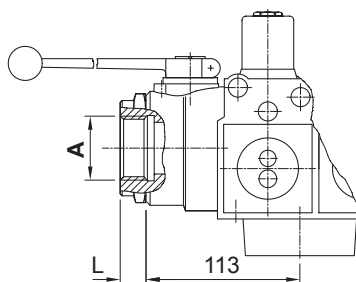
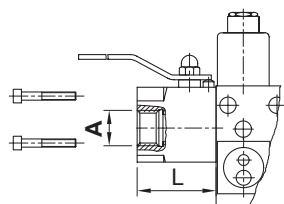


Ball Valves Robinet à boisseau

Kugelhähne Llaves esféricas

BV

Standard Connection A



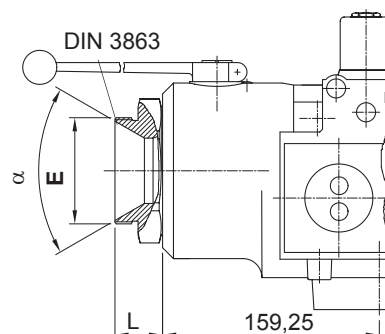
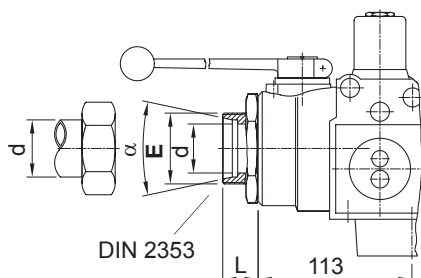
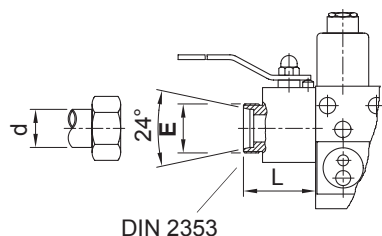
	A	L	Typ No.
* G (BSP) 1/2"	83	BG 0.5	
G (BSP) 3/4"	58	BG .75	
G (BSP) 1"	58	BG 1	
NPT 3/4"	58	BN .75	
NPT 1"	58	BN 1	

* Reduction

	A	L	Typ No.
G (BSP) 1"	19	BG 1.0	
G (BSP) 1 1/4"	21	BG 1.25	
G (BSP) 1 1/2"	24	BG 1.5	
G (BSP) 2"	30	BG 2	
NPT 1"	28	BN 1	
NPT 1 1/4"	28	BN 1.25	
NPT 1 1/2"	34	BN 1.5	
NPT 2"	34	BN 2	

	A	L	Typ No.
G (BSP) 2"	31	BG 2.0	
G (BSP) 2 1/2"	31	BG 2.5	
NPT 2"	31	BN 2.0	
NPT 2 1/2"	35	BN 2.5	

Optional Connection E



	E	L	d	Typ No.
* M 22x1.5	80	15	BM 22	
M 30x2	54	22	BM 30	
M 36x2	54	28	BM 36	

* Reduction

	E	L	α	d	Typ No.
M 36x2	23	24°	28	BM 36.0	
M 45x2	25	24°	35	BM 45	
M 52x2	25	24°	42	BM 52	

DIN 3863				
	E	L	α	Typ No.
M 65x2	26	60°	BM 65	
M78x2	37	60°	BM 78	

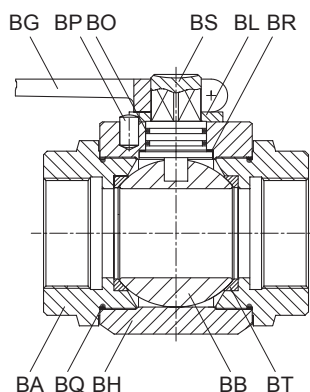
	E	L	α	Typ No.
M78x2	35	60°	BM 78.0	

Parts List

BG Grip
BP Limit Pin
BO Shaft O-Ring
BS Shaft
BL Limit Disc
BR Bearing ring
BA Connection
BQ Connection O-Ring
BH Housing
BB Ball
BT POM Seal

Stückliste

BG Griff
BP Kerbstift
BO O-Ring Schaltwelle
BS Schaltwelle
BL Arretierscheibe
BR Gleitring
BA Adapter
BQ O-Ring Adapter
BH Gehäuse
BB Kugel
BT POM Dichtung



Liste des pièces

BG Levier
BP Goupille
BO O-Ring axe du levier
BS Axe du levier
BL Disque d'arrêt
BR Anneau de glissement
BA Connexion
BQ Connexion O-Ring
BH Cage
BB Boule
BT POM joint

Lista de piezas

BG Brazo
BP Pasador tope
BO Junta eje de giro
BS Eje de giro
BL Retén
BR Anillo deslizante
BA Conexión salida
BQ Junta conexión salida
BH Cuerpo
BB Bola
BT Junta de POM

TH

Tank Heater Chauffage pour cuve à huile

Tankheizung Calefacción para depósitos



EN ISO 9001



Type TH
250 Watt
1.9 kg



100 - 130 AC Option
200 - 240 AC Standard

20° - 25° C Standard (68° - 77° F)
15° - 20° C Option (59° - 68° F)
25° - 30° C Option (77° - 86° F)
35° - 40° C Option (95° - 104° F)

Application

(GB)

TH tank heaters are intended primarily for application in hydraulic control systems for machine tools, presses, hydraulic elevators, servo systems, etc. where overnight conditions or periods of non-operation cause the temperature of the hydraulic fluid to fall below desirable levels.

The heater is designed to maintain up to approximately 500 litres (130 US gals) of oil in an unheated room at a temperature of +20° C to +25° C (68° F to 77° F).

Construction

Through the large heat dissipation area of the housing, the heaters surface temperature remains under +50° C (120° F) and thereby avoids oxidation or premature aging of the oil. The built-in thermostat switches the heating element on at an oil temperature of approximately +20° C (68° F) and off again when the oil temperature has risen to approximately +25° C (77° F).

Should the heater in an unsubmerged state be subject to an ambient temperature of under 20° C (68° F), it will switch on for a short period before switching off again as heat is conducted through the housing to the thermostat. Under this condition, the surface temperature of the heater will not exceed 90° C (190° F).

Application

(F)

Ce chauffage est spécialement destiné aux installations à commande oléodynamique telles que machines-outils, presses, ascenseurs oléodynamiques, etc. pour lesquelles la température de l'huile ne doit pas descendre au-dessous d'un minima donné.

Ce chauffage est conçu de manière à maintenir des cuves à l'huile d'une capacité d'environ 500 ltr à une température moyenne de +20° C à +25° C dans des locaux non chauffés.

Construction

Pour éviter une détérioration prématurée de l'huile, le carter du chauffage présente une grande surface de radiation qui maintient la température de celui-ci au dessous de +50° C. Le thermostat incorporé assure un fonctionnement autonome. Température de mise en service +20° C, température de coupure +25° C.

Dans le cas d'une mise en service occidentale de l'appareil à air libre, la température de ce dernier étant inférieure à 20° C, il ne résulterait aucun dommage pour le chauffage, car après la coupure d'alimentation au moyen du thermostat, la température du surface du carter ne surpassera pas 90° C.

Anwendung

(D)

TH-Tankheizungen sind für hydraulische Steueranlagen, für Werkzeugmaschinen, Pressen, hydraulische Aufzüge etc. vorgesehen, bei welchen über Nacht oder nach längerem Stillstand der Maschine die Öltemperatur des Steuersystems unter den erwünschten Wert zurückfällt.

Die Tankheizung ist so ausgelegt, dass Behälter bis ca. 500 Liter Öl-Inhalt in unbeheizten Räumen auf eine Mindesttemperatur von +20° C bis +25° C gehalten werden.

Konstruktion

Das Gehäuse der Öltankheizung besitzt eine große Wärmeabstrahlungsfläche, so dass die Außentemperatur des Gehäuses nicht über +50° C steigt und ein Verbrennen bzw. vorzeitiges altern des Öls vermieden wird. Das eingebaute Thermostat schaltet die Heizung bei ca. +20° C ein und bei ca. +25° C Öltemperatur selbsttätig ab und erfordert keine sonstige Überwachung.

Falls sich die Heizung bei Lufttemperatur von unter 20° C in nicht eingetauchtem Zustand einschaltet, kommt es zu keinerlei Schaden, da nach automat. Wiederabschaltung des Thermostats die Gehäusefläche eine Temperatur von höchstens 90° C erreicht.

Aplicación

(E)

Las calefacciones TH, han sido previstas para plantas hidráulicas de mando, máquinas-herramienta, prensas, ascensores hidráulicos, etc. y aplicables en aquellos casos, en los que la temperatura del aceite del sistema de mando desciende por debajo del valor deseado durante la noche, o cuando la máquina lleva parada durante mucho tiempo.

La calefacción para depósitos está diseñada de tal manera, que puede mantener a temperatura mínima de +20° C hasta +25° C, los recipientes de capacidad máx. de 500 litros de aceite en locales que no disponen de calefacción.

Construcción

La carcasa de la calefacción para depósitos de aceite tiene una gran superficie de irradiación térmica, hasta tal punto, que la temperatura exterior de la carcasa no sube más de +50° C, con lo que así se evita que el aceite se combusione o se envejezca prematuramente. El termostato incorporado conecta la calefacción a unos +20° C y la desconecta automáticamente, sin requerir ningún otro control o vigilancia, cuando el aceite alcanza unos +25° C de temperatura.

Si la calefacción, en estado no sumergido, se pone en marcha a temperatura del aire inferior a 20° C, no se ocasionará daño alguno, porque después de la desconexión automática del termostato, la temperatura superficial de la carcasa no se elevará por encima de 90° C.

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GmbH

Designer and Manufacturer of the highest
quality control valves & safety components
for hydraulic elevators



Tank Heater Chauffage pour cuve à huile

(GB)

Installation

The heater is supplied with 2.5 metres (98 inches) of electrical cable sheathed over a length of 1.2 metres (48 inches) by an oil resistant hose. The cable outside the hose should not be submerged in the oil.

Two powerful magnets are fitted underneath the heater so that the installation in a normal sheet tank is simply a matter of placing the heater at the bottom surface of the tank. Alternatively, the heater can be mounted through means of two available M6 holes. The magnets also extract unavoidable fine metal particles from the oil thereby protecting the pump from wearing out.

Since the heated oil convects upwards, oil below the heater remains cold. It is therefore important to mount the heater low down in the tank.

Due to the location of the thermostat, the heater should be mounted such that the cable inlet is directed upwards (see diagram below).

(F)

Installation

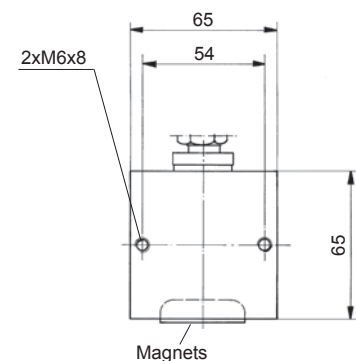
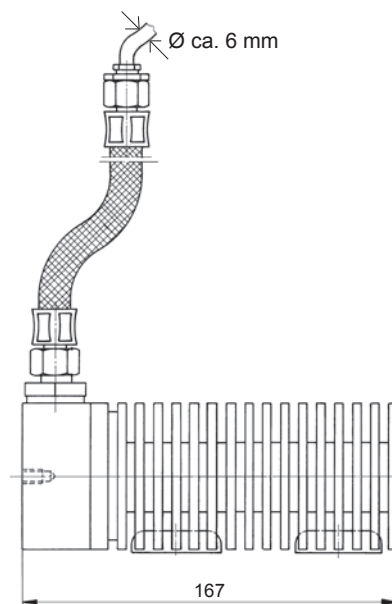
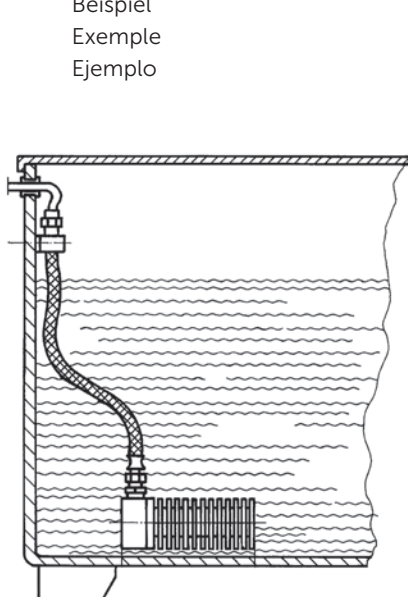
Le chauffage est équipé d'un câble électrique de 2,5 m dont la partie inférieure est protégée par un tuyau hydraulique sur une longueur de 1,2 m. La partie du câble non protégée ne doit pas être immergée.

L'installation dans la cuve s'effectue au moyen de deux aimants puissants situés sur la face inférieure du carter, de sorte qu'il suffit d'apposer le chauffage sur le fond de la cuve. Dans les cas où la cuve ne serait pas en acier ou bien s'il était nécessaire de monter différemment le chauffage, on dispose pour cela de 2 fixations filetées M6 situées sur le couvercle du carter. Les aimants attirent les fines particules métalliques se trouvant inévitablement dans l'huile, et de ce fait protègent la pompe contre l'usure.

Du fait de la convection vers le haut de la chaleur l'huile se trouvant en dessous du chauffage demeure froide. C'est pourquoi il est recommandé d'installer le chauffage aussi bas que possible dans la cuve.

Il est nécessaire, en raison de la position du thermostat, que le raccord de câble sortant du couvercle soit dirigé vers le haut.

Example
Beispiel
Exemple
Ejemplo



Tankheizung Calefacción para depósitos

TH

(D)

Installation

Die Öltankheizung wird mit 2,5 m langem Kabel ausgerüstet, wovon die unteren 1,2 m von einem hydraulischen Schlauch geschützt sind. Das Kabel außerhalb des hydraulischen Schlauches soll nicht in Öl getaucht werden.

Das Anbringen im Behälter erfolgt durch 2 unten am Gehäuse befindliche starke Magnete, so dass die Heizung lediglich an den Boden des Behälters angelegt werden muss. Falls der Behälter nicht aus Stahl ist, oder die Heizung sonst anders montiert werden soll, sind zwei M 6-Befestigungsgewinde am Deckel des Gehäuses vorhanden. Die Magnete ziehen die unvermeidbaren feinen metallischen Teile aus dem Öl und schützen somit die Pumpe vor Abnutzung.

Durch die Wärmekonvektion nach oben bleibt das Öl unterhalb der Heizung kalt. Deshalb soll die Heizung möglichst tief im Behälter montiert werden.

Der Kabelanschluss der Tankheizung muss nach oben gerichtet werden (siehe Skizze).

(E)

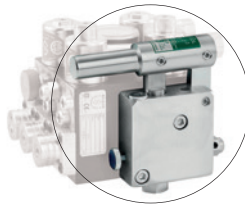
Instalación

La calefacción para depósitos de aceite va equipada con un cable de conexión de 2,5 m de largo, de los que 1,2 m van protegidos por el tubo flexible hidráulico. El cable no protegido por el tubo flexible hidráulico no debe sumergirse en el aceite.

Su colocación en el depósito se hace por medio dos imanes potentes que se encuentran debajo de la carcasa, de manera que sólo hay que colocar la calefacción en el suelo del recipiente. Si el recipiente no es de acero, o si la calefacción debe ser montada en otro lugar, en estos casos, emplear las dos roscas de fijación M 6 que se encuentran en la tapa de la carcasa. Los imanes retiran del aceite las finas e inevitables partículas metálicas, al tiempo que protegen a la bomba de un prematuro desgaste.

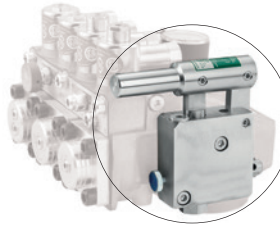
Debido a la confección térmica que tiende hacia arriba, el aceite permanece frío debajo de la calefacción. Para evitar tal efecto se recomienda colocar la calefacción lo más profunda posible.

Por la posición del termostato, la conexión de cable que sobresale de la tapa debe dirigirse hacia arriba.



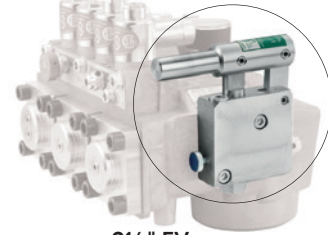
H 11

3/4" EV



H 11

1 1/2" EV



H 11

2 1/2" EV

Description

(GB)

H 11 and H 12 pumps are for application with hydraulic lifting or pressing equipment, for emergency operation of hydraulic elevators and for the pressure testing of hydraulic systems in general. The H 11 is constructed for side mounting. The H 12 is fitted with a base plate for separate application.

The built-in relief valve should be adjusted to prevent unintentional high pressure being applied to the system. A built-in manual valve for releasing pressure from the system is available as an option.

Description

(F)

Les pompes à main H 11 et H 12 sont prévues pour utilisation avec des installations hydrauliques de levage et de pression, pour la commande d'urgence des ascenseurs hydrauliques, ainsi que pour le contrôle de pression des systèmes hydrauliques en général. La pompe à main H11 est construite pour le montage latéral. La H 12, pourvue d'une plaque de montage est prévue pour l'application séparée.

Pour éviter une surpression inopinée dans le système, une soupape de surpression est incorporée. Une soupape manuelle permettant la dépressurisation du système est livrable sur demande.

Beschreibung

(D)

H 11 und H 12 Handpumpen sind geeignet für den Einsatz mit hydraulischen Hebe- und Pressanlagen, für die Notbetätigung von hydraulischen Aufzügen, sowie zur Druckprüfung von hydraulischen Systemen im allgemeinen. Die H 11 Handpumpe ist für die seitliche Montage konstruiert. Die H 12 Handpumpe ist mit einer Montageplatte ausgerüstet und für den separaten Einsatz vorgesehen.

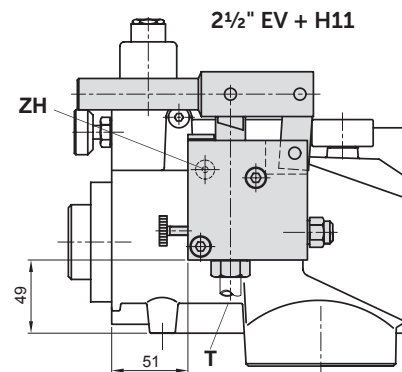
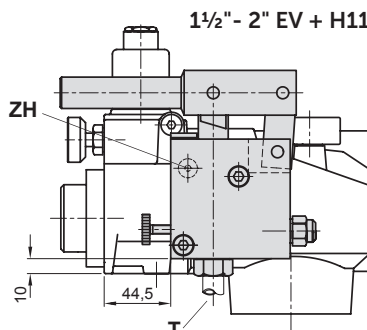
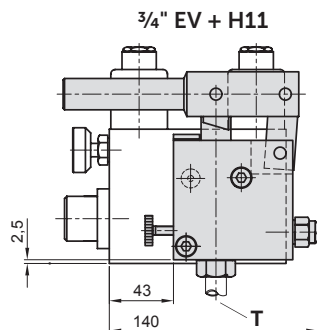
Um zu verhindern, dass unbeabsichtigt ein zu hoher Druck in das System gesetzt wird, ist ein Überdruckventil eingebaut. Auf Wunsch ist auch ein eingebautes Zylinderdruck-Entlastungsventil lieferbar.

Descripción

(E)

Las bombas a mano H 11 y H 12 son aptas para ser empleadas en instalaciones hidráulicas de elevación y prensado, para el accionamiento de emergencia de elevadores hidráulicos, así como para comprobar la presión de sistemas hidráulicos en general. La bomba a mano H 11 está concebida para ser montada lateralmente a la electroválvula EV, mientras que la H 12 se instala sobre una placa de montaje de forma independiente.

Para evitar, una presión demasiado elevada en el sistema, hay que regular convenientemente la válvula de sobrecarga incorporada. Con el fin de aligerar la presión del sistema, se puede suministrar una válvula de evacuado, sobre pedido.



See also EV prospect.

Siehe auch EV Prospekt.

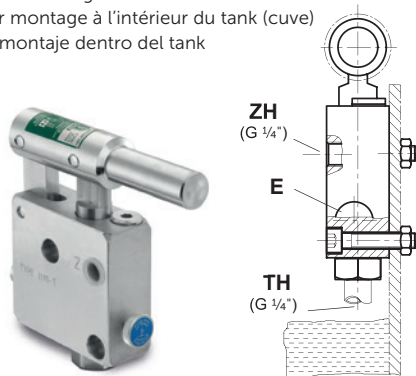
Autres dates voir prospectus EV.

Veáse también prospecto EV.

H 11T

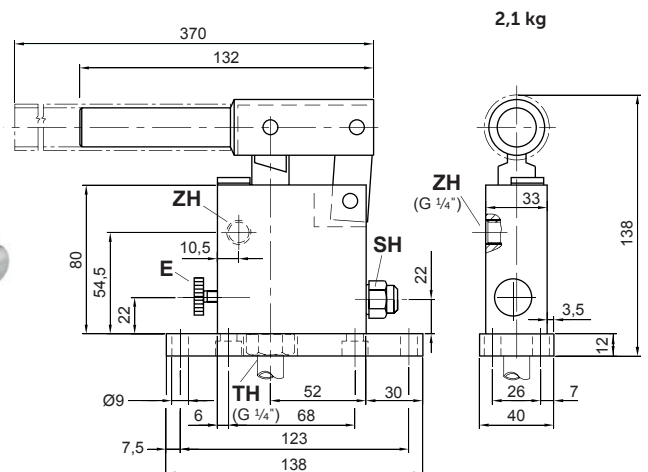
For mounting inside tank
Für die Montage am Tankinneren
Pour montage à l'intérieur du tank (cuve)
Por montaje dentro del tank

1,7 kg



H 12

For separate installation
Für die separate Montage
Pour installation séparée
Por instalación separado





Hand pump Pompe à main

(GB)



Warning: Only qualified personnel should adjust or service valves. Unauthorised manipulation may result in injury, loss of life or damage to equipment. Prior to servicing internal parts, ensure that the electrical power is switched off and residual pressure in the valve is reduced to zero.

Installation

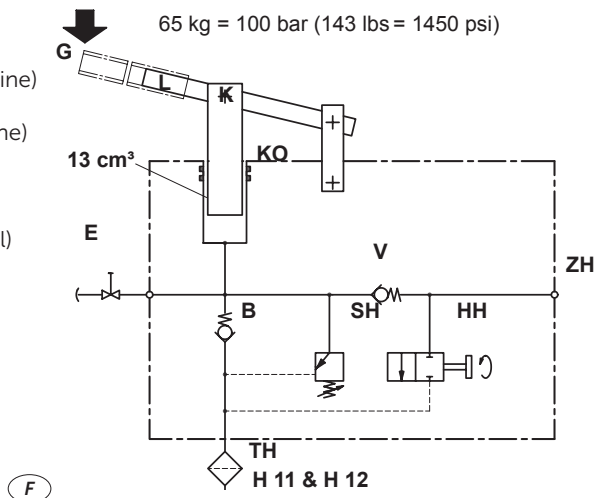
The inside diameter of the suction line should not be less than 8 mm diameter (5/16"). The connection of the suction line to the hand pump must be a perfect seal. A filter fitted to the bottom end of the suction line is recommended.

Air Bleed

If the operation of the pump arm does not produce a build up of system pressure, it may be necessary to release trapped air out of the hand pump by opening the air bleed screw **E** half a turn and pumping several strokes until oil appears at the bleed screw thread.

Elements

- V** Check-Valve (Pressure line)
- VO** Check-Valve (O-Ring)
- B** Check-Valve (Suction line)
- K** Piston
- KO** O-Ring - Piston
- SH** Relief-Valve
- HH** Pressure bleed (optional)
- E** Air bleed
- L** Lever
- G** Lever extension
- ZH** Pressure Port
- ZO** O-Ring - Port
- TH** Suction Port



Avertissement: seul les techniciens qualifiés en ascenseur sont en droit d'effectuer de nouveaux réglages et des travaux d'entretien. Une utilisation non autorisée risque d'entraîner des blessures, des accidents mortels ou des dommages matériels. Avant tous travaux de maintenance sur des pièces internes, il faut s'assurer que le conduit de cylindre est bien fermé, que le courant de l'ascenseur est coupé et que la pression dans la valve au dessus de la soupape de vidange a bien été réduite à zéro.



Installation

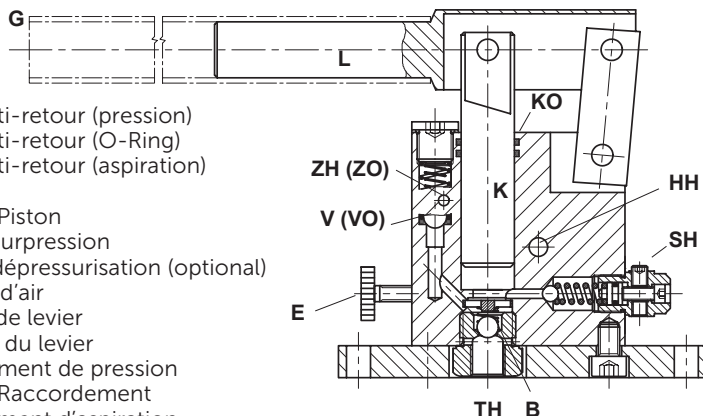
Le diamètre intérieur de la conduite d'aspiration doit être de 8 mm au minimum. Le raccordement de la conduite d'aspiration à la pompe à main doit être d'une étanchéité parfaite. Il est recommandé de munir d'un filtre l'extrémité inférieure de conduite d'aspiration.

Purge de l'air

Au cas où l'utilisation du rallonge de levier ne cause pas la montée en pression du système il sera nécessaire de purger; pour cela ouvrir d'un demi tour la vis de purge **E** et pomper jusqu'à l'apparition d'huile à la vis de purge.

Éléments

- V** Clapet anti-retour (pression)
- VO** Clapet anti-retour (O-Ring)
- B** Clapet anti-retour (aspiration)
- K** Piston
- KO** O-Ring - Piston
- SH** Valve de surpression
- HH** Valve de dépressurisation (optional)
- E** Vis purge d'air
- L** Rallonge de levier
- G** Extension du levier
- ZH** Raccordement de pression
- ZO** O-Ring - Raccordement
- TH** Raccordement d'aspiration



Handpumpe Bomba a mano

(D)



Warnung: Neueinstellungen und Wartung dürfen nur durch qualifiziertes Aufzugspersonal durchgeführt werden. Nicht autorisierte Bedienung kann Verletzungen, tödliche Unfälle oder materielle Schäden zur Folge haben. Vor der Wartung innerer Teile ist sicher zustellen, dass die Zylinderleitung geschlossen, der elektrische Strom des Aufzuges abgeschaltet und der Druck im Ventil über das Notablassventil auf Null reduziert worden ist.

Installation

Der Durchmesser der Ansaugleitung sollte mindestens 8 mm haben. Der Anschluss der Saugleitung an der Handpumpe muss einwandfrei dicht sein. Ein Sieb, angebracht am unteren Ende der Saugleitung, ist empfohlen.

Entlüftung

Falls die Betätigung des Pumpenhebels zu keinem Aufbau des Systemdruckes führt, muss die Handpumpe entlüftet werden. Dazu die Entlüftungsschraube **E** ½ Umdrehung öffnen und den Hebel mehrmals betätigen, bis Öl aus dem Gewinde der Entlüftungsschraube kommt. Entlüftungsschraube wieder schließen.

Elemente

- V** Rückschlagventil (Druckleitung)
- VO** Rückschlagventil (O-Ring)
- B** Rückschlagventil (Saugleitung)
- K** Druckkolben
- KO** O-Ring - Druckkolben
- SH** Überdruckventil
- HH** Druckentlastung (auf Wunsch)
- E** Entlüftungsschraube
- L** Hebel
- G** Hebelverlängerung
- ZH** Druckanschluss
- ZO** O-Ring - Anschluss
- TH** Sauganschluss

Aviso: El ascensor sólo debe ser reajustado y mantenido por personal calificado. Un manejo no autorizado puede producir lesiones, accidentes mortales y daños materiales. Antes de efectuar el mantenimiento, asegurar siempre que la línea del cilindro esté cerrada, que la alimentación de corriente del ascensor esté desconectada, y que la presión de la válvula haya sido reducida a cero a través de la bajada manual.



Instalación

La tubería de aspiración deberá tener, como mínimo 8 mm de diámetro interior. La conexión, entre tubería de aspiración y bomba, debe ser de una hermeticidad perfecta. Es recomendable colocar un filtro en el extremo inferior de la tubería de aspiración.

Purga de aire

Si al accionar la palanca de la bomba no se consigue aumentar la presión del sistema, será necesario evacuar el posible aire que pueda encontrarse en la bomba. Para ello, se dará media vuelta al tornillo de descarga **E** y se accionará varias veces la palanca, hasta que se aprecie aceite en la rosca del tornillo de descarga.

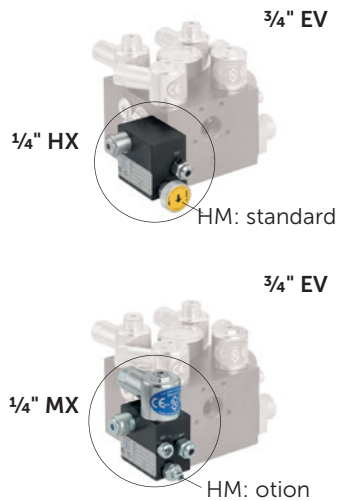
Elementos

- V** Válvula de antiretorno (presión)
- VO** Válvula de antiretorno (O-Ring)
- B** Válvula de antiretorno (aspiración)
- K** Pistón
- KO** Anillo O - Pistón
- SH** Válvula de seguridad (aspiración)
- HH** Descarga de presión (opcional)
- E** Purga de aire
- L** Palanca
- G** Alargadera de palanca
- ZH** Conexión de presión
- ZO** Anillo O - Conexión
- TH** Conexión de aspiración

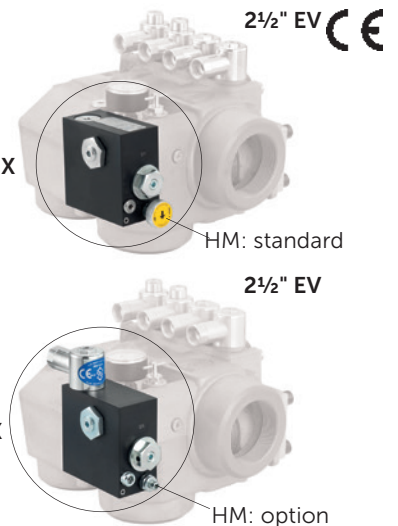
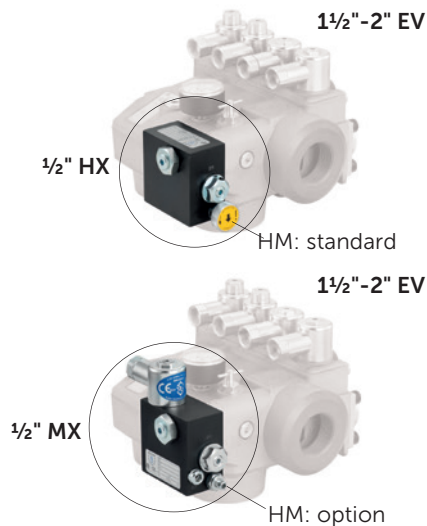
**HX
MX**

Down Valves Soupape de descente

Senkventile Válvulas de bajada



(GB)



(D)

HX are manually operated down valves, adjustable in their down speed. They close automatically upon release. They can be used for emergency manual lowering or in combination with the EV down valve to achieve an overspeed of the elevator for testing the pipe rupture valve.

MX are solenoid operated down valves, adjustable in their acceleration, down speed and deceleration. They can be used for the revision or inspection travel of the elevator or as a particularly slow down speed valve in addition to the two down speeds of the EV valve to obtain extremely exact floor stops.

HX sind handbetätigte Senkventile, die in der Senkgeschwindigkeit einstellbar sind. Beim Loslassen des Griffes schließt sich das Ventil automatisch. Es kann als Notablassventil oder kombiniert mit dem Senkventil des EV Blockes benutzt werden, um eine Übergeschwindigkeit des Aufzuges zu erreichen, damit das Rohrbruchventil geprüft werden kann.

MX sind magnetbetätigte Senkventile, die in der Beschleunigung, Senkgeschwindigkeit und Abbremsung einstellbar sind. Sie können entweder für die Revisions- oder Inspektionsfahrt eines Aufzuges benutzt werden oder mit dem EV Ventil als dritte, besonders langsame Senkgeschwindigkeit, um ein sehr präzises Halten des Aufzuges zu ermöglichen.

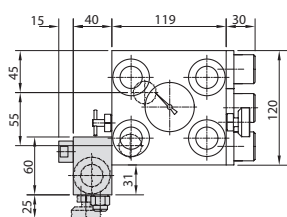
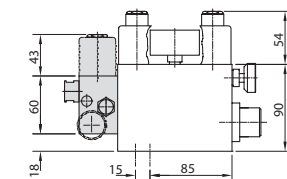
HX sont des soupapes de descente à commande manuelle dont la vitesse de descente est réglable. Les soupapes ferment automatiquement dès qu'on lâche le bouton de commande. Elles peuvent être utilisées comme descente de secours ou être combinées avec la soupape de descente du bloc EV pour obtenir une survitesse de descente afin de tester la valve de parachute pour ascenseur.

MX sont des soupapes de descente à commande électromagnétique dont l'accélération, la vitesse et le freinage en descente sont réglables. Elles peuvent être utilisées pour la révision ou l'inspection de l'ascenseur, ou être réglées comme une 3-ième vitesse de descente particulièrement lente pour permettre un arrêt très précis.

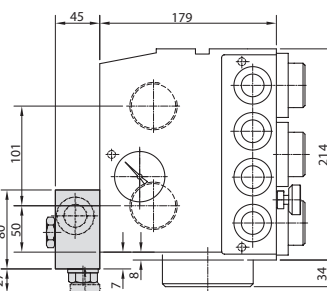
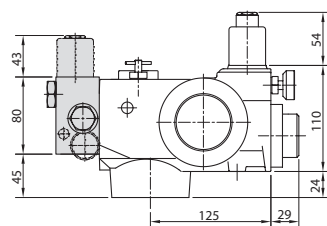
Las HX son válvulas de bajada accionadas a mano cuya velocidad se puede regular. Al soltar el manipulador la válvula se cierra automáticamente. Puede ser utilizada como válvula de descarga de emergencia junto con la válvula de bajada del EV, al objeto de conseguir una hipervelocidad, para que la válvula para caídas para ascensor pueda ser controlada.

Las MX son válvulas de bajada accionadas por imán, regulables en la aceleración, velocidad de bajada y deceleración. Pueden emplearse para la marcha de la revisión o inspección del ascensor o como tercera, muy lenta velocidad de bajada, posibilitan una exactísima parada del ascensor.

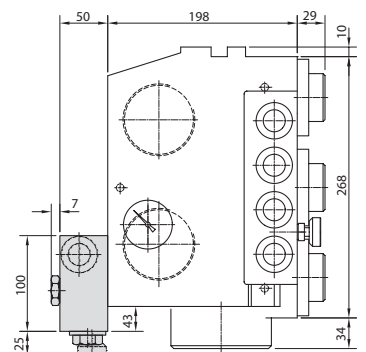
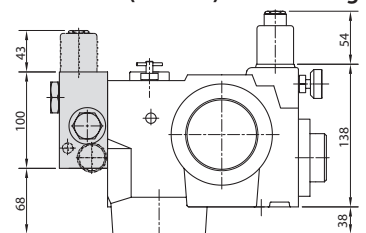
1/4" HX (3/4" EV) 0.50 kg
1/4" MX



1/2" HX (1 1/2"-2" EV) 0.90 kg
1/2" MX



3/4" HX (2 1/2" EV) 1.50 kg
3/4" MX



See also EV prospect.

Siehe auch EV Prospekt.

Autres dates voir prospectus EV.

Véase también prospecto EV.

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Designer and Manufacturer of the highest quality control valves & safety components for hydraulic elevators

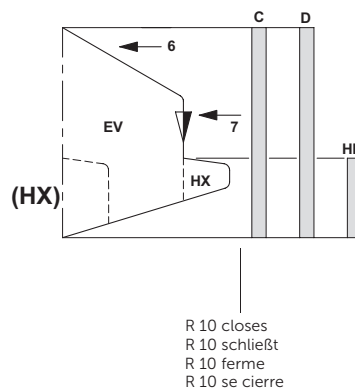
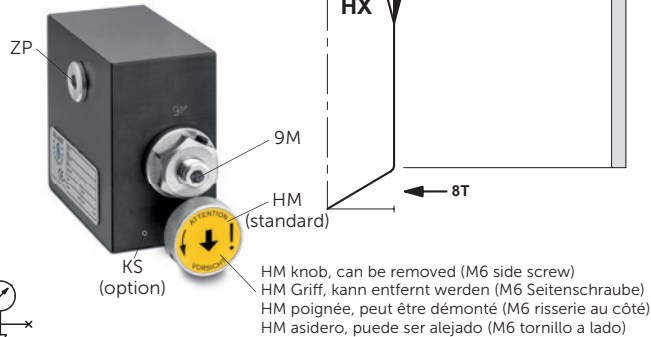
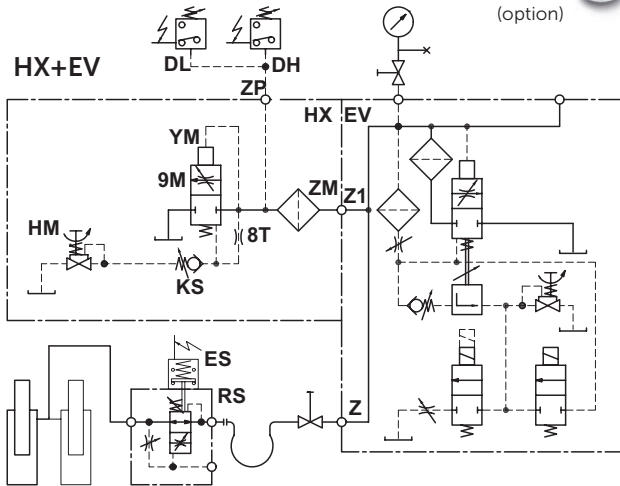
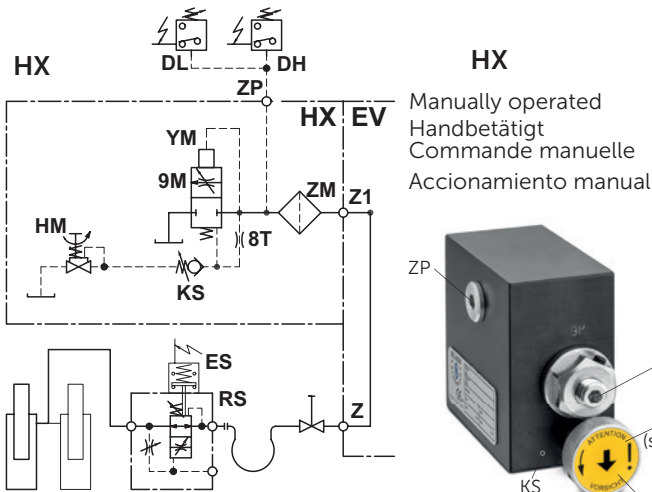
GmbH



Hydraulic Circuit
Hydraulisches Steuerschema
Schéma commande hydraulique
Esquemas del mando hidráulico

Electrical Sequence
Elektrisches Schaltdiagramm
Diagramme de commande électrique
Diagramas de conexiones eléctricas

HX
MX



(GB)

Adjustments

- 6M** Down Acceleration
- 8M** Down Deceleration
- 8T** Fixed orifice
- 9M** Down Speed
- DM** Solenoid
- HM** Manual Lowering
- YM** Down Valve
- ZP** Connection - Pressure Switch

Optional Equipment:

- KS** Slack Rope Valve
- RS** Pipe Rupture Valve
- ES** Pipe Rupture Valve End Switch
- DH** High Pressure Switch
- DL** Low Pressure Switch

For EV control elements refer to EV literature.

(D)

Einstellungen

- 6M** Anfahrddrossel 'ab'
- 8M** Abbremsdrossel 'ab'
- 8T** Düse
- 9M** Senkfahrteinstellung
- DM** Magnetventil
- HM** Handablass
- YM** Senkventil
- ZP** Anschluss - Druckschalter

Optionale Erweiterungen:

- KS** Kolbensicherung
- RS** Rohrbruchventil
- ES** Rohrbruchventil Endschalter
- DH** Druckschalter Hochdruck
- DL** Druckschalter Niederdruck

Für EV Steuerelemente siehe EV Prospekt.

(F)

Réglages

- 6M** Vis d'adjustment 'descente'
- 8M** Étrangleur de ralentissement 'descente'
- 8T** Tuyère
- 9M** Réglage de grande vitesse
- DM** Solénoïde
- HM** Descente manuelle
- YM** Soupape 'descente'
- ZP** Connection interrupteur à pression

Équipement sur demande:

- KS** Sécurité contre mou de câble
- RS** Soupape parachute
- ES** Interrupteur de soupape parachute
- DH** Contact de surpression
- DL** Contact de sous-pression

Pour les éléments de réglage de l'EV voir documentation EV.

(E)

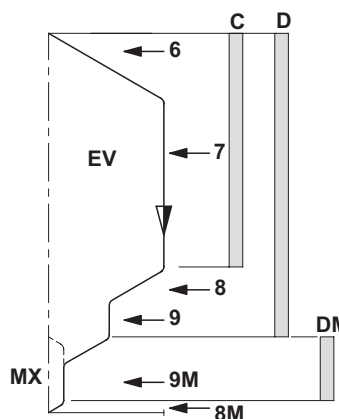
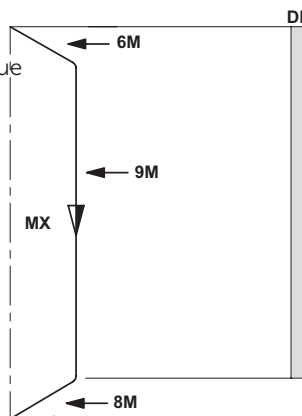
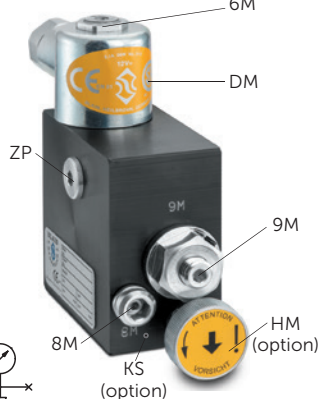
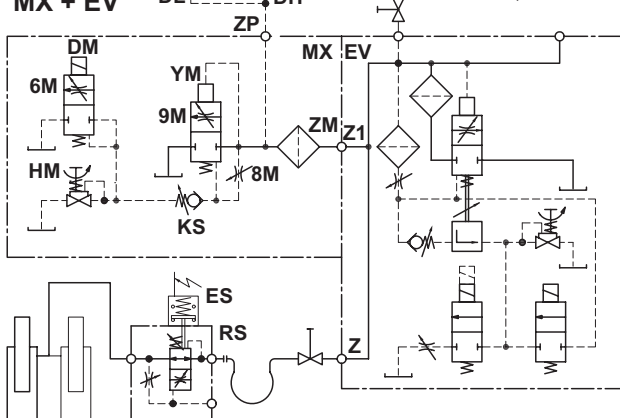
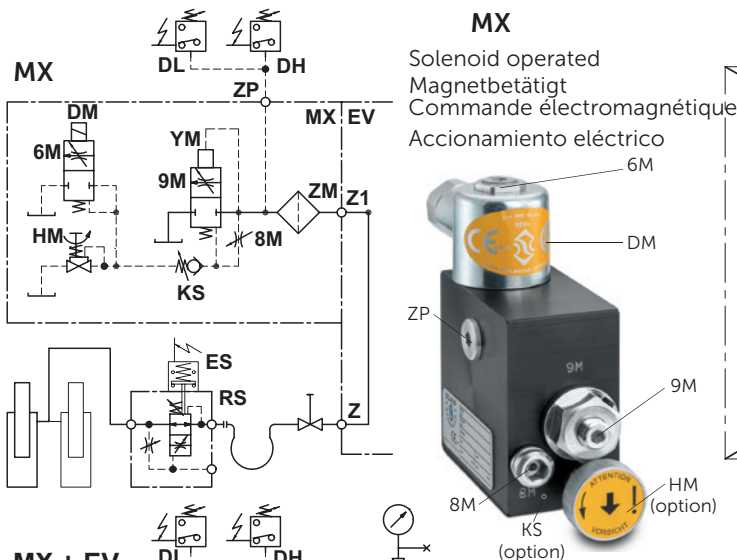
Ajustes

- 6M** Aceleración 'bajada'
- 8M** Frenado en 'bajada'
- 8T** Tobera
- 9M** Velocidad 'bajada'
- DM** Válvula magnética
- HM** Bajada manual
- YM** Válvula bajada
- ZP** Conexión - Tornillo de presión

Implementos opcionales:

- KS** Válvula aflojamiento cables
- RS** Válvula paracaída
- ES** Interruptor final válv. paracaída
- DH** Interruptor de alta presión
- DL** Interruptor de baja presión

Para elementos de mando de la EV veáse el prospecto de la EV.





(GB)



Warning: Only qualified personnel should adjust or service valves.

HX Manual: HM Manual Lowering: Turning 'out' (c-clockwise) opens the valve and allows the car to be lowered.

9M Down Speed: The down speed of the car is according to the setting of adjustment **9M**. Turning 'in' (clockwise) provides a slower, turning 'out' (c-clockwise) a faster down speed.

Stop: Upon release, **HM** closes automatically. The car will stop according to the built in damping.

MX Solenoid: 6M Down Acceleration: When coil **DM** is energized, the car will accelerate downwards according to the setting of adjustment **6M**. Turning 'in' (clockwise) provides a softer, turning 'out' (c-clockwise) a quicker down acceleration. Preadjustment: Turn adj. **6M** all the way 'in' (clockwise) and then energize coil **DM**. Turn **6M** slowly back 'out' (c-clockwise) until the car accelerates downwards.

9M Down Speed: See **9M** above (**HX**).

8M Stop: With coil **DM** de-energized, the car will decelerate according to the setting of adjustment **8M**. Turning 'in' (clockwise) provides a softer, turning 'out' (c-clockwise) a quicker deceleration.

KS Slack Rope Valve: All down solenoids must be de-energized! The **KS** is adjusted with a 3 mm Allen key. With **K** turned all the way 'in', then half a turn back out, the unloaded car should descend when Manual Lowering **H** is opened. Should the car not descend, **K** must be turned out until the car just begins to descend, then turned out a further half turn to ensure that with cold oil, the car can still be lowered as required.

(F)



Avertissement: seul les techniciens qualifiés en ascenseur sont en droit d'effectuer de nouveaux réglages et des travaux d'entretien.

HX Manuelle: HM Descente manuelle: Ouvrir la valve et l'ascenseur descend à petite vitesse.

9M Vitesse en descente: La vitesse de l'ascenseur en descente est fonction de l'étrangleur **9M**. Pour ralentir la vitesse de descente 'visser' l'étrangleur pour l'accélérer, le 'dévisser'.

Stop: Dès qu'on lâche **HM**, l'ascenseur s'arrête en fonction de l'amortissement incorporé.

MX Électromagnétique: 6M Démarrage descente: A la mise sous tension de **DM**, l'ascenseur accélère en descente en fonction du réglage de l'étrangleur **6M**. Pour obtenir un démarrage descente en douceur, 'visser', pour en obtenir un plus brusque, 'dévisser'. Préréglage: Fermer entièrement le réglage **6M** et mettre **DM** sous tension. Dévisser lentement **6M** jusqu'à ce que l'ascenseur démarre en descente.

9M Vitesse descente: Voir **9M** ci-dessus (**HX**).

8M Stop: Solénoïde **DM** étant sous courant, l'ascenseur ralentit en fonction du réglage de l'étrangleur **8M**. Pour obtenir un ralentissement en douceur, 'visser' l'étrangleur, pour en obtenir un brusque, le 'dévisser'.

KS Sécurité contre le mou de câble: Tous les solénoïdes étant sans courant! Elle est réglée au moyen d'un tournevis à empreinte. Veuillez faire baisser l'asc. sans charge et avec la vis **K** entièrement serrée, puis desserrée un demi tour, en même temps gardant la descente de secours **H** est ouverte. Si l'ascenseur s'arrête, la vis à réglage **K** doit être desserrée, jusqu'au point où l'asc. bouge à peine, puis veuillez desserrer la vis à réglage **K** encore un demi tour, ainsi que l'asc. continuellement peut baisser aussi avec d'huile froide.

(D)



Warnung: Neueinstellungen und Wartung dürfen nur durch qualifiziertes Aufzugspersonal durchgeführt werden.

HX Handbetätigung: HM Handablass: Das Drehen nach links öffnet das Ventil und der Aufzug fährt abwärts.

9M Senkgeschwindigkeit: Sie ergibt sich entsprechend der Drossel **9M**. 'Hinein' (Uhrzeigersinn) bewirkt eine langsamere, 'heraus' eine schnellere Senkgeschwindigkeit.

Halt: Beim Loslassen schließt **HM** automatisch und der Aufzug hält entsprechend der eingebauten Dämpfung an.

MX Magnetbetätigung: 6M Anfahrt abwärts: Mit Spule **DM** unter Strom beschleunigt der Aufzug entsprechend der Drossel **6M** abwärts. 'Hinein' (Uhrzeigersinn) bewirkt eine weichere, 'heraus' eine härtere Anfahrt abwärts. Voreinstellung: Einstellung **6M** ganz zudrehen und dann Magnetventil **DM** unter Strom setzen. **6M** langsam herausdrehen bis der Aufzug abwärts beschleunigt.

9M Senkgeschwindigkeit: Siehe **9M** oben (**HX**).

8M Halt: Mit Spule **DM** stromlos, wird der Aufzug entsprechend der Drosseleinstellung **8M** abgebremst. 'Hinein' (Uhrzeigersinn) bewirkt eine weichere, 'heraus' eine härtere Abbremsung.

KS Kolbensicherung: Alle Senk-Magnetventile müssen stromlos sein! Mit **K** ganz hineingedreht, dann eine halbe Umdrehung zurück, soll der unbeladene Aufzug abwärts fahren, während Notablass **H** geöffnet ist. Bleibt der Aufzug noch stehen, so muss die Einstellschraube **K** herausgedreht werden bis der Aufzug gerade noch fährt, dann eine halbe Umdrehung herausdrehen, damit sich der Aufzug auch noch bei kaltem Öl absenken lässt.

(E)



Aviso: El ascensor sólo debe ser reajustado y mantenido por personal calificado.

HX Manual: HM Bajada manual: Al girar a la izquierda de forma manual se abre la válvula y el ascensor baja.

9M Velocidad en bajada: La velocidad en bajada del ascensor resulta según la regulación del ajuste **9M**. Girándolo a la derecha se consigue una velocidad en bajada más lenta, y en sentido contrario una más rápida.

Parada: Al soltar de **HM**, el ascensor se para de conformidad con la amortiguación incorporada.

MX Electromagnética: 6M Aceleración bajada: Si la bobina **DM** está bajo corriente, el descenso del ascensor se acelera conforme al reglaje del tornillo de regulación **6M**. Girándolo a la derecha se obtiene una aceleración bajada suave y a la izquierda una brusca. Reglaje preventivo: Girar el tornillo **6M** hasta el tope y después poner la válvula electromagnética **DM** bajo corriente. A continuación girar lentamente el tornillo **6M** en dirección contraria hasta que el ascensor acelere en descenso.

9M Velocidad de bajada: Ver **9M** arriba (**HX**).

8M Parada: Con la bobina **DM** sin corriente, el ascensor se frenará según la regulación del ajuste **8M**. Girándolo a la derecha, se consigue un frenado más suave, y en sentido contrario uno más brusco.

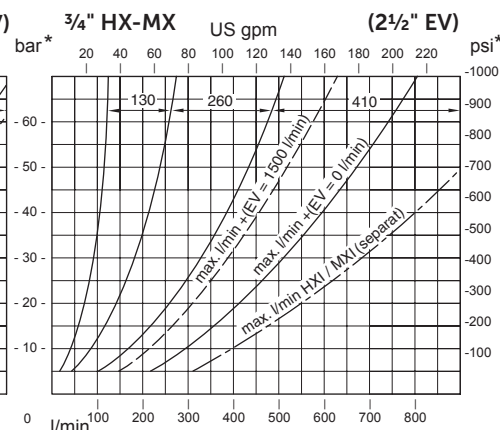
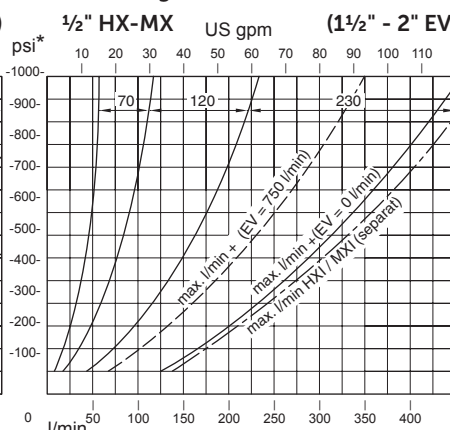
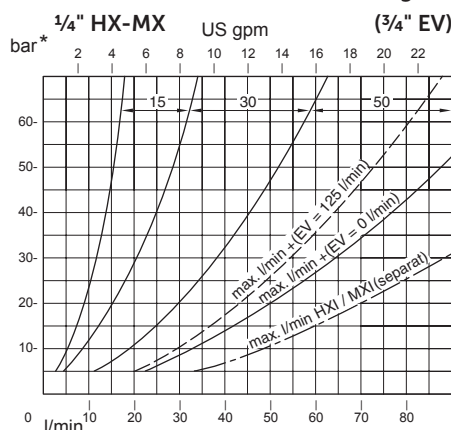
KS Válvula aflojamiento cables: Todas las bobinas sin corriente! Es ajustada con un 3 mm llave Allen. Con **K** girado del todo y entonces una media vuelta atrás, el ascensor vacía debe marchar abajo mientras la bajada manual **H** está abierta. Si el ascensor todavía queda quieta, el tornillo reg. **K** debe ser desgirado hasta el momento en que el ascensor arranca y entonces **K** debe ser desgirado otra vez una media vuelta hasta el momento en que el ascensor arranca y entonces **K** debe ser desgirado otra vez una media vuelta para asegurar que con el aceite frío todavía se pueda bajar el ascensor.

Insert Selection

Diagramme für Einsatzgrößen

Sélection de cartouches

Selección del modelo



* Pressure with empty car * Druckangaben bei unbeladenem Fahrkorb * Pression avec cabine vide * Presión con cabina vacía



Parts List Liste des pièces de rechange

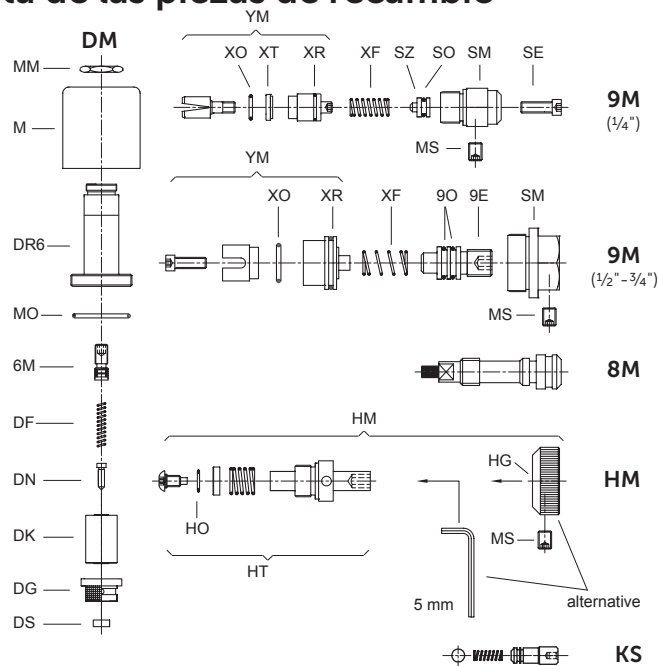
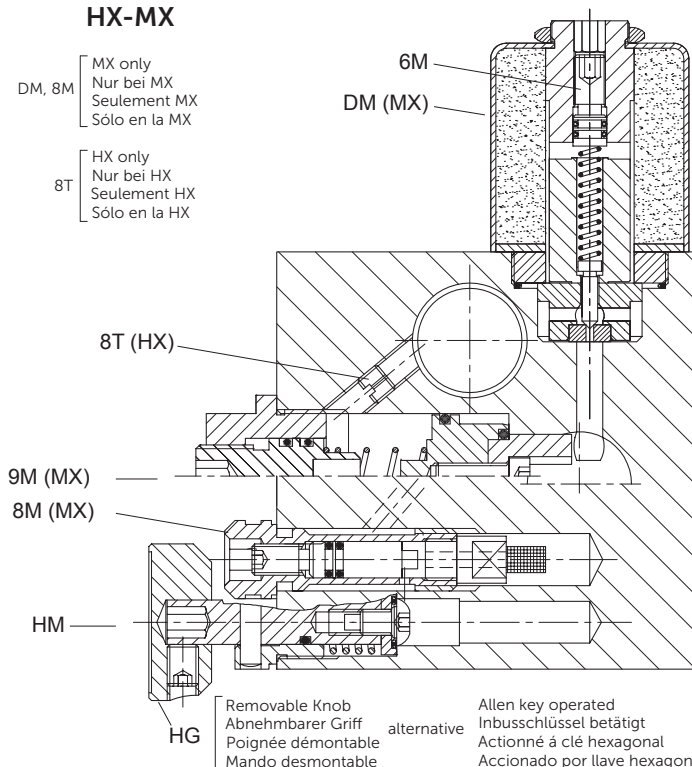
Ersatzteil-Liste Lista de las piezas de recambio

**HX
MX**

(GB)	(D)
Pos. No. Item	Pos. Nr. Benennung
MM Nut - Solenoid	MM Mutter - Magnetventil
M Coil - Solenoid (indicate voltage)	M Magnetspule (Spannung angeben)
DR6 Tube - Solenoid 'Down'	DR6 Rohr - Magnetventil 'ab' mit Einst. 6
MO O-Ring - Solenoid	MO O-Ring Magnetventil
DM 6M Adjustment - 'Down' Acceleration	DM 6M Anfahrddrossel 'ab'
DF Spring - Solenoid 'Down'	DF Feder - Magnetventil 'ab'
DN Needle - 'Down'	DN Nadel Magnetventil 'ab'
DK Core Solenoid	DK Kern - Magnetventil 'ab'
DG Seat Housing w. Screen Sol.'Down'	DG Sitzhalter mit Sieb - Magnetv. 'ab'
DS Seat Solenoid 'Down'	DS Sitzscheibe - Magnetventil 'ab'
8M 8M Adjustment - 'Down' Deceleration	8M 8M Abbremsdrossel 'ab'
ZS ZS Connecting Screw	ZS ZS Anschlussschraube
FZ FZ Screwed Filter	FZ FZ Einschraubfilter
MS Locking Screw	MS Sicherungsschraube
HM HG Knurled Knob (self-closing)	HM HG Griff-Handablass - selbstschließend
HO Seal Manual Lowering	HO Dichtung - Handablass
MS Locking Screw	MS Sicherungsschraube
9E Adjustment - 'Down' Leveling	9E Einstellschraube - Schleichfahrt
9O O-Ring - Adjustment	9O O-Ring - Einstellschraube
XF Spring - 'Down' Valve	XF Feder - Senkventil
XR O-Ring - 'Down' Valve	XR O-Ring - Senkkolben
9M YM 'Down' Valve	9M YM Senkkolben
XO Seal - 'Down' Valve	XO Dichtung - Senkkolben
XT Disc - O-Ring - Centering	XT Teller - O-Ringzentrierung
SE Adjustment Screw	SE Einstellschraube
SM Hexagonal	SM Sechskant
SO O-Ring Nipple	SO O-Ring - Zapfen
SZ Nipple	SZ Zapfen

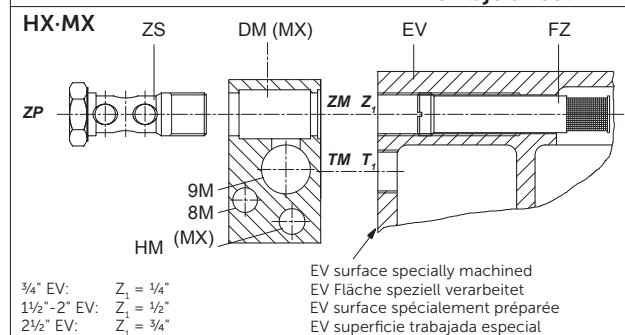
(F)	(E)
Pos. No. Dénomination	Pos. No. Denominación
MM Ecrou de l'électro - vanne	MM Tuerca - válvula magnética
M Bobine (indiquer tension)	M Bobina magnética (indicar tensión)
DR6 Tube de l'électro - vanne 'descente'	DR6 Tubo - válvula magnética 'bajada'
MO O-Ring de l'électro - vanne	MO Anillo O - válvula magnética
DM 6M Vis d'ajustement - 'descente'	DM 6M Tornillo arranque en 'bajada'
DF Ressort de l'électro - vanne 'desc.'	DF Muelle - válvula magnética 'bajada'
DN Pointeau de l'électro - vanne 'desc.'	DN Aguja - válvula magnética 'bajada'
DK Noyau de l'électro - vanne 'desc.'	DK Núcleo - válvula magnética 'bajada'
DG Support de siège avec filtre - 'desc.'	DG Soporte - asiento con tamiz 'baj.'
DS Disque de siège d'électro-vanne 'de.'	DS Disco asiento - válv. mag. 'bajada'
8M 8M Etrangleur de ralentissement 'desc.'	8M 8M Tornillo frenado en 'bajada'
ZS ZS Vis de raccordement	ZS ZS Tornillo de conexión
FZ FZ Filtre insérable	FZ FZ Filtro para atornillar
MS Vis de fixation	MS Tornillo de afianzamiento
HM HG Poignée de 'descente' manuelle	HM HG Mando accionamiento desc. manual
HO Joint étanche de 'descente'	HO Junta - descarga manual
MS Vis de fixation	MS Tornillo de afianzamiento
9E Vis d'ajustage de petite vitesse	9E Tornillo regulación
9O O-Ring de la vis d'ajustage	9O Anillo O - Tornillo regulación
XF Ressort de la valve 'descente'	XF Muelle - válvula de 'bajada'
XR O-Ring - Piston de 'descente'	XR Anillo O - émbolo 'bajada'
9M YM Piston de 'descente'	9M YM Émbolo de 'bajada'
XO Joint étanche du piston de 'desc.'	XO Junta - émbolo de 'bajada'
XT Disque - O-Ring de centrage	XT Platillo centrage junta O
SE Vis d'ajustage	SE Tornillo de regulación
SM Vis six pans creux	SM Hexágono
SO O-Ring du tenon	SO Junta O - pivote
SZ Tenon	SZ Pivote

HX-MX



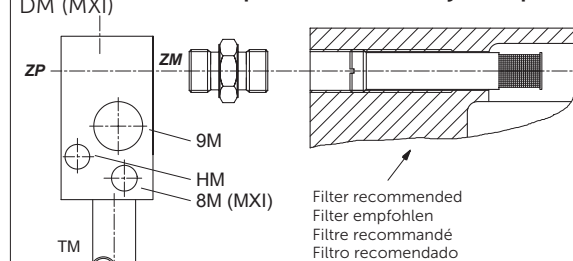
Direct mounted Montée directe

Direktanbau Montaje direct



HXI-MXI: Indirect mounted Montée indépendante

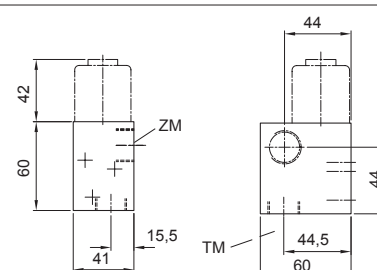
Separater Anschluss Montaje independiente



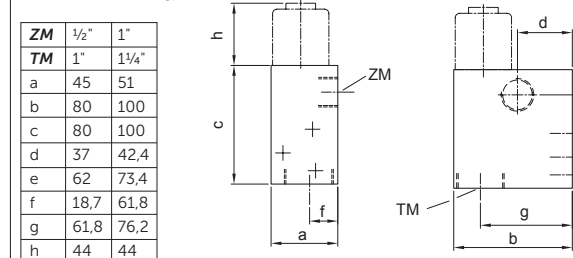
HXI-MXI: 1/4"

ZM: 1/2"

TM: 1/2"



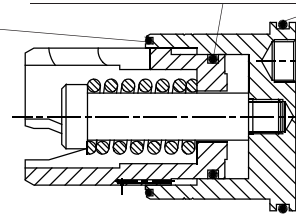
HXI-MXI: 1/2" & 3/4"





Can cause down leak:
replace complete flow guide.
Kann ein Leck verursachen:
Senkkolben ersetzen.
Peut causer une fuite:
remplacer le piston descente.
Puede causar escape:
Cambie embolo bajada.

EV 3/4" 9.00 x 1.50
EV 1/2" 25.00 x 2.00



Cannot cause down leak.
Kann kein Leck verursachen.
N'est pas la cause de fuites en descente.
No puede causar derrame.

EV 3/4" 26.00 x 2.00
EV 1/2" 39.34 x 2.62

not recommended for Pmax > 50 bar
nicht empfohlen für Pmax > 50 bar
50 bar □ n üzerinde tavsiye edilmez

CX Pressure Compensated Down Flow Guide recommended for use to maintain down speed within close limits when empty to full loading ratios of the elevator exceed 1:2.5. The CX is interchangeable with the standard flow guide and is selectable using the same chart.

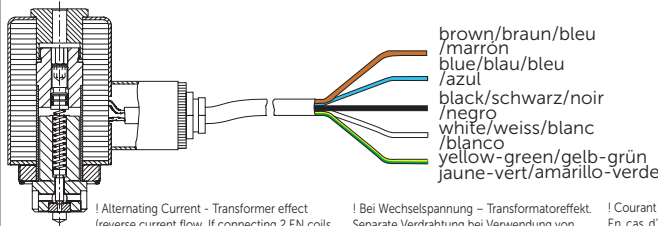
CX soupape de descente compensée par pression recommandée pour conserver une vitesse de descente constante pour des variations de charge de vide à maxi dans un rapport de 1:2.5. La CX est interchangeable avec le piston standard, et, sélectionné selon le même diagramme.

CX Druckkompensierter Senkkolben wird empfohlen, um die Senkgeschwindigkeit bei Veränderung von Leerlast zu Vollast im Verhältnis 1:2.5 konstant zu halten. Der CX ist mit dem Standardkolben austauschbar und nach dem gleichen Diagramm auswählbar.

CX Válvula de bajada compensada se recomienda, para mantener la velocidad de bajada constante, cuando se cambie la carga vacía o llena en la proporción de 1:2.5. El CX es intercambiable con el pistón estándar y es selectivo según el mismo diagrama.



EN Wiring Instructions
EN instruction des fils conducteurs électriques



Hinweise zur Verdrahtung von EN
EN instrucción cableado

Standard Cable

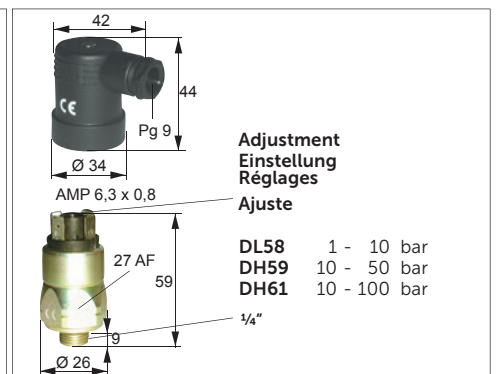
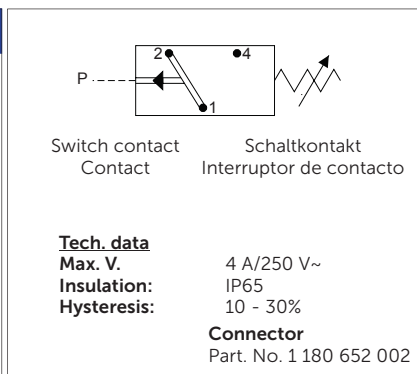
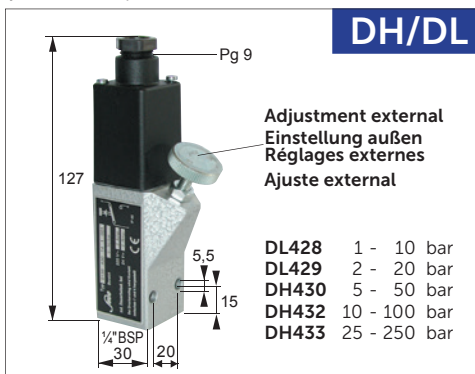
- Main coil / Hauptspule / Bobine principal / Bobina principal
- Zero wire / Null Leiter / Conducteur neutre / Conductor neutro
- Emergency coil / Notstromspule / Bobine descente de secours / Válvula de emergencia
- Zero wire / Null Leiter / Conducteur neutre / Conductor neutro
- + Ground/Erde / Terre / Tierra

EN Emergency Power Coil. Should there be an interruption of the main power to the elevator, the emergency lowering coil EN, fed by 12 V DC or 24 V DC supply, enables a command to be given from the car or elsewhere to lower the car to the floor below. When ordering please state main and emergency voltages. Is the main Voltage below 80 V, all emergency coils have 4 wires (black, blue, brown, white). Is the main Voltage above 80 V, a 5th wire (green-yellow) is added for the ground.

EN bobine descente de secours. En cas de coupure du courant principal de l'ascenseur, la bobine EN alimentée en courant de secours 12 V DC ou 24 V DC, permet, à la suite d'une commande l'évacuation de la cabine à l'étage inférieur désiré. A la commande préciser s'il vous plait la tension nominale ainsi que la tension de secours. Si la tension principale est d'en bas de 80 V toutes les bobines descente de secours ont 4 câbles (noir, bleu, brun, blanc). Si la tension principale est sur 80 V s'ajoute encore un 5. câble (jaune-vert) pour la terre.

EN Notstromspule. Im Falle einer Stromunterbrechung in der Hauptstromleitung zum Aufzug, ermöglicht die Notstromspule EN, die durch 12 V = oder 24 V = angetrieben wird, ein Signal aus der Kabine oder woanders her zu geben, um die Kabine zur nächsten Etage abzusinken. Bei Bestellung bitte die Haupt- sowie die Notstromspannung angeben. Unter 80 V Hauptspannung besitzen alle Notstromspulen 4 Leiter (schwarz, blau, braun, weiss). Über 80 V Hauptspannung kommt noch ein 5. Leiter (grün-gelb) für die Erde hinzu.

EN Bobina de corriente de emergencia. En caso de interrupción del conductor principal de corriente para el ascensor, la válvula de emergencia EN, accionamiento por voltajes de 12 V DC o 24 V DC, posibilita que se pueda dar un comando de abajo de la cabina o otro sitio. Cuando pidan, por favor, nos den el voltaje principal y el de emergencia. De menos de 80 V de voltaje principal todas las bobinas de emergencia tienen 4 conducciones (negro, azul, marrón, blanco). Más de 80 V de voltaje principal se sobreviene aún una 5. dirección (amarillo-verde) por la tierra.



DL, DH Pressure Switches

DL switches are used to signal a drop in hydraulic pressure below the normal operating pressure. DH switches are used to signal hydraulic pressure above the normal operating pressure.

DL, DH Interrupteur à pression

DL indique une baisse de pression par rapport à la pression de référence. DH indique une augmentation de pression par rapport à la pression de référence.

DL, DH Druckschalter

DL Schalter signalisieren ein Absinken des hydraulischen Druckes unter gewünschte Werte. DH Schalter signalisieren ein Ansteigen des hydraulischen Druckes über gewünschte Werte.

DL, DH Interruptores de presión

DL Interruptores signalizan una disminución de la presión hidráulica por debajo de valores deseados. DH Interruptores signalizan una aumentación de la presión hidráulica por encima de valores deseados.

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Designer and Manufacturer of the highest
quality control valves & safety components
for hydraulic elevators

Pressure Compensated Down Valve for EV Valves

Druckkompensiertes Senkventil für EV Ventile



Standard Down Valve X
Standard Senkventil X



Compensated Down Valve CX
Kompensiertes Senkventil CX

Down valves X and CX are interchangeable

Senkventile X und CX sind untereinander austauschbar

'X' Advantages

Smoother operation
Shorter travel time with load
Only the o-rings need to be serviced
Lower cost

'X' Vorteile

Weichere Fahreigenschaften
Kürzere Fahrtzeit mit Zuladung
Nur die O-Ringe müssen bei Wartungen getauscht werden
Geringere Kosten

'CX' Advantages

No overspeeding with excessive load

'CX' Vorteile

Keine überhöhte Geschwindigkeit bei übermäßiger Zuladung

Application

We recommend using the compensated 'CX' down valve once the maximum pressure is more than 2.5 times of the minimum pressure.

Compensated down valves are available for all EV sizes.

Anwendung

Wir empfehlen den Einsatz des kompensierten Senkventils 'CX', sobald der Gesamtdruck der beladenen Kabine das 2,5-fache der leeren Kabine überschreitet.

Kompensierte Senkkolben sind für alle Größen des EV erhältlich.

Performance comparison

Standard Down Valve X:

An increase in load of 100% will cause an increase in down speed of approximately 60%.

Compensated Down Valve:

An increase in load of 100% will cause a change of down speed within $\pm 10\%$.

Not recommended when system pressure exceeds above 50 bar.

Leistungsvergleich

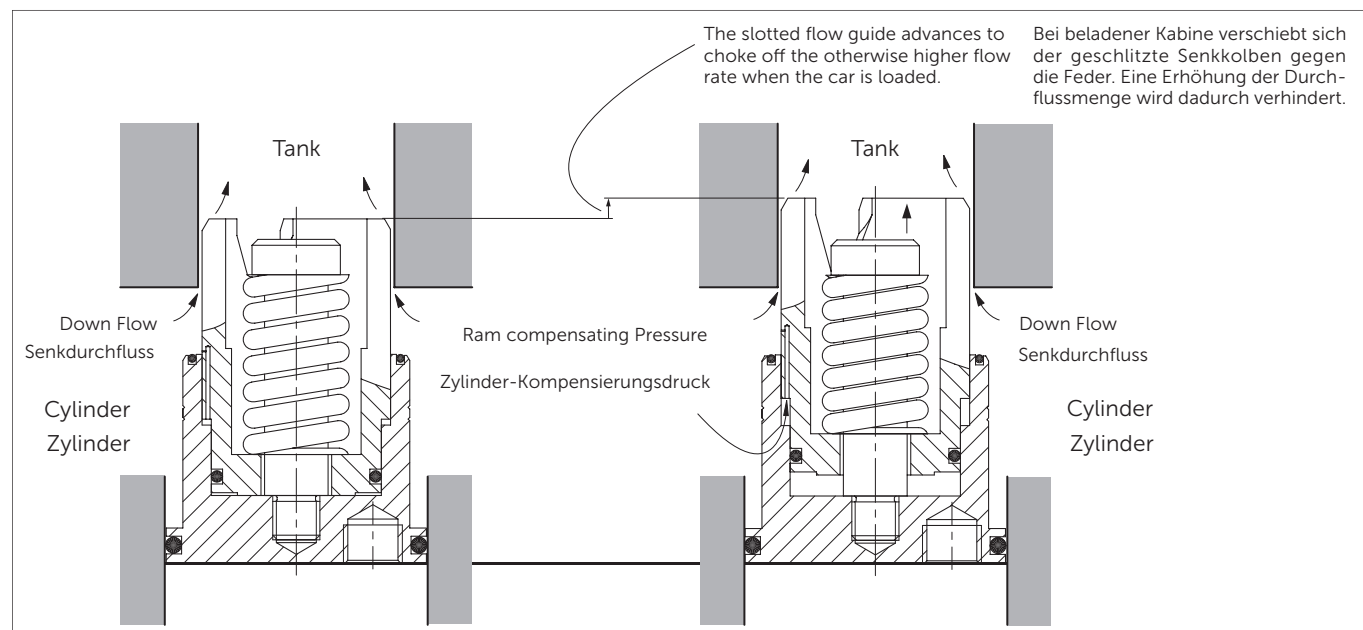
Standard Senkventil X:

Eine Erhöhung der Zuladung von 100% verursacht eine Erhöhung der Senkgeschwindigkeit um etwa 60%.

Kompensiertes Senkventil:

Eine Erhöhung der Zuladung von 100% verursacht eine Veränderung der Senkgeschwindigkeit von $\pm 10\%$.

Eine Kompensierung erfolgt bis 50bar.



KS Slack Rope Valve (KS) option for EV, KV and L10

GB

Purpose

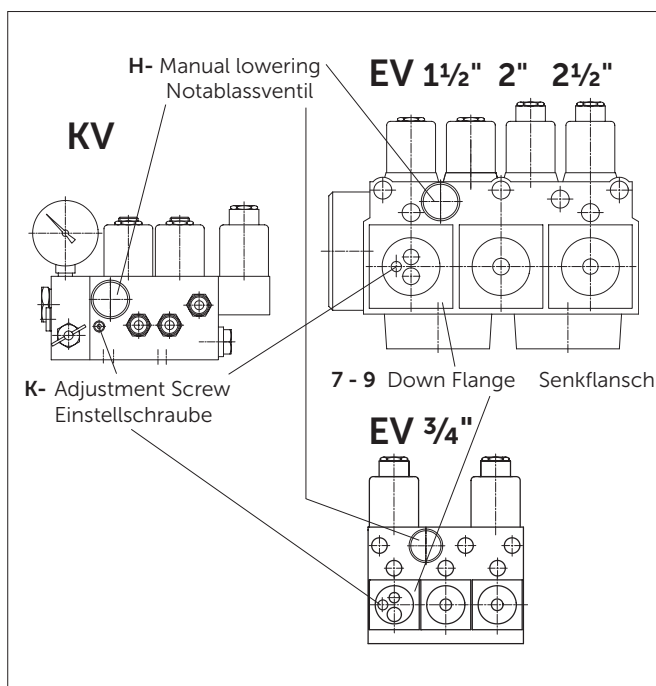
In the case of the operation of the safeties in a 2:1 hydraulic lift system where the weight of the car is no longer carried by the ropes, the electrical supply to the elevator should automatically be switched off. As the ram comes to a stop, usually after about 60 cm, a limited slack rope condition will occur. The **KS** Slack Rope Valve avoids the RAM being lowered by the opening of the manual lowering valve which would otherwise cause additional slack rope. The **KS** Slack Rope Valve prevents the pressure holding up the RAM from being evacuated through the manual lowering valve. The **KS** adjustment is next to adjustment **7** and **9** on the Down Flange on the EV valve and on the left hand side directly under the Manual Lowering on the KV valve.

Function

The **KS** valve is adjusted to a pressure just above the pressure produced by the weight of the ram. When under normal operating conditions, the weight of the car acts upon the ram through the 2:1 roping, the resulting pressure is sufficient to open the poppet of the **KS** valve when the manual lowering **H** is opened, allowing the car to descend as required. When however the 'safeties' have operated and only the weight of the ram and sheave block are acting upon the hydraulic system, the resulting pressure is too low to open the **KS** valve. The ram and sheave block can not be lowered.

Adjustment

The **KS** is adjusted with a 3 mm Allen key by turning the screw **K** 'in' for higher pressure and 'out' for lower pressure. With **K** turned all the way 'in', then half a turn back out, the unloaded car should descend when the **D** coil alone is energized. Should the car not descend, **K** must be turned out until the car just begins to descend, then turned out a further half turn to ensure that with cold oil, the car can be lowered as required.



Kolbensicherung (KS) Option für EV, KV und L10



D

Zweck

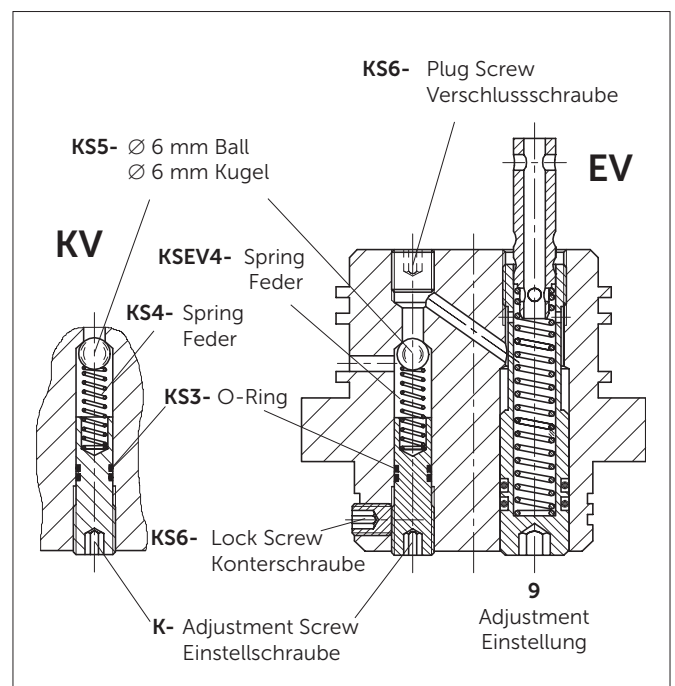
Bei einem 2:1-Aufzug, der in Fang geht, kommt der unbeladene Kolben erst nach ca. 60 cm zum Stillstand, sodass ein gewisses 'Schlaffseil' entsteht. Um zu verhindern, dass beim Öffnen des Notablassventils der Aufzugs-Kolben absinkt und die Seile schlaff werden, ist die Kolbensicherung **KS** vorzusehen. Diese befindet sich neben den Einstellungen **7** und **9** im Senkventilflansch am EV Ventil und direkt links unter dem Notablass am KV.

Funktion

Durch den Druck, verursacht vom Gesamtgewicht des Zylinderkolbens mit Seil-Joch einerseits und der Aufzugskabine andererseits, öffnet sich die Kolbensicherung beim Betätigen des Notablassventils **H**. Dies verursacht das erwünschte Absinken des Aufzuges. Wenn jedoch die Aufzugskabine im "Fang" ist, wirkt lediglich das Gewicht von Aufzugs-Kolben mit Seil-Joch auf das hydraulische System, wodurch zu wenig Druck entsteht, um die Kolbensicherung **KS** zu öffnen. Der Aufzugs-Kolben mit Seil-Joch bleibt stehen.

Einstellung

Zur Einstellung der Kolbensicherung (mit einem 3 mm Inbusschlüssel) vorher die Konterschraube lösen (nur bei EV). Anschließend, je nach Notwendigkeit, die Einstellschraube **K** hinein- (höherer Druck) oder herausdrehen (niederer Druck). Mit **K** ganz hineingedreht, dann eine halbe Umdrehung zurück, soll der unbeladene Aufzug abwärts fahren, während nur Spule **D** unter Strom steht. Bleibt der Aufzug noch stehen, so muss die Einstellschraube **K** herausgedreht werden bis der Aufzug gerade noch fährt, dann eine weitere halbe Umdrehung herausdrehen, damit sich der Aufzug auch bei kaltem Öl absenken lässt. Konterschraube wieder sichern.





Sécurité contre le mou de câble 'KS' pour EV, KV et L10 d'ascenseur

(F)

Usage

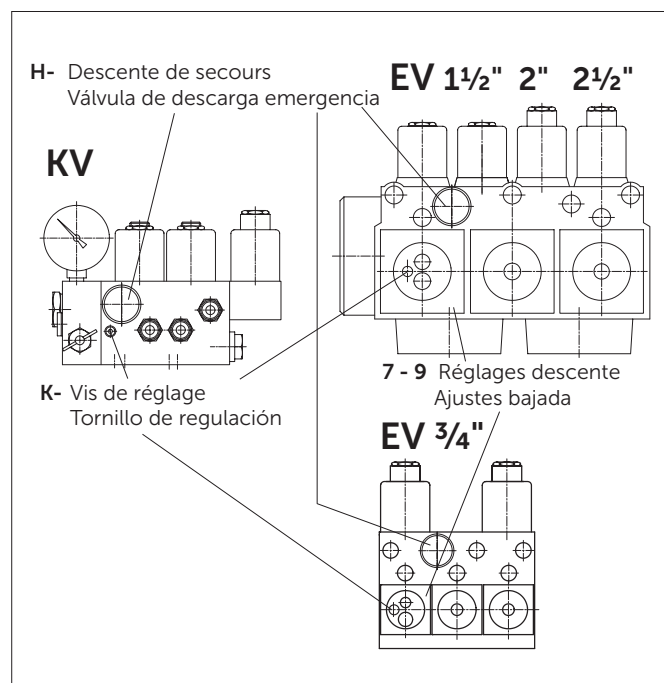
La valve de piston **KS** est conçue pour empêcher à un système d'élévation 2:1 en prise de sécurité que lors de l'ouverture du descente de secours, le piston d'élévation descende et que les câbles se détendent. Cette valve se trouve à côté des réglages **7** et **9** dans le manchon de la valve de descente sur la valve EV et directement à gauche sous le descente de secours du KV.

Fonction

La valve du piston s'ouvre sous l'effet de la pression, causée par le poids total du piston cylindrique avec joug du câble d'une part et de la cabine d'élévation d'autre part, si l'on actionne descente de secours **H**. Ceci provoque la descente souhaitée de l'ascenseur. Cependant si la cabine de l'ascenseur est en „prise de sécurité”, seul le poids du piston d'élévation avec joug du câble agira sur le système hydraulique, ce qui aura pour conséquence que la pression pour ouvrir le sécurité contre le mou de câble **KS** sera trop faible. Le piston d'élévation et le joug du câble resteront immobiles.

Réglage

On pourra ajuster le sécurité contre le mou de câble avec une clé mâle coudée de 3 mm en desserrant le contre-écrou (seulement à EV) et en faisant entrer (pression plus élevée) ou sortir (pression plus faible) la vis de réglage **K**. Alors que **K** est entièrement entrée et qu'on la dévisse ensuite d'une demi-rotation, l'ascenseur sans charge devra descendre alors que seule la bobine **D** est active. Si l'ascenseur reste immobile, il faut dévisser la vis de réglage **K** jusqu'à ce que l'ascenseur se mette en mouvement puis dévisser encore d'une autre demi-rotation afin que l'ascenseur descende même si l'huile est froide.



Válvula de aflojamiento cables 'KS' para válvulas de control de ascensor EV, KV y L10

(E)

Objetivo

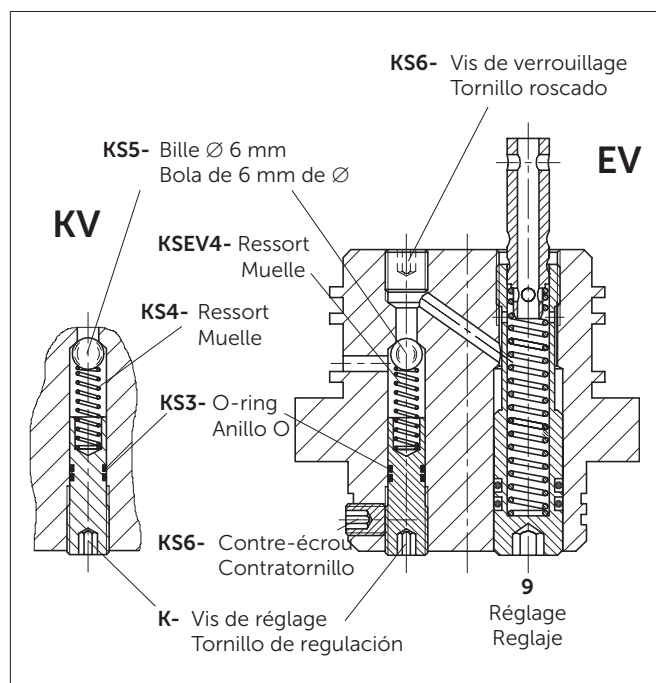
La válvula de aflojamiento cables **KS** sirve para evitar que en un ascensor con tracción 2:1 y el émbolo cual está detenido por maniobra de socorro manual, descienda al abrir la bajada y se aflojen los cables. Está ubicado junto a los ajustes **7** y **9** en la brida de la válvula de bajada de la válvula EV y justo a la izquierda debajo de la bajada maunal de KV.

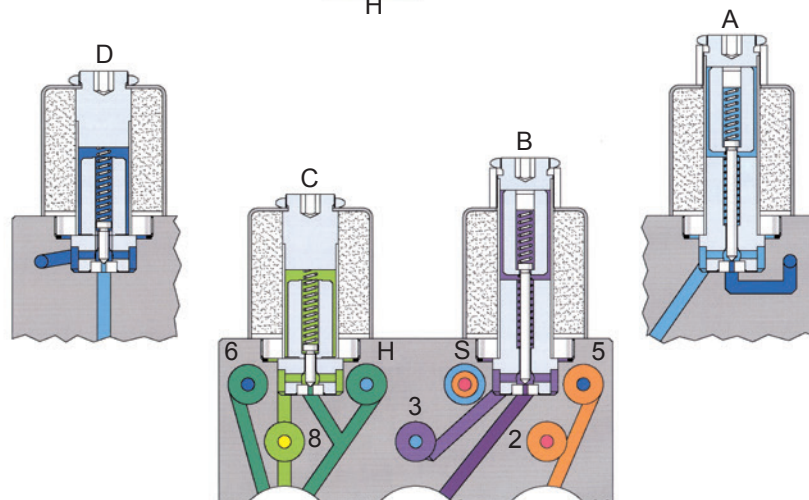
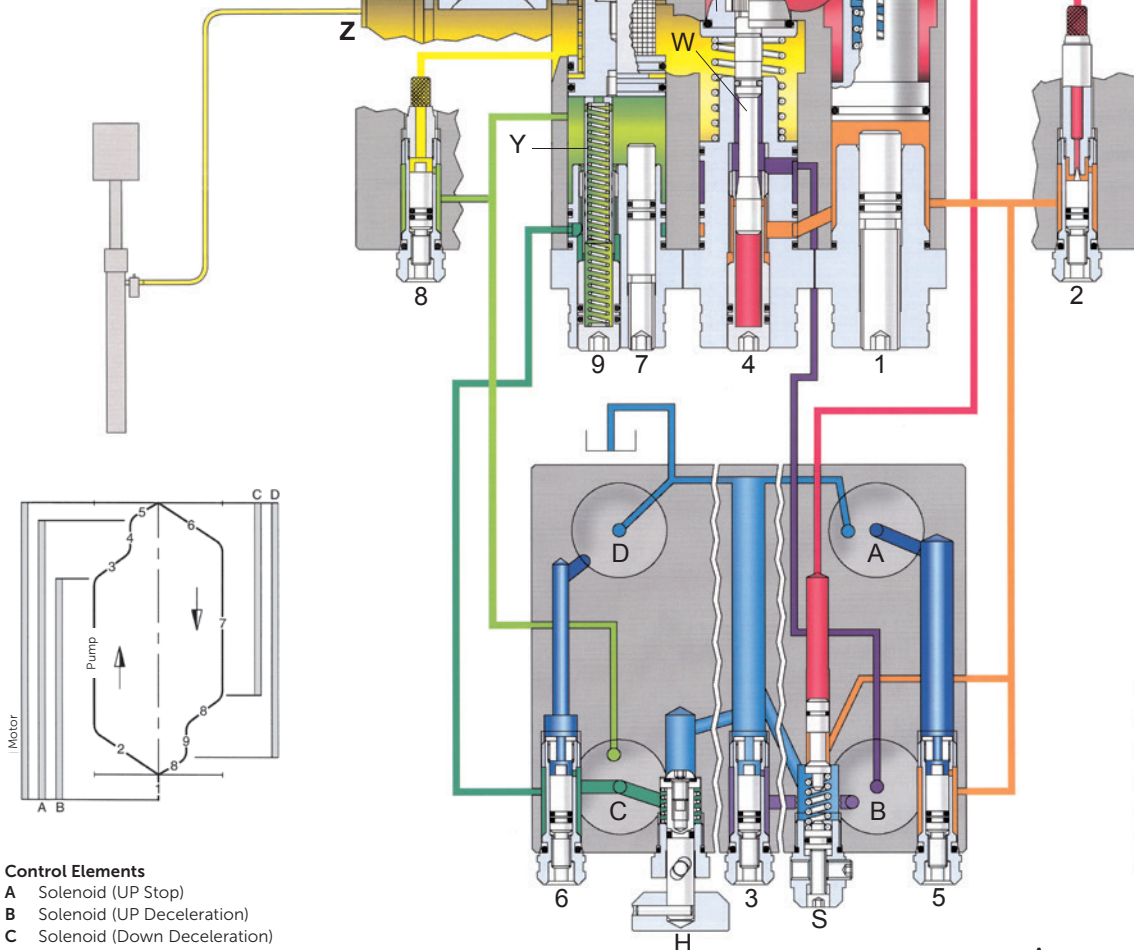
Función

La presión originada por el peso total del pistón del cilindro con la polea de los cables y por la cabina del ascensor provoca la apertura de la válvula de aflojamiento cables al accionar la válvula de bajada manual **H**. Así se consigue el descenso deseado del ascensor. Sin embargo, cuando la cabina del ascensor se encuentra detenida por maniobra de socorro manual, sobre el sistema hidráulico sólo ejerce peso el pistón del ascensor con la polea de los cables, por lo que la presión generada no es suficiente para abrir la válvula de aflojamiento cables **KS**. Consecuentemente, el pistón del ascensor con el yunque de los cables se queda detenido.

Ajuste

La válvula de aflojamiento cables se regula con una llave Allen de 3 mm soltando el contratornillo (sólo EV) y enroscando (más presión) o desenroscando (menos presión) el tornillo de regulación **K**. Cuando **K** se halla completamente enroscado y se da media vuelta en sentido inverso, el ascensor sin carga tiene que bajar, estando sólo bajo corriente la bobina **D**. Si el ascensor aún sigue parado, hay que desenroscar el tornillo de regulación **K** de tal forma que el ascensor aún se mueva; a continuación, desenroscar el tornillo una media vuelta más, para que el ascensor también pueda descender cuando el aceite esté frío.



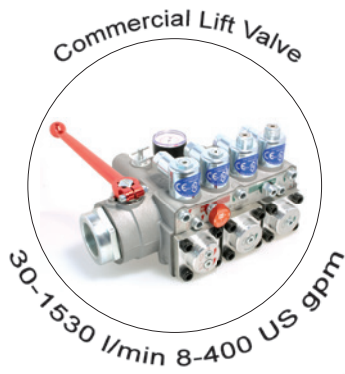


The finest elevator control valves

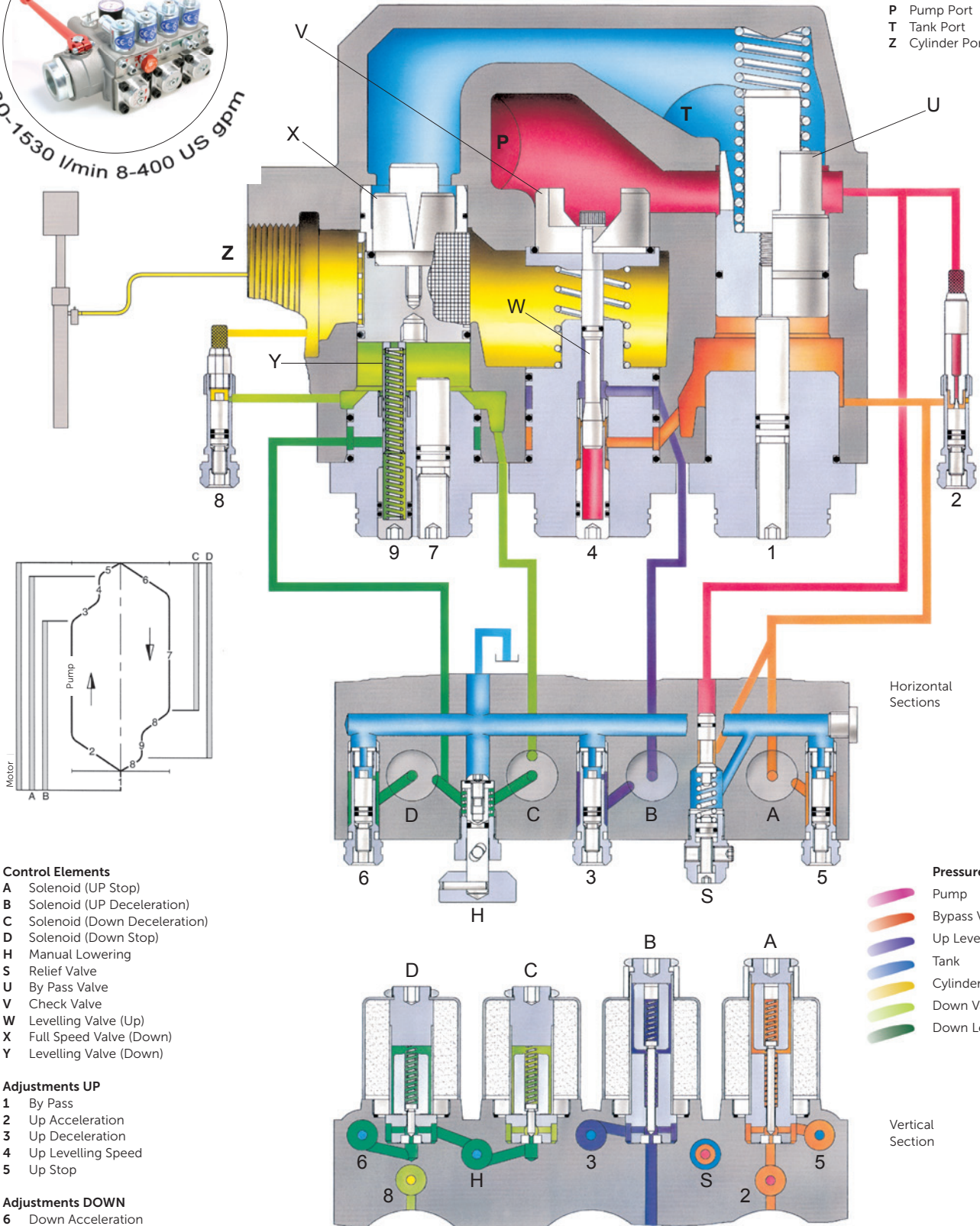
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Elevator Control Valves

1½" - 2½"



P Pump Port
T Tank Port
Z Cylinder Port



Control Elements

- A Solenoid (UP Stop)
- B Solenoid (UP Deceleration)
- C Solenoid (Down Deceleration)
- D Solenoid (Down Stop)
- H Manual Lowering
- S Relief Valve
- U By Pass Valve
- V Check Valve
- W Levelling Valve (Up)
- X Full Speed Valve (Down)
- Y Levelling Valve (Down)

Adjustments UP

- 1 By Pass
- 2 Up Acceleration
- 3 Up Deceleration
- 4 Up Levelling Speed
- 5 Up Stop

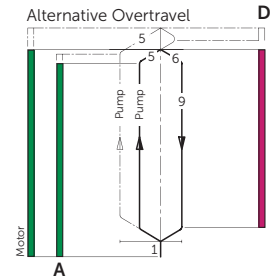
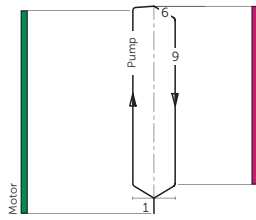
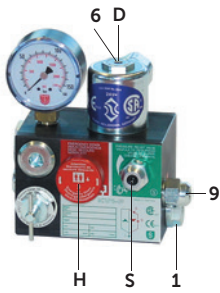
Adjustments DOWN

- 6 Down Acceleration
- 7 Down Full Speed
- 8 Down Deceleration
- 9 Down Levelling Speed

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Control Valves for Small Lifts



Control Elements

- A Solenoid Up
- D Solenoid Down
- F Main Filter
- H Manual Lowering
- L Gauge Shut Off Cock
- U Bypass Valve
- V Check Valve
- Y Down Levelling Valve

Adjustments

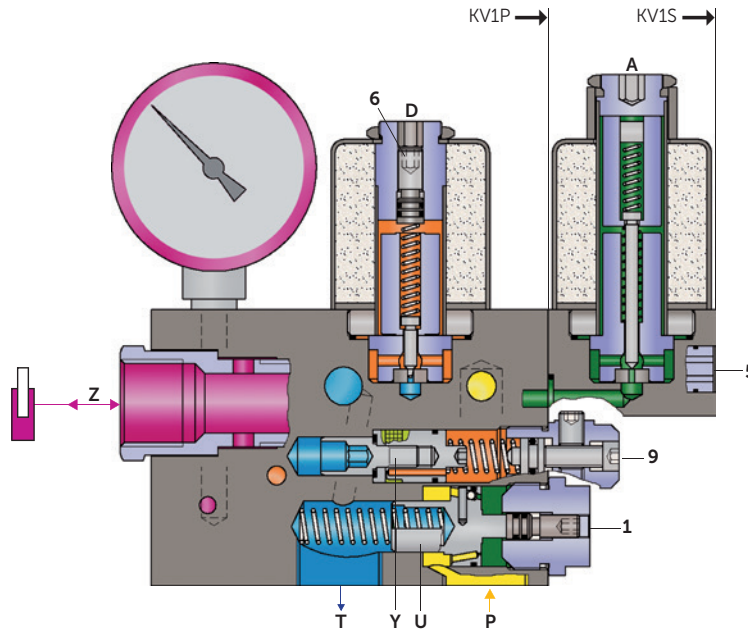
- 1 Bypass
- 5 Up Soft Stop
- 6 Down Acceleration
- 9 Down Speed
- S Relief Valve

Connections

- P Pump
- T Tank
- Z Cylinder

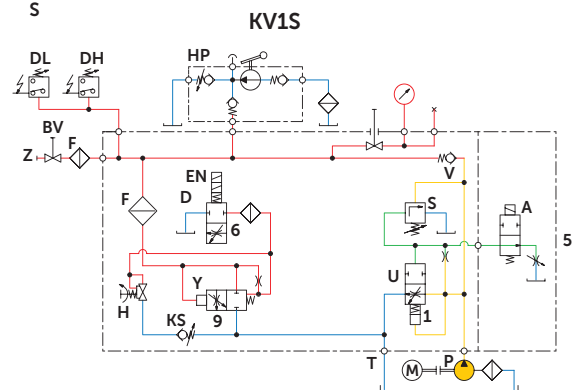
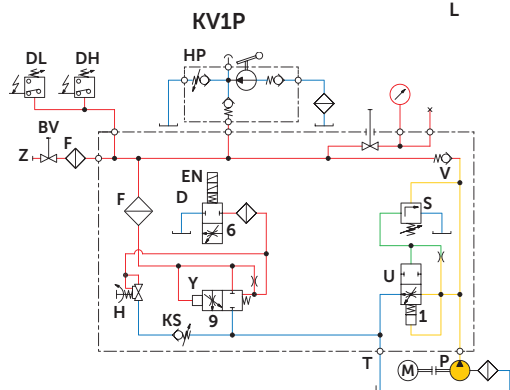
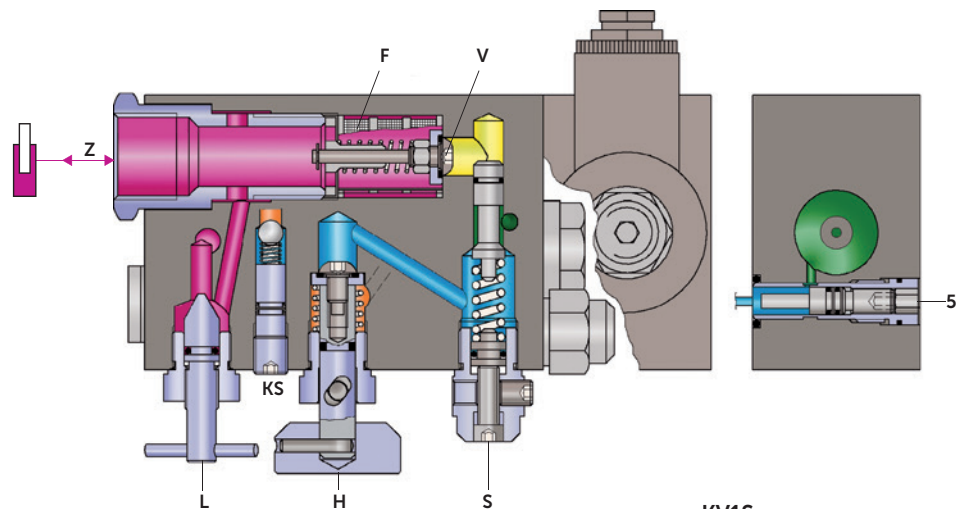
Options

- BV Ball Valve built in
- EN Emergency Power Solenoid
- HP Hand Pump H 13
- KS Slack Rope Valve
- DH Pressure Switch 10-100 bar
- DL Pressure Switch 1-10 bar
- CSA CSA Solenoids
- HA Manual Down Speed Adj.
- RS Pipe Rupture Valve
- ES Pipe Rupture Valve Switch



Pressures

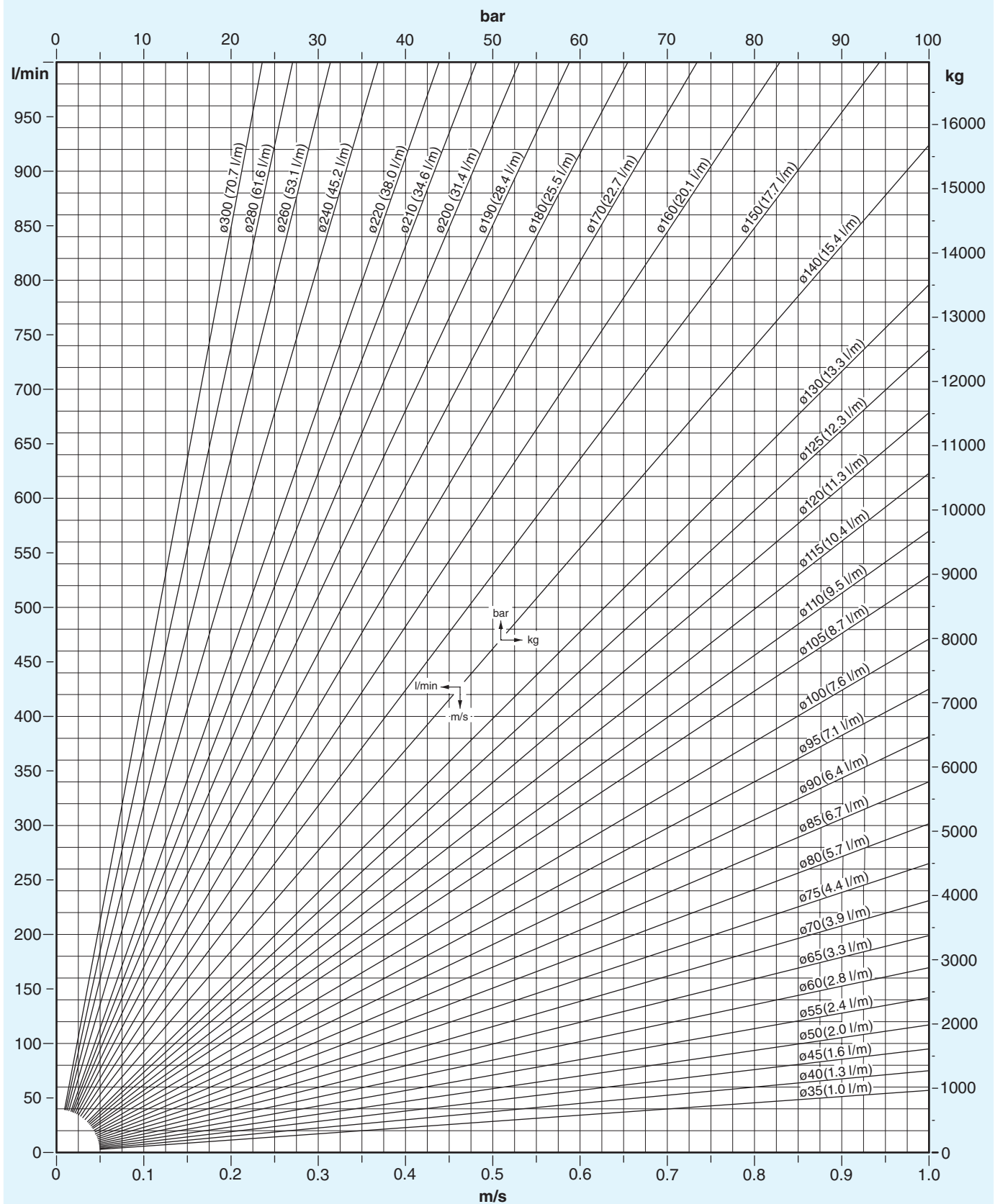
- Pump
- Pilot Up
- Tank
- Cylinder
- Pilot Down



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Cylinder-Pump Selection Chart



Flow - Pressure Tables



EN ISO 9001

Ram Ø • Area • Speed • Flow

m/s	0.05	0.10	0.15	0.20	0.25	0.30	0.35	0.40	0.45	0.50	0.55	0.60	0.70	0.80	0.90	1.00
Ø mm cm²	l/min															
35 9.6	2.9	5.8	8.7	11.5	14	17	20	23	26	29	32	35	40	46	52	58
40 12.6	3.8	7.5	11.3	15.1	19	23	26	30	34	38	41	45	53	60	68	75
45 15.9	4.8	9.5	14.3	19.1	24	29	33	38	43	48	52	57	67	76	86	95
50 19.6	5.9	11.8	17.7	23.6	29	35	41	47	53	59	65	71	82	94	106	118
55 23.8	7.1	14.3	21.4	28.5	36	43	50	57	64	71	78	86	100	114	128	143
60 28.3	8.5	17.0	25.4	33.9	42	51	59	68	76	85	93	102	119	136	153	170
65 33.2	10.0	19.9	29.9	39.8	50	60	70	80	90	100	110	119	139	159	179	199
70 38.5	11.5	23.1	34.6	46.2	58	69	81	92	104	115	127	139	162	185	208	231
75 44.2	13.3	26.5	39.8	53.0	66	80	93	106	119	133	146	159	186	212	239	265
80 50.3	15.1	30.2	45.2	60.3	75	90	106	121	136	151	166	181	211	241	271	302
85 56.7	17.0	34.0	51.1	68.1	85	102	119	136	153	170	187	204	238	272	306	340
90 63.6	19.1	38.2	57.3	76.3	95	115	134	153	172	191	210	229	267	305	344	382
95 70.9	21.3	42.5	63.8	85.1	106	128	149	170	191	213	234	255	298	340	383	425
100 78.5	23.6	47.1	70.7	94.2	118	141	165	188	212	236	259	283	330	377	424	471
105 86.6	26.0	52.0	77.9	103.9	130	156	182	208	234	260	286	312	364	416	468	520
110 95.0	28.5	57.0	85.5	114.0	143	171	200	228	257	285	314	342	399	456	513	570
115 103.9	31.2	62.3	93.5	124.6	156	187	218	249	280	312	343	374	436	499	561	623
120 113.1	33.9	67.9	101.8	135.7	170	204	238	271	305	339	373	407	475	543	611	679
125 122.7	36.8	73.6	110.4	147.3	184	221	258	295	331	368	405	442	515	589	663	736
130 132.7	39.8	79.6	119.5	159.3	199	239	279	319	358	398	438	478	557	637	717	796
140 153.9	46.2	92.4	138.5	184.7	231	277	323	369	416	462	508	554	647	739	831	924
150 176.7	53.0	106.0	159.0	212.1	265	318	371	424	477	530	583	636	742	848	954	1060
160 201.1	60.3	120.6	181.0	241.3	302	362	422	483	543	603	664	724	844	965	1086	1206
170 227.0	68.1	136.2	204.3	272.4	340	409	477	545	613	681	749	817	953	1090	1226	1362
180 254.5	76.3	152.7	229.0	305.4	382	458	534	611	687	763	840	916	1069	1221	1374	1527
190 283.5	85.1	170.1	255.2	340.2	425	510	595	680	766	851	936	1021	1191	1361	1531	1701
200 314.2	94.2	188.5	282.7	377.0	471	565	660	754	848	942	1037	1131	1319	1508	1696	1885
210 346.4	103.9	207.8	311.7	415.6	520	623	727	831	935	1039	1143	1247	1455	1663	1870	2078
220 380.1	114.0	228.1	342.1	456.2	570	684	798	912	1026	1140	1254	1368	1597	1825	2053	2281
240 452.4	135.7	271.4	407.2	542.9	679	814	950	1086	1221	1357	1493	1629	1900	2171	2443	2714
260 530.9	159.3	318.6	477.8	637.1	796	956	1115	1274	1434	1593	1752	1911	2230	2548	2867	3186
280 615.8	184.7	369.5	554.2	738.9	924	1108	1293	1478	1663	1847	2032	2217	2586	2956	3325	3695
300 706.9	212.1	424.1	636.2	848.2	1060	1272	1484	1696	1909	2121	2333	2545	2969	3393	3817	4241

Ram Ø • Area • Load • Pressure

kg	500	750	1000	1500	2000	2500	3000	3500	4000	4500	5000	6000	7000	8000	9000	10000
Ø mm cm²	bar															
35 9.6	51	76	102	153	204	255	306	357	408	459	510	612	714	816	918	1020
40 12.6	39	59	78	117	156	195	234	273	312	351	390	468	546	625	703	781
45 15.9	31	46	62	93	123	154	185	216	247	278	308	370	432	493	555	617
50 19.6	25	38	50	75	100	125	150	175	200	225	250	300	350	400	450	500
55 23.8	21	31	41	62	83	103	124	145	165	186	206	248	289	330	372	413
60 28.3	17	26	35	52	69	87	104	121	139	156	173	208	243	278	312	347
65 33.2	15	22	30	44	59	74	89	103	118	133	148	177	207	237	266	296
70 38.5	13	19	26	38	51	64	76	89	102	115	127	153	178	204	229	255
75 44.2	11	17	22	33	44	56	67	78	89	100	111	133	155	178	200	222
80 50.3	9.8	15	20	29	39	49	59	68	78	88	98	117	137	156	176	195
85 56.7	8.6	13	17	26	35	43	52	61	69	78	86	104	121	138	156	173
90 63.6	7.7	12	15	23	31	39	46	54	62	69	77	93	108	123	139	154
95 70.9	6.9	10	14	21	28	35	42	48	55	62	69	83	97	111	125	138
100 78.5	6.2	9.4	13	19	25	31	38	44	50	56	62	75	87	100	112	125
105 86.6	5.7	8.5	11	17	23	28	34	40	45	51	57	68	79	91	102	113
110 95.0	5.2	7.7	10	16	21	26	31	36	41	47	52	62	72	83	93	103
115 103.9	4.7	7.1	9.4	14	19	24	28	33	38	43	47	57	66	76	85	94
120 113.1	4.3	6.5	8.7	13	17	22	26	30	35	39	43	52	61	69	78	87
125 122.7	4.0	6.0	8.0	12	16	20	24	28	32	36	40	48	56	64	72	80
130 132.7	3.7	5.5	7.4	11	15	19	22	26	30	33	37	44	52	59	67	74
140 153.9	3.2	4.8	6.4	9.6	13	16	19	22	26	29	32	38	45	51	57	64
150 176.7	2.8	4.2	5.6	8.3	11	14	17	19	22	25	28	33	39	44	50	56
160 201.1	2.4	3.7	4.9	7.3	9.8	12	15	17	20	22	24	29	34	39	44	49
170 227.0	2.2	3.2	4.3	6.5	8.6	11	13	15	17	19	22	26	30	35	39	43
180 254.5	1.9	2.9	3.9	5.8	7.7	9.6	12	14	15	17	19	23	27	31	35	39
190 283.5	1.7	2.6	3.5	5.2	6.9	8.6	10	12	14	16	17	21	24	28	31	35
200 314.2	1.6	2.3	3.1	4.7	6.2	7.8	9.4	11	13	14	16	19	22	25	28	31
210 346.4	1.4	2.1	2.8	4.2	5.7	7.1	8.5	9.9	11	13	14	17	20	23	26	28
220 380.1	1.3	1.9	2.6	3.9	5.2	6.5	7.7	9.0	10.3	12	13	16	18	21	23	26
240 452.4	1.1	1.6	2.2	3.3	4.3	5.4	6.5	7.6	8.7	9.8	11	13	15	17	20	22
260 530.9	0.9	1.4	1.8	2.8	3.7	4.6	5.5	6.5	7.4	8.3	9.2	11	13	15	17	19
280 615.8	0.8	1.2	1.6	2.4	3.2	4.0	4.8	5.6	6.4	7.2	8.0	9.6	11	13	14	16
300 706.9	0.7	1.0	1.4	2.1	2.8	3.5	4.2	4.9	5.6	6.2	6.9	8.3	9.7	11	13	14

in² = 6,45 cm²

1 in = 25,4 mm

1 m/s = 197 ft/min

1 Imp. gpm = 4,55 l/min

1 US gmp = 3,79 l/min

1 kg = 2,2 lbs

1 bar = 14,5 psi



Anja Blain CEO

Since half a century Blain Hydraulics has focused and specialized in flow control valves for hydraulic elevators. Blain is the largest supplier of elevator valves in the world with a large global footprint. At Blain safety, reliability and quality of our products are of utmost importance. As a pioneer, Blain has been building products which are above and beyond the standards. With product support in multiple languages and across different time zones, more than a million valves in operation worldwide endorse us a leading supplier of key elevator components.

At Blain, flow control is in our DNA, we don't just manufacture a valve, we engineer it.

Anja Blain (Managing Director/CEO)

BLAIN HYDRAULICS is the leading manufacturer of high quality hydraulic elevator products for five decades. Blain products have proven their safety and quality by possessing more than one third of the global market share and one million valves in operation in more than 75 countries worldwide.





ABOUT US

A brief history of Blain Hydraulics – incorporated in 1971 by Roy W. Blain



Roy W. Blain 1932-2014

Roy W. Blain was born in Salford, Manchester, in May 1932 and studied mechanical engineering at Salford Royal Technical College in northern England.

Following two years as a sailor in the Merchant Navy, he served in the Army for a further two years. He then pursued a career in industrial hydraulics, working first in England and then in Switzerland, Spain, and the USA. Upon arriving in Germany, he founded Blain Hydraulics. Roy W. Blain was a true pioneer in the field of hydraulic lifts. A genuine gentleman and kind-hearted visionary, he worked in the field of hydraulic lift technology for more than five decades.

1971-1980 During this decade Blain Hydraulics GmbH was founded in Heilbronn. With a modest infrastructure as well as man power, elevator control valves like EV & KV started rolling out initially with 1 person and eventually with 5 people on the outskirts of Heilbronn. To cater for growing demand, the factory was relocated within Heilbronn and steadily expanded.

1981-1990 During this period, Blain adds new KV models (small lift valves) to its range, to meet demand for small lifts in commercial and private settings. In addition, the L10 pressure relief valve is introduced as an additional safety valve. This product is now known as the UCM-A3 valve. Blain obtains CSA certification for the export of goods to North America. The company expands its infrastructure to meet growing customer demand.

1991-2000 Over the course of this decade, the company has modernised its machinery to make production more cost-effective and profitable. Blain obtained ISO 9001 certification and successfully obtained type approval for its pipe rupture valves in accordance with EU directives. The company's product range was expanded. Blain introduces the SEV electronic servo valve and other products, such as the MD fine levelling drive. Additional equipment, such as ball valves, is also added to the product range.

2001-2010 Blain is the first company worldwide to bring

explosion-proof MEX solenoid valves for lifts to market. The company becomes the world's largest manufacturer of lift control valves in terms of production capacity and installations. Additionally, new pipe rupture valves were introduced during this decade.

2011-2020 Blain launches the EV4 (VVVF control valve) in collaboration with YASKAWA as part of a joint venture. Exports of Blain products reach a new record high. In 2015, Blain expanded its presence in India by setting up Blain India. A partnership with Brazilian company DAIKEN ELEVADORES is formed, enabling Blain to increase its presence in the hydraulic lift market in South America. In summer 2017, Blain Hydraulics introduces the L20 as a new UCM-A3 valve. This enables existing installations to be modernised with less effort and at lower cost.

In 2018, Blain's presence is expanded to the Middle East and Africa with the establishment of a location in Turkey. With the founding of Blain Hydraulics USA in 2018, the company sends a strong signal of customer proximity in North America. The direct presence in the North American market enables faster, more personal and more efficient customer service. In 2019, Blain presents the next generation of smart valves, which offer users easy installation and monitoring via smartphone or tablet. This new product line consists of the SEV servo electronic valve and the EV40 VVVF valve. With these products, Blain Hydraulics is the first company to introduce smart technology to the hydraulic lift industry, once again demonstrating Blain Hydraulics' innovative strength.

2021-today Blain Hydraulics is investing in the sustainable design of its production site. The company's roofs are being renovated and fitted with photovoltaic systems. This enables the company to generate a large proportion of its own electricity. The workstations in the assembly department are being designed ergonomically. With the purchase of automatic storage systems more space at the site is created. In 2024, the product family will be expanded to include another safety component. The new iL 10-S is also TÜV tested and approved. While it differs visually from its predecessor, the iL 10, it has the same function. The company now has an international customer base spanning more than 75 countries. Blain currently employs around 120 people from over 14 countries.



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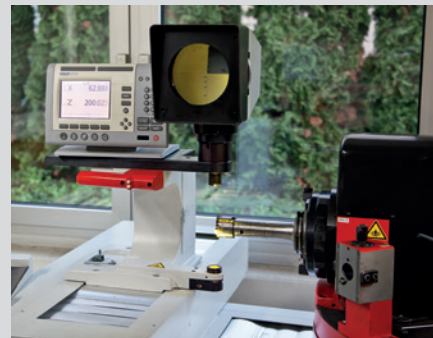
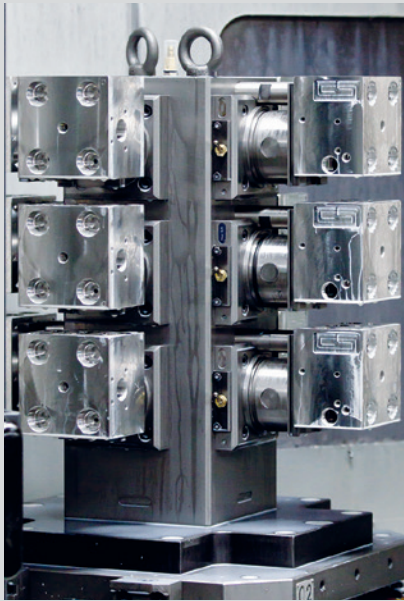
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BLAIN HYDRAULICS

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